Roll No.: | | | | | | | | | | | |

Test Date: 28-12-2025

Aakash
Medical | IIT-JEE | Foundations

B
Booklet Code

Aakash Delhi/NCR Zone

AIATS
ALL INDIA AAKASH TEST SERIES

for

Medical Entrance Exam - 2026

National Eligibility-cum-Entrance Test (NEET)

TEST No. 5

(XII Studying Students)

INSTRUCTIONS FOR CANDIDATES

1. Read each question carefully.
2. It is mandatory to use Blue/Black Ball Point Pen to darken the appropriate circle in the answer sheet.
3. Mark should be dark and should completely fill the circle.
4. Rough work must not be done on the answer sheet.
5. Do not use white-fluid or any other rubbing material on answer sheet. No change in the answer once marked is allowed.
6. Student cannot use log tables and calculators or any other material in the examination hall.
7. Before attempting the question paper, student should ensure that the test paper contains all pages and no page is missing.
8. Before handing over the answer sheet to the invigilator, candidate should check that Roll No. and Centre Code have been filled and marked correctly.
9. Immediately after the prescribed examination time is over, the answer sheet to be returned to the invigilator.
10. (i) The question paper consists of 180 compulsory questions (45 questions each in Physics and Chemistry and 90 questions in Biology (Botany & Zoology)). The total duration of the examination will be 180 minutes (3 Hrs). (ii) Each correct answer carries four marks. One mark will be deducted for each incorrect answer from the total score.

Note : It is compulsory to fill **Roll No.** and **Test Booklet Code** on answer sheet, otherwise your answer sheet will not be considered.

Test No. 5

TOPICS OF THE TEST

Physics	Laws of Motion, Work, Energy and Power, System of Particles & Rotational Motion
Chemistry	Thermodynamics, Equilibrium
Botany	Biological Classification, Morphology of Flowering Plants
Zoology	Breathing and Exchange of Gases, Body Fluids and Circulation, Excretory Products and Their Elimination

Aakash Medical|IIT-JEE|Foundations (AR-001)

MM : 720

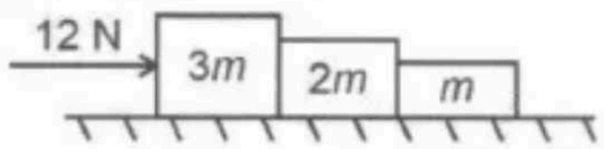
TEST - 05

Time : 3 Hrs.

[PHYSICS]

Choose the correct answer:

1. Three blocks of masses $3m$, $2m$ and m are pushed by a force of 12 N as shown in figure given below. The ratio of force which $2m$ exerts on m, to that exerted by $3m$ on $2m$ is



(1) 1 : 3 (2) 1 : 4

(3) 1 : 6 (4) 1 : 2

2. Consider the following statements A and B and identify the correct answer:

A : A force is said to be conservative, if work done by the force in moving a body from one position to another, depends on path followed between these two positions.

B : Force of friction is a conservative force.

(1) A is true but B is false

(2) A is false but B is true

(3) Both A and B are true

(4) Both A and B are false

3. A ball of mass 20 kg is thrown in air with speed $v_1 = 3$ m/s from a height of 20 cm and at a height of 40 cm its speed becomes $v_2 = 2$ m/s. The work done on the ball by air resistance is

(1) 25 J

(2) 20 J

(3) –15 J

(4) –10 J

4. A force $F = (6 + 2x)$ acts on a particle along x-direction where F is in newton and x in metre. The work done by this force during displacement of the particle from $x = 2$ m to $x = 5$ m, is

(1) 49 J (2) 59 J

(3) 69 J (4) 39 J

5. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : A safe turn by a cyclist should neither be fast nor be sharp.

Reason (R) : The bending angle of the cyclist from the vertical would decrease with increase in velocity.

In the light of the above statements, choose the **correct** answer from the options given below.

(1) **A** is true but **R** is false

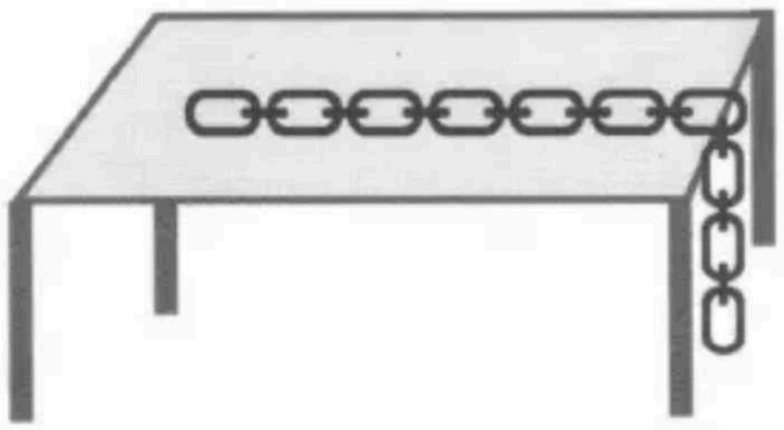
(2) **A** is false but **R** is true

(3) Both **A** and **R** are true and **R** is the correct explanation of **A**

(4) Both **A** and **R** are true but **R** is **not** the correct explanation of **A**

6. A chain of mass M and length 4 m is placed on a horizontal table such that, some part of it is hanging outside the table. If coefficient of friction between chain and table is 0.1, then the maximum length of chain which can be hung outside the table so that the chain does not slide is

Space for Rough Work



(1) 0.46 m (2) 0.36 m

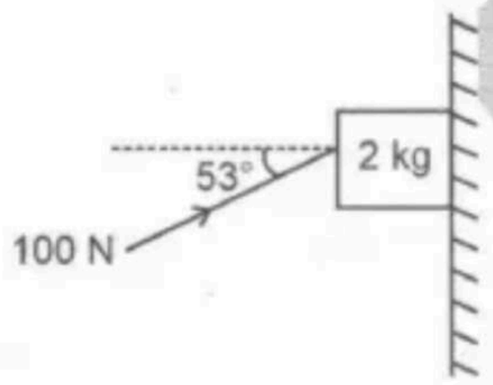
(3) 0.56 m (4) 0.86 m

7. A block of metal of mass 3 kg is resting on a frictionless horizonal plane. It is struck by a jet releasing water at a rate of 2 kg/s and at a speed of 5 m/s, in a direction perpendicular to the surface of block but along the horizontal plane. The water comes to rest relative to block after impact. The initial acceleration of the block is

(1) 5.33 m/s^2 (2) 20 m/s^2

(3) 10 m/s^2 (4) 3.33 m/s^2

8. A block of mass 2 kg is acted upon by a force F = 100 N at an angle 53° with horizontal against a vertical wall as shown below. The coefficient of static friction (μ_s) between the block and wall is 0.5 and coefficient of kinetic friction (μ_k) between block and wall is 0.4. The acceleration of the block is



(1) 20 m/s^2 (2) 45 m/s^2

(3) 10 m/s^2 (4) 18 m/s^2

9. A block of mass 2 kg is moving with velocity of 2 m/s on a horizontal rough surface, for which coefficient of friction is $\frac{3}{4}$, then distance travelled by the block before coming to rest is

(1) $\frac{15}{4}$ m (2) $\frac{4}{15}$ m

(3) $\frac{6}{11}$ m (4) $\frac{5}{11}$ m

10. The coefficient of static friction between a block and an inclined plane is $\frac{1}{\sqrt{3}}$, then the angle of repose for the inclined plane is

(1) 60° (2) 45°

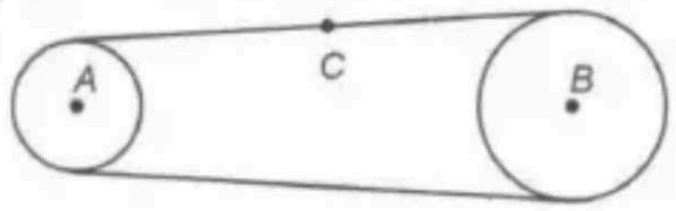
(3) 30° (4) 90°

11. The power output of an automobile engine is advertised to be 200 HP at 6000 rpm. The corresponding torque generated is nearly

(1) 237.5 N m (2) 337.5 N m

(3) 400 N m (4) 113.5 N m

12. A belt moves without slipping over two pulleys A and B which are mounted on two fixed horizontal axles as shown in the figure. Radii of the pulleys A and B are 25 cm and 50 cm respectively. Pulley A is driven at constant angular acceleration of 0.2 rad/s^2. The acceleration of a point C on the belt and angular acceleration of pulley B are respectively are



(1) 0.2 m/s^2, 0.2 rad/s^2

(2) 0.3 m/s^2, 0.2 rad/s^2

(3) Zero, zero

(4) 0.05 m/s^2, 0.1 rad/s^2

Space for Rough Work

13. A solid sphere is rotating in free space. If the sphere is shrinking keeping its mass constant, then the physical quantity that will not be affected is

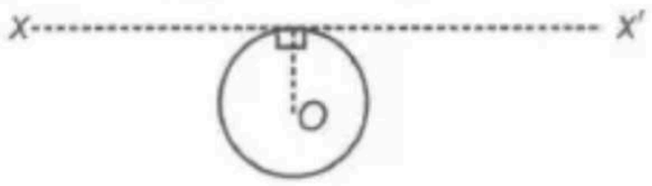
(1) Angular velocity

(2) Moment of inertia

(3) Angular momentum

(4) Rotational kinetic energy

14. A thin wire of length 2 m and uniform linear mass density 3 g/m is bent into a circular loop with centre *O* as shown. The moment of inertia of the loop about the axis *xx'* is [take $\pi^2 = 10$]



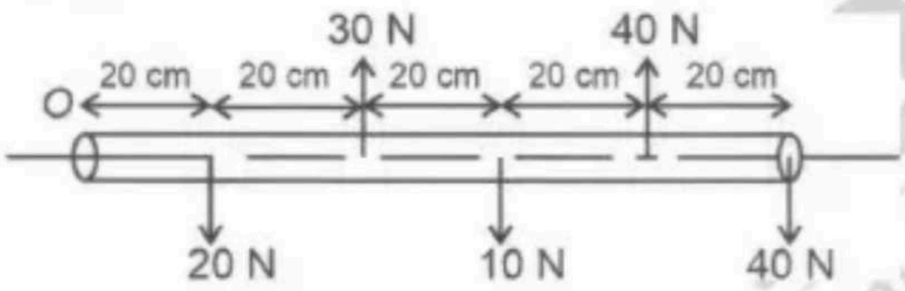
(1) 9×10^2 kg m^2

(2) 0.9 kg m^2

(3) 0.9×10^{-3} kg m^2

(4) 0.9×10^{-2} kg m^2

15. Five forces are acting perpendicular to the axis of a rod as shown in figure at distances of 20 cm, 40 cm, 60 cm, 80 cm and 100 cm respectively from one end of the rod. Under the influence of these forces, the rod



(1) Is at rest

(2) Undergoes rotational motion only

(3) Undergoes translatory motion only

(4) Undergoes translational as well as rotational motion

16. A string is wrapped around the rim of a wheel of moment of inertia 4 kg m^2 and radius 1 m. The wheel is free to rotate about its axis. Initially wheel is at rest and the string is then pulled tangentially by a force $F = 40$ N. The angular velocity of the wheel after 5 s is

(1) 100 rad/s

(2) 200 rad/s

(3) 50 rad/s

(4) 250 rad/s

17. Choose the correct statement among the following.

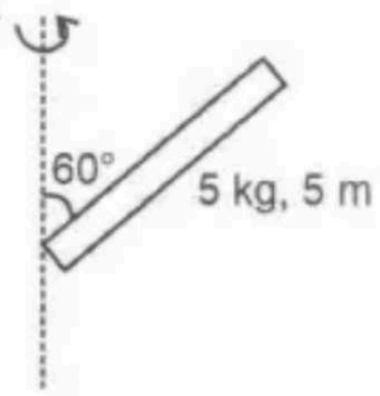
(1) In a rigid body, angular velocity of any point on the body with respect to any other point on the body keeps changing and is different than the angular velocity of the body

(2) Radius of gyration (k) for an annular disc of mass M with inner and outer radius as R_1 and R_2 respectively, about an axis perpendicular to disc and passing through center is $\sqrt{R_1^2 - R_2^2}$

(3) Radius of gyration does not depend on distribution of mass of body

(4) Torque is an axial vector

18. The moment of inertia of a rod about an axis inclined at an angle of 60° with the rod and passing through one end as shown in figure, is



(1) 125 kg m^2

(2) 4 kg m^2

(3) $\frac{125}{4}$ kg m^2

(4) 225 kg m^2

Space for Rough Work

19. Two discs of moment of inertia 2 kg m^2 and 4 kg m^2 having angular speeds 4 rad/s and 2 rad/s respectively are rotating in same sense about a common axis passing through their centre of mass and perpendicular to their plane. If the two are slowly brought in contact with each other and then they start rotating together along same axis, then the final rotational kinetic energy of system will be

(1) $\frac{64}{3}$ J (2) $\frac{84}{3}$ J

(3) $\frac{94}{3}$ J (4) Zero

20. A disc is rotating with angular velocity $\vec{\omega}$. A force $\vec{F}$ acts at a point on the disc whose position vector with respect to axis of rotation is $\vec{r}$. The power associated with torque due to the force is given by

(1) $(\vec{F}\times\vec{r})\times\vec{\omega}$

(2) $(\vec{\omega}\times\vec{F})\cdot\vec{r}$

(3) $(\vec{r}\times\vec{F})\cdot\vec{\omega}$

(4) $(\vec{r}\cdot\vec{F})\cdot\vec{\omega}$

21. Two objects of masses 0.2 kg and 0.5 kg possess velocities of $5\hat{i}$ m/s and $(10\hat{i}+10\hat{j})$ m/s respectively. The velocity of their centre of mass is (in m/s)

(1) $\frac{20\hat{i}+30\hat{j}}{7}$ (2) $\frac{30\hat{i}+20\hat{j}}{7}$

(3) $\frac{60\hat{i}+50\hat{j}}{7}$ (4) $\frac{50\hat{i}+60\hat{j}}{7}$

22. Choose the incorrect statement among the following.

(1) Linear momentum of system of colliding bodies remains conserved in all types of collisions

(2) In inelastic collision all the forces involved in the collision are not conservative

(3) When a heavy body collides elastically against a light body at rest, the heavy body slightly slows down whereas light body moves with the speed nearly double of heavier body

(4) When body *A* collides elastically against body *B* of equal mass at rest, then both bodies *A* and *B* come to rest

23. A shell at rest on a smooth horizontal surface explodes into two fragments of masses 1 kg and 6 kg. If just after explosion, the 1 kg fragment moves with speed 2 m/s, then work done by internal forces during the explosion is

(1) $\frac{8}{3}$ J

(2) $\frac{7}{3}$ J

(3) 3 J

(4) 5 J

24. A body is displaced from position $x = 6$ m to $x = 3$ m along the *x*-axis. For the forces mentioned in column-I, match the corresponding work done (W) by each force on the body in column-II.

	Column-I		Column-II
a.	$\vec{F} = 5\hat{i}$ N	(i)	Positive
b.	$\vec{F} = 8\hat{j}$ N	(ii)	Negative
c.	$\vec{F} = -5\hat{i}$ N	(iii)	Zero
d.	$\vec{F} = (-8\hat{i}+5\hat{j})$ N	(iv)	$W = 24$ J

(1) a(ii), b(iii), c(i), d(i,iv)

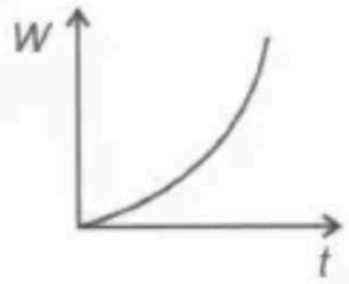
(2) a(i), b(ii), c(iii), d(i,iv)

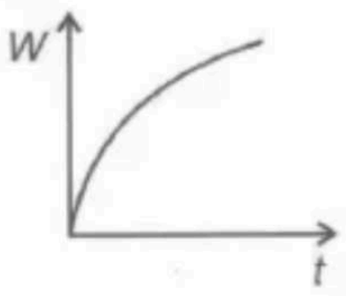
(3) a(iv), b(ii), c(iii), d(i)

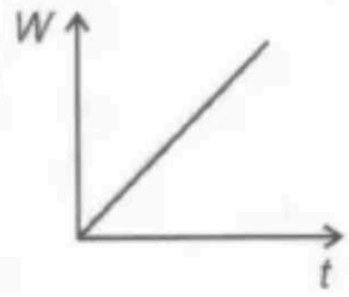
(4) a(ii), b(iv), c(iii), d(ii,iv)

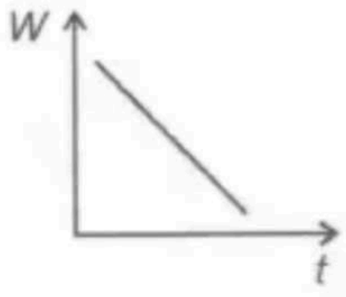
Space for Rough Work

25. An object of mass 5 kg is moving along x-axis such that its position x varies with time t as $x^2 = 25t^3$, with x in metre and t in second. Work done (W) on the object in time t can be shown as

(1) 

(2) 

(3) 

(4) 

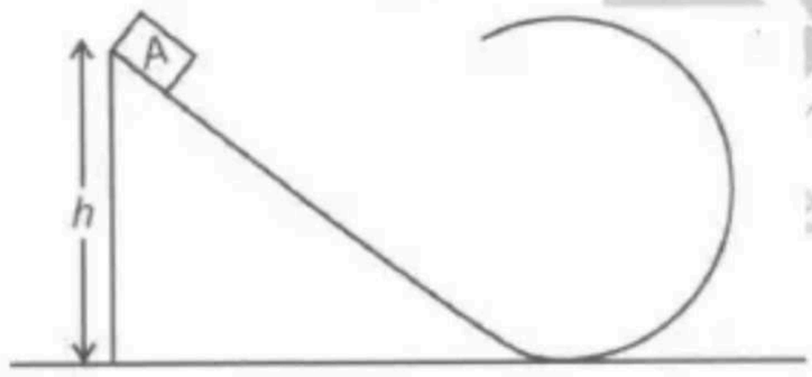
26. A car starts from rest on a 500 m long horizontal bridge. The coefficient of friction between the tyre and bridge is 1. The minimum possible time taken to cross the bridge is

(1) 2 s (2) 5 s

(3) 4 s (4) 10 s

27. A block is released from rest at the top of an inclined plane which later curves smoothly into a circular track of radius 1 m as shown in figure. The minimum height h from which it should be released so that it is able to complete the circle, is

(Friction is absent everywhere)



(1) 5 m (2) 2.5 m

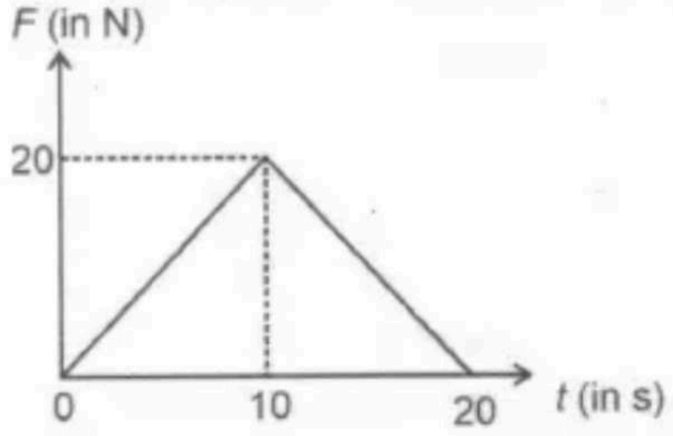
(3) 1.5 m (4) 0.5 m

28. A ball of mass 500 g strikes a surface with speed 40 m/s at an angle of 60° from normal and rebounds with same speed in 0.4 s. The average force applied by surface on ball is

(1) 500 N (2) 200 N

(3) 50 N (4) 60 N

29. For the given force-time (F-t) graph, the impulse from $t = 0$ to $t = 20$ s and the final momentum respectively are (initial momentum is 10 kg-m/s)



(1) 200 N s, 210 kg m s^{-1}

(2) 210 N s, 200 kg m s^{-1}

(3) Zero, 200 kg m s^{-1}

(4) 110 N s, 100 kg m s^{-1}

30. The correct statement describing the action-reaction pair of forces under Newton's third law of motion is

(1) They are equal in magnitude

(2) They may have same direction

(3) They may not act on different bodies

(4) All of these

31. A particle moves along a circle with speed $v = 2t$, where t is time. The total acceleration of the particle at a time when it has traced $\left(\frac{1}{4}\right)^{th}$ of the circumference is

(1) $2(\sqrt{1+\pi^2})$ (2) $\sqrt{1+\pi}$

(3) $\sqrt{1+2\pi}$ (4) Zero

Space for Rough Work

32. A particle has instantaneous velocity of $(2\hat{i}-3\hat{j}+4\hat{k})$ m/s under the influence of a constant force of $(2\hat{i}+3\hat{j}+5\hat{k})$ N. The instantaneous power supplied is

(1) 20 W (2) 15 W

(3) 25 W (4) 26 W

33. In a case where a 2 kg body collides head-on elastically with another body of mass m, the kinetic energy transferred is maximum when (m is at rest initially)

(1) m = 4 kg (2) m = 2 kg

(3) m = 1 kg (4) m = 5 kg

34. Consider the given statements:

Statement A : Potential energy of a system can be positive, negative or zero.

Statement B : Work done by tension in the string, connected to a block performing uniform circular motion, is zero.

Statement C : The area under force-displacement curve gives distance.

Choose the correct answer from the options given below.

(1) Only statement A is true

(2) Only statements A & B are true

(3) Only statements B & C are true

(4) Only statement C is true

35. A particle is moving with constant speed of 2 m/s along the line y = 2 in positive x-direction. The magnitude of its angular velocity about origin when its position makes 30° with x-axis is

(1) 0.25 rad/s (2) 0.35 rad/s

(3) 0.5 rad/s (4) 0.75 rad/s

36. The potential energy function of an object of mass 5 kg moving in xy-plane in a conservative field is given by $U = (3x + 4y)$ joule, where x and y are coordinates of object (in metre). The magnitude of its acceleration is

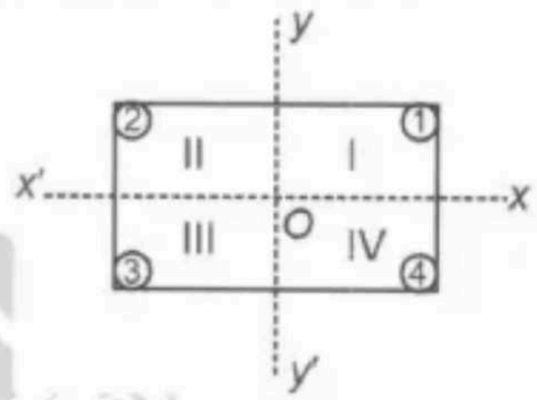
(1) 2 m/s^2

(2) 3 m/s^2

(3) 4 m/s^2

(4) 1 m/s^2

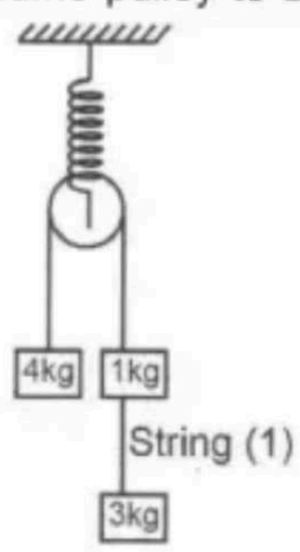
37. Figure shows a uniform rectangular plate from which identical small circular portions are removed from the corners. After circles 1, 2 and 3 are removed, the C.O.M. of remaining portion will be



(1) At O (2) In III quadrant

(3) In IV quadrant (4) On Oy' axis

38. Consider the given spring-pulley system as shown in figure. Then acceleration of pulley immediately after string (1) is cut, will be (Assume pulley to be ideal)



(1) $\frac{g}{2}$

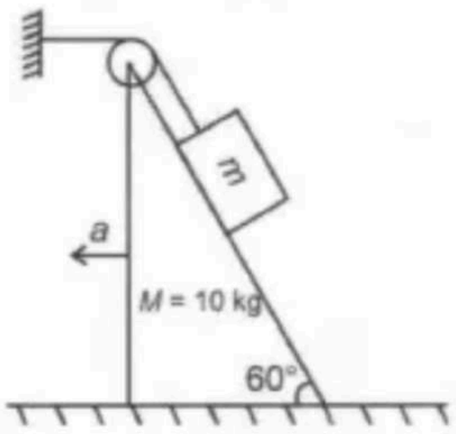
(2) $\frac{3g}{2}$

(3) g

(4) Zero

Space for Rough Work

39. In the given figure if acceleration of wedge (of mass $M = 10$ kg) with respect to ground is $a = 2$ m/s^2, then the acceleration of $m = 2$ kg block with respect to M and ground respectively, are



(1) 2 m/s^2, 4 m/s^2 (2) $2\sqrt{3}$ m/s^2, 2 m/s^2

(3) 2 m/s^2, 2 m/s^2 (4) $2\sqrt{3}$ m/s^2, 4 m/s^2

40. A wheel is rotating with a speed of 420 rev/minute. After being switched off, it comes to rest in 10 s. Assuming constant angular deceleration, the number of revolutions made by it before coming to rest, will be

(1) 37

(2) 35

(3) 45

(4) 47

41. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The potential energy of a system increases if work is done against a conservative force.

Reason (R) : Work done by a conservative force on a system is equal to negative of change in potential energy.

In the light of the above statements, choose the most appropriate answer from the options given below.

(1) (A) is true but (R) is false

(2) (A) is false but (R) is true

(3) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(4) Both (A) and (R) are true and (R) is the correct explanation of (A)

42. A constant power P is applied to a car of mass $m = 2500$ kg. The velocity of the car increases from $v = 4$ m/s to $v = 6$ m/s when car travels a distance of 250 m. The value of power P is nearly

(1) 507 W (2) 605 W

(3) 400 W (4) 100 W

43. If the magnitude of resultant of two forces $(F_1 + F_2)$ and $(F_1 - F_2)$ is given as $\sqrt{(F_1^2 + F_2^2)}$, then the angle between these two forces is

(1) $\theta = \sin^{-1}\left(\frac{F_1 - F_2}{F_1 + F_2}\right)$

(2) $\theta = \cos^{-1}\left(\frac{F_1 - F_2}{F_1 + F_2}\right)$

(3) $\theta = \cos^{-1}\left[\frac{-(F_1^2 + F_2^2)}{2(F_1^2 - F_2^2)}\right]$

(4) $\theta = \sin^{-1}\left[\frac{F_1^2 + F_2^2}{F_1^2 - F_2^2}\right]$

44. Given below are two statements :

Statement A : When a particle moves in a vertical circle with the help of a string, then its tangential acceleration is always variable.

Statement B : When a particle moves in a vertical circle with the help of a string, then its radial acceleration is always variable.

Space for Rough Work

In the light of the above statements, choose the **most appropriate** answer from the options given below.

(1) Both statements are false

(2) Statement A is true but statement B is false

(3) Statement A is false but statement B is true

(4) Both statements are true

45. If a vector $\vec{A} = (2\hat{i} + 3\hat{j} - 5\hat{k})$ and vector $\vec{B} = (3\hat{i} - 2\hat{j} + 5\hat{k})$, then $|\vec{A} \times \vec{B}|$ is

(1) $\sqrt{119}$ units (2) $\sqrt{819}$ units

(3) $\sqrt{117}$ units (4) $\sqrt{911}$ units

[CHEMISTRY]

46. Identify the correct relation between C_p and C_v for two moles of an ideal gas. (C_p and C_v are molar heat capacity at constant pressure and constant volume respectively, for an ideal gas)

(1) $C_p + C_v = R$

(2) $C_p - C_v = 2R$

(3) $C_p = 2R - C_v$

(4) $C_p - C_v = R$

47. Two moles of an ideal gas at 1 bar and 298 K compressed at constant temperature by applying constant external pressure of 4 bar. The work done on the system is

(1) 14.86 kJ (2) 49.88 kJ

(3) 1.486 kJ (4) 4.98 kJ

48. Four moles of an ideal gas at 1 bar and 298 K is expanded into vacuum to double the volume. The work done will be

(1) $-5RT\,(V_2 - V_1)$ (2) $-4RT\ 2.303 \log 2$

(3) $-5RT \ln \frac{4V_2}{V_1}$ (4) Zero

49. Among the following, the set which represents path functions, is

(A) $H - TS$ (B) $U + PV$

(C) q (D) W

Choose the correct option.

(1) (A), (B) and (D) only

(2) (C) and (D) only

(3) (B), (C) and (D) only

(4) (A), (C) and (D) only

50. At 60°C, $N_2O_4(g)$ is 50% dissociated as given

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

The value of $\Delta G°$ at 60°C and at 3 atm pressure is

(1) –3.84 kJ/mol (2) –2.73 kJ/mol

(3) –1.06 kJ/mol (4) –7.41 kJ/mol

51. Given below are two statements:

Statement I : For irreversible isothermal process $\Delta U = 0$ (for an ideal gas)

Statement II : Both for reversible and irreversible expansion of an ideal gas under isothermal conditions, ΔS_{total} is zero.

In the light of the above statements, choose the **correct** answer from the options given below.

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

(3) Statement I is incorrect but Statement II is correct

(4) Both Statement I and Statement II are correct

Space for Rough Work

52. Match List-I with List-II

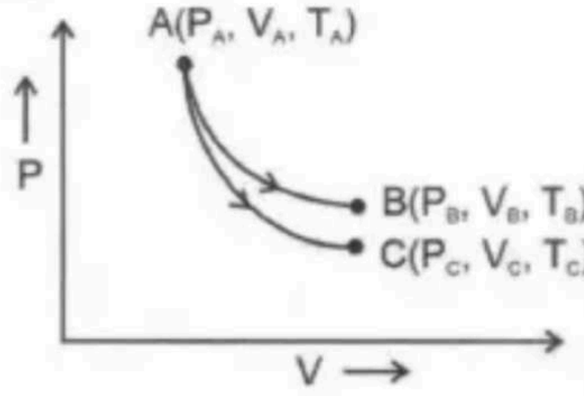
	List-I (Condition)		List-II (Spontaneity of reaction)
a.	$\Delta_r H^\ominus < 0$ and $\Delta_r S^\ominus > 0$	(i)	Spontaneous only at low temperature
b.	$\Delta_r H^\ominus < 0$ and $\Delta_r S^\ominus < 0$	(ii)	Spontaneous at all temperatures
c.	$\Delta_r H^\ominus > 0$ and $\Delta_r S^\ominus < 0$	(iii)	Spontaneous only at high temperature
d.	$\Delta_r H^\ominus > 0$ and $\Delta_r S^\ominus > 0$	(iv)	Non-spontaneous at all temperatures

Choose the correct option.

(1) a(ii), b(iv), c(i), d(iii) (2) a(ii), b(i), c(iv), d(iii)

(3) a(iii), b(i), c(iv), d(ii) (4) a(iii), b(iv), c(i), d(ii)

53. Reversible expansion of an ideal gas under isothermal and adiabatic conditions are shown in the figure



Which of the following options is correct?

(1) $T_A = T_C$

(2) AC $\rightarrow$ Isothermal expansion

(3) $W_{isothermal} > W_{adiabatic}$

(4) $\Delta S_{adiabatic} > \Delta S_{isothermal}$

54. Predict in which of the following processes, entropy increase(s).

(a) $2NaHCO_3(s) \longrightarrow Na_2CO_3(s) + CO_2(g) + H_2O(g)$

(b) $4Fe(s) + 3O_2(g) \longrightarrow 2Fe_2O_3(s)$

(c) $H_2(g) \longrightarrow 2H(g)$

(d) $N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$

The correct option(s) is/are

(1) (a) and (c) only

(2) (b) and (d) only

(3) (c) only

(4) (a), (b) and (c) only

55. Consider the following statements

(a) The energy of an isolated system is constant.

(b) Heat capacity and temperature are intensive properties.

(c) Standard enthalpy of formation of $S_{(s)}$, rhombic and $P_{(s)}$, white are zero.

The correct statement(s) is/are

(1) (a) only (2) (a) and (c) only

(3) (b) and (c) only (4) (a), (b) and (c)

56. Identify the incorrect statement(s) among the following.

(A) For a system of constant volume, heat involved directly changes to internal energy.

(B) According the third law of thermodynamics, the entropy of any pure crystalline substance approaches zero as the temperature approaches absolute zero.

(C) For adiabatic free expansion of an ideal gas, $w \neq 0$, $q = 0$ and $\Delta U \neq 0$.

Choose the correct option.

(1) (A) and (B) only (2) (C) only

(3) (B) and (C) only (4) (A) only

Space for Rough Work

57. At standard conditions, if the change in the enthalpy for the following reaction is –109 kJ mol^{-1}

$H_2(g) + Br_2(g) \longrightarrow 2HBr(g)$

The bond energy of $Br_2(g)$ and HBr(g) is 192 kJ mol^{-1} and 368 kJ mol^{-1} respectively. The bond energy of H_2 (in kJ mol^{-1}) is

(1) 518 (2) 259

(3) 435 (4) 736

58. One mole of an ideal gas is expanded isothermally and reversibly to half of its initial pressure. The change in entropy for the process is

(1) 5.74 J K^{-1} mol^{-1} (2) 3.94 J K^{-1} mol^{-1}

(3) 7.96 J K^{-1} mol^{-1} (4) 4.38 J K^{-1} mol^{-1}

59. Given below are two statements:

Statement I : During stretching of rubber, entropy decreases.

Statement II : If the standard enthalpies of formation of $CO_2(g)$, $H_2O(l)$ and glucose(s) at 25°C are –400 kJ mol^{-1}, –300 kJ mol^{-1} and –1300 kJ mol^{-1} respectively, then the standard enthalpy of combustion per gram of glucose at 25°C is –16.11 kJ g^{-1}.

In the light of the above statements, choose the **correct** answer from the options given below.

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

(3) Statement I is incorrect but Statement II is correct

(4) Both Statement I and Statement II are correct

60. Equal volumes of two diatomic gases, A and B are mixed at same temperature and pressure. The ratio of specific heats $\left(\frac{C_p}{C_v}\right)$ of the mixture will be

(1) 1.66 (2) 1.33

(3) 1.40 (4) 0.83

61. Consider the following processes,

	ΔH(kJ mol^{-1})
$\frac{1}{2}A \longrightarrow B$	+150
$3B \longrightarrow 2C + D$	+125
$E + A \longrightarrow 2D$	+350

For $B + 3D \longrightarrow 2C + 2E + A$, ΔH will be

(1) 775 kJ mol^{-1} (2) 275 kJ mol^{-1}

(3) –275 kJ mol^{-1} (4) –775 kJ mol^{-1}

62. The equilibrium constant for the reversible reactions are given as,

(I) $2A(g) \rightleftharpoons B(g) + 2C(g)$; K_1

(II) $\frac{3}{2}A(g) \rightleftharpoons \frac{3}{4}B(g) + \frac{3}{2}C(g)$; K_2

Then K_1 and K_2 are related as

(1) $K_1 = \sqrt{K_2}$

(2) $K_2 = \sqrt{K_1}$

(3) $K_1 = (K_2)^{\frac{3}{4}}$

(4) $K_2 = (K_1)^{\frac{3}{4}}$

63. In which one of the following equilibria, $K_p \neq K_c$?

(1) $\frac{1}{2}N_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons NO(g)$

(2) $2C(s) + O_2(g) \rightleftharpoons 2CO(g)$

(3) $NO_2(g) + SO_2(g) \rightleftharpoons NO(g) + SO_3(g)$

(4) $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$

Space for Rough Work

64. Two solids dissociate as follows

$A(s) \longrightarrow B(g) + C(g)$; $K_{p_1} = x$

$D(s) \longrightarrow C(g) + E(g)$; $K_{p_2} = y$

The total pressure when both the solids are kept in same container and dissociate simultaneously is

(1) $\sqrt{x+y}$ atm

(2) (x + y) atm

(3) $\frac{\sqrt{x+y}}{2}$ atm

(4) $2\sqrt{x+y}$ atm

65. The standard Gibbs energy change at 300 K for reaction

$2A(g) \rightleftharpoons B(g) + C(g)$

is 2500 J. At a given time, composition of reaction mixture is $[A] = \frac{1}{2}$ mol L^{-1}, [B] = 2 mol L^{-1} and [C] = 2 mol L^{-1}. The reaction proceeds in

[Antilog(–0.436) = 0.36]

(1) Backward direction ; $Q_c > K_c$

(2) Backward direction ; $Q_c < K_c$

(3) Forward direction ; $Q_c > K_c$

(4) Forward direction ; $Q_c < K_c$

66. K_{sp} of following salts have been given as

Salts	K_{sp}
Ag_2CrO_4	1.1×10^{-12}
AgCl	1.8×10^{-10}
AgBr	5×10^{-13}
AgI	8.3×10^{-17}

Identify the salt which will precipitate last if $AgNO_3$ solution is added to the solution containing equal moles of NaCl, NaBr, NaI and Na_2CrO_4.

(1) Ag_2CrO_4 (2) AgI

(3) AgCl (4) AgBr

67. pH of a saturated solution of $Ba(OH)_2$ is 11. The value of solubility product (K_{sp}) of $Ba(OH)_2$ is

(1) 5×10^{-9}

(2) 5×10^{-10}

(3) 5×10^{-7}

(4) 5×10^{-8}

68. Which of the following expressions is not applicable to the following hydrolysis equilibrium?

$CN^- + H_2O \rightleftharpoons HCN + OH^-$

(1) $K_h = \frac{K_w}{K_a(HCN)}$

(2) $h = \sqrt{\frac{K_h}{C}}$

(3) $pH = 7 - \frac{1}{2}(pK_a(HCN) + \log C)$

(4) $[H^+] = \sqrt{\frac{K_w \times K_a}{C}}$

69. For the equilibrium, $NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$, the equilibrium constant at 25°C is

(Given : $NH_4^+ + H_2O \rightleftharpoons NH_4OH + H^+$;

$K_h = 5.5 \times 10^{-10}$)

(1) 1.8×10^{-4}

(2) 1.8×10^{-6}

(3) 1.8×10^{-5}

(4) 1.8×10^{-7}

Space for Rough Work

70. Calculate the pH of 0.05 M $(CH_3COO)_2Ca$ solution. (Given pK_a (CH_3COOH) = 4.74)

(1) 10.51 (2) 8.87

(3) 6.42 (4) 13.19

71. Which of the following solutions will undergo cationic hydrolysis?

(1) Na_2CO_3 (2) K_2SO_4

(3) NaCN (4) $(NH_4)_2SO_4$

72. Match List-I with List-II

	List-I (Salt)		**List-II (Degree of hydrolysis)**
a.	CH_3COONa	(i)	No hydrolysis
b.	NH_4Cl	(ii)	$\sqrt{\frac{K_w}{C \times K_a}}$
c.	NaCl	(iii)	$\sqrt{\frac{K_w}{C \times K_b}}$
d.	CH_3COONH_4	(iv)	$\sqrt{\frac{K_w}{K_a \times K_b}}$

Choose the correct option.

(1) a(ii), b(iii), c(i), d(iv) (2) a(ii), b(iii), c(iv), d(i)

(3) a(iii), b(ii), c(i), d(iv) (4) a(iii), b(iv), c(i), d(ii)

73. Given below are two statements

Statement I : $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) + heat$; the equilibrium shifts in backward direction by increasing temperature.

Statement II : Both BF_3 and CH_4 molecules behave as Lewis acid.

In the light of above statements, choose the correct answer from the options given below.

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

(3) Statement I is incorrect but Statement II is correct

(4) Both Statement I and Statement II are correct

74. A solution contains a mixture of Ag^+ (0.1 M) and Hg_2^{2+} (0.1 M) which are to be separated by selective precipitation. The maximum concentration of iodide ion at which only one of them gets precipitated is [Given K_{sp} (AgI) = 8.5×10^{-17} , K_{sp} (Hg_2I_2) = 2.5×10^{-26})]

(1) 8.5×10^{-6} M

(2) 5×10^{-13} M

(3) 1.7×10^{-4} M

(4) 6×10^{-5} M

75. The pH of one litre of a buffer solution is 8.26. The amount of ammonium chloride to be dissolved in 0.2 M ammonia solution to make one litre of the buffer is

(pK_b (NH_4OH) = 4.74)

(1) 107 g

(2) 87.5 g

(3) 72.3 g

(4) 126 g

76. Given below are two statements :

Statement I : Phosphoric acid has more than one ionizable proton per molecule of the acid.

Statement II : Ionisation constant K_{a_1} is higher than K_{a_2} for carbonic acid.

In the light of the above statements, choose the **correct** answer from the options given below.

Space for Rough Work

(1) Both Statement I and Statement II are incorrect
(2) Statement I is correct but Statement II is incorrect
(3) Statement I is incorrect but Statement II is correct
(4) Both Statement I and Statement II are correct

77. The percentage of pyridine that forms pyridinium ion in a 0.20 M aqueous pyridine solution is (K_b for $C_5H_5N = 1.7 \times 10^{-9}$)
(1) 0.006%
(2) 0.0092%
(3) 0.77%
(4) 0.004%

78. The pK_b of dimethylamine and pK_a of acetic acid are 3.27 and 4.77 respectively at T(K). The pH of dimethylammonium acetate solution is
(1) 6.25
(2) 8.50
(3) 5.50
(4) 7.75

79. The heat required to raise the temperature of 54.0 g of aluminium from 45°C to 65°C is (Molar heat capacity of Al is 24 $J\ K^{-1}\ mol^{-1}$)
(1) 960 J (2) 96 kJ
(3) 480 J (4) 48 kJ

80. Which among the following salt solutions is most basic in nature?
(1) Ammonium acetate (2) Ammonium cyanide
(3) Potassium nitrate (4) Calcium cyanide

81. If K_a for monobasic acid HA is 10^{-5} and concentration of A^- is thrice than that of HA. The pH of the solution is
(1) 6.12 (2) 4.32
(3) 5.48 (4) 5.15

82. Match List-I with List-II
[Given pK_a (CH_3COOH) = 4.76, pK_b(NH_4OH) = 4.75]

	List-I (Solution)		**List-II (Type of buffer, pH)**
a.	50 mL of 0.1 M HCl + 100 mL of 0.1 M CH_3COONa	(i)	Basic, 9.25
b.	100 mL of 0.2 M NH_4OH + 50 mL of 0.2 M HCl	(ii)	Acidic, 4.76
c.	200 mL of 0.2 M CH_3COOH + 50 mL of 0.2 M NaOH	(iii)	Basic, 9.85
d.	100 mL of 0.2 M NH_4OH + 20 mL of 0.1 M H_2SO_4	(iv)	Acidic, 4.28

Choose the correct option.
(1) a(ii), b(i), c(iv), d(iii) (2) a(i), b(ii), c(iii), d(iv)
(3) a(iv), b(iii), c(ii), d(i) (4) a(iii), b(iv), c(ii), d(i)

83. Given below are two statements:
Statement I : A mixture of benzoic acid and sodium benzoate can act as a buffer solution.
Statement II : On 50% neutralisation of CH_3COOH with NaOH, pH of the solution is independent of CH_3COONa concentration.
In the light of the above statements, choose the **correct** answer from the options given below.
(1) Both Statement I and Statement II are incorrect
(2) Statement I is correct but Statement II is incorrect
(3) Statement I is incorrect but Statement II is correct
(4) Both Statement I and Statement II are correct

Space for Rough Work

84. For the reaction, $A_2O_4(l) \rightleftharpoons 2AO_2(g)$

$\Delta U = 1.1$ kJ, $\Delta S = 11$ J K^{-1} at 300 K hence, ΔG is

(1) –2.78 kJ (2) –5.56 kJ

(3) 5.56 kJ (4) 2.78 kJ

85. In the following reaction

$$HC_2O_4^- + PO_4^{3-} \rightleftharpoons HPO_4^{2-} + C_2O_4^{2-}$$

Which pair can act as Bronsted base only?

(1) $HC_2O_4^-$ and PO_4^{3-} (2) $C_2O_4^{2-}$ and HPO_4^{2-}

(3) $HC_2O_4^-$ and HPO_4^{2-} (4) $C_2O_4^{2-}$ and PO_4^{3-}

86. The conjugate base of hydrazoic acid (N_3H) is

(1) N^{3-} (2) N_3^-

(3) N_2^- (4) HN_3^-

87. Given below are two statements:

Statement I : Upon addition of oxalic acid to the following reaction mixture:

$$Fe^{3+}(aq) + SCN^-(aq) \rightleftharpoons [Fe(SCN)]^{2+}(aq)$$

The intensity of deep red colour of the solution decreases.

Statement II : Addition of aqueous $HgCl_2$ to the following reaction mixture.

$$Fe^{3+}(aq) + SCN^-(aq) \rightleftharpoons [Fe(SCN)]^{2+}(aq)$$

The equilibrium shifts from left to the right.

In the light of the above statements, choose the **correct** answer from the options given below.

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

(3) Statement I is incorrect but Statement II is correct

(4) Both Statement I and Statement II are correct

88. $N_2O_4(g) \rightleftharpoons 2NO_2(g)$

At a fixed temperature, the volume of reaction container is halved.

Choose the correct statement regarding the equilibrium constant (K_p) and degree of dissociation (α).

(1) K_p decreases and α increases

(2) Neither K_p nor α changes

(3) K_p remains constant and α decreases

(4) K_p increases and α decreases

89. The following reaction occurs in a blast furnace where iron ore is reduced to iron metal:

$$Fe_2O_3(s) + 3CO(g) \rightleftharpoons 2Fe(l) + 3CO_2(g)$$

Which one of the following changes will not affect the state of equilibrium?

(A) Removal of $CO_2(g)$

(B) Addition of an inert gas at constant volume

(C) Addition of Fe_2O_3

(D) Addition of an inert gas at constant pressure

(E) Addition of $CO(g)$

Choose the correct option.

(1) (A) and (E) only

(2) (B), (C) and (D) only

(3) (B), (C) and (E) only

(4) (C), (D) and (E) only

90. In which of the following reactions, an increase in the volume of the container will favour the equilibrium shifts in forward direction?

(1) $2NO_2(g) \rightleftharpoons 2NO(g) + O_2(g)$

(2) $4NH_3(g) + 5O_2(g) \rightleftharpoons 4NO(g) + 6H_2O(l)$

(3) $3O_2(g) \rightleftharpoons 2O_3(g)$

(4) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

Space for Rough Work

[BOTANY]

91. Pinnately compound leaves can have all the following features, **except**
(1) Leaflets are attached to rachis
(2) Possess a bud in its axil
(3) Leaflets may be attached at the tip of petiole
(4) Stipules may be present at the base of leaf

92. Read the statements (A-D) and select the **correct** option.
(A) All protozoans are heterotrophic and live as predators or parasites
(B) All amoeboid protozoans have silica shells on their surface
(C) *Entamoeba* possess thousands of cilia for locomotion in their freshwater habitat.
(D) Sporozoans have an infectious spore like stage in their life cycle.
(1) Only (A), (B) and (D) are correct
(2) Only (C) and (D) are correct
(3) Only (A) and (D) are correct
(4) Only (B) and (D) are correct

93. Choose the **correct** match.

(1)	Testa	–	The inner protective covering of seed
(2)	Hilum	–	Scar on the seed coat through which developing seeds are attached to the fruit
(3)	Coleoptile	–	Sheath enclosing radicle in monocot seed
(4)	Aleurone layer	–	Outer covering of embryo that separates the endosperm from a proteinaceous layer

94. In tomato
(a) Petals show valvate aestivation
(b) Ovary is bilocular
(c) Stamens are epiphyllous
(d) Placenta is present at periphery of ovary
In the light of above statements select the **correct** option.
(1) Only (a) and (d) are true
(2) Only (a) and (b) are true
(3) (b), (c) and (d) are true
(4) (a), (b) and (d) are true

95. Guava, Peach, Cucumber, ray florets of Sunflower, rose, grass, onion, Tulip, Mulaithi, Belladonna, *Petunia*, Tobacco, Chilli, *Aloe*, China rose, Mustard, Plum
How many of the above given plants have superior ovary?
(1) 15 (2) 11
(3) 10 (4) 12

96. State **True(T)** or **False(F)** and select the correct option.
(A) Infectious agents discovered by T.O. Diener are devoid of DNA.
(B) According to R.H. Whittaker, heterotrophs with tissue/organ/organ system level of body organisation have higher orders of complexity than the multicellular heterotrophs with cell wall made of chitin.
(C) *Claviceps* is a source of many alkaloids and LSD.
(D) Pseudopodia are locomotory and feeding structures in sporozoans.

Space for Rough Work

	A	B	C	D
(1)	T	F	F	T
(2)	T	T	T	F
(3)	F	F	T	T
(4)	T	F	T	F

97. In six kingdom classification, how many kingdoms have organisms with autotrophic mode of nutrition?

(1) Five

(2) Two

(3) Four

(4) Three

98. How many statements are **correct** w.r.t. member of Monera?

(A) Bacteria reproduce sexually by adopting a primitive type of DNA transfer

(B) Eubacteria such as cyanobacteria are characterised by presence of a rigid cell wall and if motile, a flagellum

(C) Some of the cyanobacteria can fix atmospheric nitrogen in specialised wall less cells called heterocyst

(D) Chemosynthetic autotrophic bacteria oxidise various inorganic substances such as nitrates, nitrites and ammonia

(E) Heterotrophic bacteria are helpful in making curd from milk and fixing nitrogen in legume

(1) 2 (2) 4

(3) 3 (4) 5

99. Mustard and China rose show similarities in

(a) Placentation

(b) Floral symmetry

(c) Adhesion of stamen

(d) Aestivation of corolla

(e) Union of carpels

(f) Type of fruit

(g) Number of carpels

In the light of above statements, select the **correct** option.

(1) (b), (c) and (f) only (2) All except (d)

(3) (b), (e) and (g) only (4) (b) and (e) only

100. Choose the odd one w.r.t. modification of stem

(1) Tendril of watermelon

(2) Tuber of potato

(3) Thorn of citrus

(4) Spine of cactus

101. Select the **incorrectly** matched pair.

	Types of Placentation	–	**Example**
(1)	Marginal	–	Pea
(2)	Axile	–	*Primrose*
(3)	Basal	–	Sunflower
(4)	Free central	–	*Dianthus*

102. Select **odd** one out w.r.t. non-endospermic seeds.

(1) Castor

(2) Pea

(3) Bean

(4) Gram

Space for Rough Work

103. A large shield shaped cotyledon found in maize seeds is known as

(1) Micropyle (2) Scutellum

(3) Epiblast (4) Tegmen

104. Match **Column-I** with **Column-II** and select the **correct** option.

	Column-I		Column-II
a.	Leaf base become swollen	(i)	Ashwagandha
b.	Tricarpellary and syncarpous gynoecium	(ii)	*Argemone*
c.	Presence of persistent outermost whorl of the flower and swollen placenta	(iii)	*Asparagus*
d.	The ovules develop on the inner wall of the ovary or peripheral part	(iv)	Pea

(1) a(iv), b(iii), c(i), d(ii)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(iii), b(iv), c(ii), d(i)

105. State **True(T)** or **False(F)** and select the correct option.

(A) A sterile stamen is called staminode.

(B) Stamens are attached to the petals in lily.

(C) Epiphyllous stamens and persistent calyx are seen in brinjal.

(D) Petals are usually brightly coloured to attract insects for pollination.

	A	B	C	D
(1)	T	T	F	F
(2)	F	F	T	T
(3)	T	F	F	T
(4)	T	T	T	F

106. Read the following statements and select the correct option.

Statement A: W.M. Stanley showed that viruses could be crystallised and crystals consist largely of proteins.

Statement B: Viruses are obligate parasites.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

107. All are sexual spores, **except**

(1) Oospores

(2) Ascospores

(3) Basidiospores

(4) Conidia

108. Which of the following pair of fungi called smut and rust fungi respectively?

(1) *Agaricus* and *Ustilago*

(2) *Ustilago* and *Neurospora*

(3) *Ustilago* and *Puccinia*

(4) *Puccinia* and *Ustilago*

Space for Rough Work

109. Multicellular organism with loose tissue body organisation having heterotrophic mode of nutrition is
(1) Plantae (2) Fungi
(3) Monera (4) Protista

110. Which of the following statements is **correct** for dinoflagellates?
(1) They have a protein rich layer called pellicle as outer covering of body which make their body flexible
(2) The cell wall has stiff cellulose plates on the outer surface
(3) In absence of sunlight, they behave like heterotrophs
(4) They are microscopic and float passively in water current

111. Identify the **correct** symbol that represent epipetalous stamens.
(1) $\overset{\frown}{\text{Epi A}_{(n)}}$
(2) $\overset{\frown}{\text{C A}}$
(3) $\overset{\frown}{\text{P A}}$
(4) $\text{A}_{(n)}$

112. Which of the following statements is true for both mango and coconut?
(1) They have fleshy edible mesocarp
(2) Pericarp is not well differentiated
(3) They develop from monocarpellary superior ovaries
(4) They have outer stony hard epicarp

113. Placentation that is found in marigold is
(1) Axile (2) Parietal
(3) Free central (4) Basal

114. In which of the following aspects mustard and, *Solanum* shows similarity?
(a) Phyllotaxy
(b) Floral symmetry
(c) Type of flower based on position of gynoecium w.r.t. other floral whorls
(d) Type of placentation
(e) Variation in the length of filament of stamens of a flower
Select the correct ones:
(1) (b), (c) and (e) only
(2) (a) and (b) only
(3) (a), (b) and (c) only
(4) (c) only

115. All are functions of roots, **except**
(1) Absorption of water and minerals from soil
(2) Provide anchorage to the plant parts
(3) Synthesis of plant growth regulators
(4) Bears and support leaves, flowers and fruits

116. The arrangements of veins and veinlets in the lamina or leaf blade is termed as
(1) Placentation
(2) Phyllotaxy
(3) Venation
(4) Inflorescence

117. Read the following statements.
(A) Flower bisexual and zygomorphic
(B) Petals five, persistent and with valvate aestivation
(C) Bicarpellary, syncarpous, superior ovary and axile placentation
(D) Fruit type capsule or berry
(E) Stem glabrous or hairy

Space for Rough Work

How many of the above features are true for the plant family to which tobacco belongs?

(1) 4

(2) 2

(3) 3

(4) 5

118. Read the features of some fungi in below given table and complete it with suitable option for A-F

Feature	*Mucor*	*Aspergillus*	Puffball	*Colletotrichum*
Type of hyphae	Coenocytic	A	Septate	Septate
Fruiting body	–	Ascocarp	Basidiocarp	B
Sexual Spore	C	Meiospore	Basidiospore	D
Asexual spore	Non motile	Conidia	E	F

(1) A-Coenocytic, D-Meiospore, F-Motile

(2) B-Oospore, D-Absent, F-Conidia

(3) B- Absent, C-Present, E-Absent

(4) A-Septate, B-Absent, F-Sporangiospore

119. Match **Column-I** with **Column-II** and select the **correct** option.

	Column-I (Cohesion of stamens)		**Column-II (Plant)**
a.	Monoadelphous	(i)	Citrus
b.	Diadelphous	(ii)	Mustard
c.	Polyadelphous	(iii)	Pea
d.	Polyandrous	(iv)	China rose

(1) a(iv), b(iii), c(i), d(ii) (2) a(iv), b(iii), c(ii), d(i)

(3) a(iii), b(iv), c(i), d(ii) (4) a(iii), b(i), c(iv), d(ii)

120. In which of the following type of aestivation, margin of petals or sepals overlap one another but not in any particular direction?

(1) Valvate

(2) Twisted

(3) Imbricate

(4) Vexillary

121. Select **odd** one out w.r.t. alternate phyllotaxy.

(1) China rose

(2) *Solanum*

(3) *Calotropis*

(4) Sunflower

122. Which of the following statement is **not** correct w.r.t. regions of the root?

(1) The apex of the root is covered by a thimble like structure which is multicellular and is made up of parenchymatous cells

(2) Some epidermal cells from the region of elongation form very fine and delicate thread like structures

(3) The cells proximal to the meristematic region towards the base of the stem are responsible for the growth of roots in length

(4) The layer which is few millimetre above the root cap divide repeatedly to produce new cells

123. Identify the **incorrectly** matched pair.

(1) Zygomorphic flower – Pea

(2) Perigynous flower – Rose

(3) Twisted aestivation of petals – Cotton

(4) Flowers have bilateral symmetry – Ashwagandha

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124. Select **odd** one out w.r.t. plants showing racemose inflorescence.

(1) Belladonna (2) *Trifolium*

(3) Sweet pea (4) *Sesbania*

125. Read the following statements(A-D).

(A) All the members of ascomycetes sexually reproduce by ascospores formed in ascocarp.

(B) White spots seen on mustard leaves are due to a fungus that have coenocytic mycelium.

(C) Protozoans are found only in fresh water.

(D) Dinoflagellates are mostly fresh water organisms and float actively in water.

In the light of above statements select the **correct** statement(s) from the options given below.

(1) Only (B)

(2) Only (B) and (C)

(3) Only (A) and (D)

(4) Only (C) and (D)

126. State True(T) or False(F) and select the **correct** option.

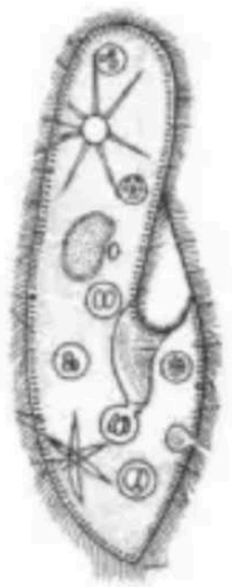
(A) During unfavourable conditions, the plasmodium of slime moulds differentiates and forms fruiting bodies bearing spores at their tips.

(B) *Rhizopus* commonly known as bread mould, produces asexual spore called zygospore.

(C) *Puccinia* asexually reproduce by conidia.

	A	B	C
(1)	T	F	F
(2)	T	F	T
(3)	F	F	T
(4)	T	T	T

127. Observe the following diagram:



Identify the above organism and select the correct statements for it.

(a) It is an actively moving organism.

(b) It captures its prey with the help of pseudopodia

(c) According to R.H. Whittaker's classification system, it is placed in the same kingdom in which *Chlorella* is placed.

(d) It possesses cell wall having stiff cellulose plates on the outer surface.

(e) It utilizes the coordinated movement of its locomotory structure to steer the food in the cavity found in the body.

(1) (b), (c) and (d) only

(2) (a), (c) and (e) only

(3) (a), (b) and (c) only

(4) (b), (d) and (e) only

128. Read the following **Assertion(A)** and **Reason(R)** and select the correct option.

Assertion (A): Both cyanobacteria and higher plants perform oxygenic photosynthesis.

Reason (R): Cyanobacteria have most of the pigments identical to higher plants.

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(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true

129. Read the following **Assertion(A)** and **Reason(R)** and select the correct option.

Assertion (A) : Viruses did not find a place in the five kingdom classification

Reason (R) : 'Viruses' are not considered truly 'living'

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

130. Match **Column-I** with **Column-II** and select the **correct** option.

	Column – I		Column – II
a.	Viruses	(i)	Mad cow disease in cattle
b.	Bacteria	(ii)	Potato spindle tuber disease
c.	Viroids	(iii)	Mosaic formation in plants
d.	Prions	(iv)	Citrus canker

(1) a(iv), b(iii), c(ii), d(i) (2) a(iii), b(iv), c(ii), d(i)
(3) a(ii), b(iii), c(i), d(iv) (4) a(i), b(iv), c(ii), d(iii)

131. Select **odd** one out w.r.t. imperfect fungi.

(1) *Alternaria* (2) *Colletotrichum*
(3) *Trichoderma* (4) *Agaricus*

132. Which of the following statements is not true w.r.t. smallest prokaryotic cell known?

(1) They are heterotrophic in nature
(2) They can survive without oxygen
(3) They have peptidoglycan cell membrane
(4) They are pathogenic to autotrophic organism as well as to heterotrophic organism

133. Read the following statements and select the **correct** option.

Statement A: Archaebacteria differ from other bacteria in having a different cell wall structure.

Statement B: Some archaebacteria are capable of converting CO_2, methanol and formic acid into methane.

(1) Only statement A is correct
(2) Only statement B is correct
(3) Both statements A and B are correct
(4) Both statements A and B are incorrect

134. Five kingdom classification was given by

(1) Aristotle (2) C. Linnaeus
(3) R. H. Whittaker (4) W.M. Stanley

135. Select the **incorrectly** matched pair.

(1) Mycoplasma – Resistant towards the antibiotics that break the cell wall
(2) *Agaricus* – Karyogamy followed by meiosis, spores are produced endogenously
(3) Diatoms – Chief producers in the ocean
(4) Lichens – Symbiotic association of algae and fungi

Space for Rough Work

[ZOOLOGY]

136. Pulmonary artery and pulmonary vein can be differentiated from one another on the basis of all of the following facts, **except**

(1) Partial pressure of O_2

(2) The blood they carry is oxygenated or deoxygenated

(3) Number of layers in their walls

(4) Partial pressure of CO_2

137. An athlete went for routine medical examination. It was observed by doctor that his heart rate is around 66 beats/min at rest. The most possible reason behind this is

(1) Greater stroke volume

(2) Hypertension

(3) Lower stroke volume

(4) Atherosclerosis

138. Match the columns w.r.t. disorders of circulatory system.

	Column I		Column II
a.	Coronary artery disease	(i)	Heart stops beating
b.	Angina	(ii)	Congestion of lungs
c.	Heart failure	(iii)	Acute chest pain
d.	Cardiac arrest	(iv)	Atherosclerosis

Select the correct option

(1) a(i), b(ii), c(iii), d(iv) (2) a(iv), b(iii), c(i), d(ii)

(3) a(iv), b(iii), c(ii), d(i) (4) a(iii), b(ii), c(i), d(iv)

139. Select the blood vessel among the following in which blood flows with maximum pressure.

(1) Pulmonary vein (2) Renal vein

(3) Hepatic vein (4) Coronary artery

140. Which of the following occurs during atrial systole?

(A) 'Dub' heart sound is produced due to closure of semilunar valves

(B) Increased flow of blood into ventricles by about 30 per cent

(C) 'Lub' heart sound is produced due to the closure of atrio-ventricular valves

(D) 'Lub' heart sound is produced due to the opening of atrio-ventricular valves

Select the **correct** option.

(1) (A) and (B) (2) Only (B)

(3) (C) and (D) (4) Only (D)

141. A student was studying about an organ that pumps blood and another organ that stores urine in humans. Which of the following statements is false?

(1) Former is characterised by the presence of nodal tissue

(2) Latter is characterised by the presence of fusiform muscle fibres

(3) Former has branched muscle fibres

(4) Latter is characterised by muscle fibres which show striated appearance when viewed under the microscope

142. Plasma without clotting factors has

(1) Antibodies (2) RBCs

(3) WBCs (4) Platelets

Space for Rough Work

143. A person with AB^{+ve} blood group can accept blood from persons with AB^{-ve} blood group and other groups because he/she has
(1) No antibodies corresponding to any blood group antigens in his/her blood plasma
(2) No antigen on his/her RBC
(3) Anti-A, Anti-B antibodies in his/her plasma
(4) Antibody against Rh antigen

144. In human heart, atrium and ventricle of same side are separated by the
(1) Thin elastic tissue
(2) Thick fibrous tissue
(3) Thin cartilaginous tissue
(4) Thick cartilaginous tissue

145. All of the following play a role in mechanism of blood coagulation in humans, **except**
(1) Thrombokinase
(2) Fibrinogen
(3) Prothrombin
(4) Serotonin

146. Consider the given features.
a. Tunica media is thicker in them as compared to arteries
b. Their lumen is narrower as compared to arteries
c. They always carry blood towards the heart
d. They always carry deoxygenated blood
How many of the above given feature(s) is/are **correct** w.r.t. veins?
(1) One
(2) Two
(3) Three
(4) Four

147. If the cardiac output of a person is 7.2 L/min, end diastolic volume of left ventricle is 150 mL and end systolic volume of left ventricle is 50 mL, then what is the heart rate of this person?
(1) 72 beats per minute
(2) 80 beats per minute
(3) 64 beats per minute
(4) 100 beats per minute

148. Choose the **incorrect** statement w.r.t. humans.
(1) Reverse ratio of heart beat rate per minute is duration of a cardiac cycle
(2) Eosinophils have bilobed nucleus
(3) Megakaryocytes are precursors of thrombocytes
(4) All leucocytes generally live for a longer duration than erythrocytes

149. Consider the given statements.
Statement (A): Exchange of O_2 and CO_2 between blood and tissues is mainly based on their pressure/concentration gradients.
Statement (B): Rate of diffusion of gases depends only on thickness of diffusion membrane and not on the solubility of the gases.
Select the **correct** option.
(1) Both statements (A) and (B) are correct
(2) Only statement (B) is correct
(3) Both statements (A) and (B) are incorrect
(4) Only statement (A) is correct

150. A healthy adult human has 'X' grams of haemoglobin in every 100 mL of blood. Numerically 'X' is equal to
(1) Percentage of neutrophils in total leucocyte count
(2) Percentage of plasma protein
(3) Percentage of monocytes in total leucocyte count
(4) Respiratory rate in healthy adult human during normal condition

Space for Rough Work

151. Select the correct set of favourable conditions required for the dissociation of oxyhaemoglobin at the tissue level in a healthy adult human.
(1) Low pO_2, high pCO_2, high H^+ and high temperature
(2) High pO_2, high pCO_2, high H^+ and high temperature
(3) Low pO_2, low pCO_2, low H^+ and low temperature
(4) High pO_2, low pCO_2, low H^+ and low temperature

152. What is the approximate partial pressure of oxygen (in mm Hg) when haemoglobin is 50% saturated with oxygen in a healthy adult human?
(1) 25-30 (2) 40-50
(3) 50-60 (4) 60-70

153. In a healthy adult man, the partial pressure of CO_2 in systemic artery is same as the partial pressure of
(1) Oxygen in blood drained by superior vena cava
(2) CO_2 in blood drained by inferior vena cava
(3) CO_2 in tissues
(4) Oxygen in atmospheric air

154. All of the following are observed during inspiration in a healthy adult human under strenuous condition, **except**
(1) Contraction of external inter-costal muscles
(2) Contraction of internal inter-costal muscles
(3) Increase in volume of thoracic chamber
(4) Contraction of diaphragm

155. Peripheral chemoreceptors that help in regulation of breathing are present in
(a) Aortic arch
(b) Carotid artery
(c) Medulla oblongata
Choose the **correct** option.
(1) (a) only
(2) (b) only
(3) (a) and (b) only
(4) (a) and (c) only

156. Choose the option that correctly represents the path of air flow in respiratory passage of humans during inspiration.
(1) Nasal chamber → Pharynx → Larynx → Trachea → Bronchi → Terminal bronchioles → Alveoli
(2) Nasal chamber → Larynx → Trachea → Pharynx → Bronchi → Terminal bronchioles → Alveoli
(3) Nasal chamber → Trachea → Pharynx → Larynx → Bronchi → Terminal bronchioles → Alveoli
(4) Nasal chamber → Trachea → Larynx → Pharynx → Bronchi → Terminal bronchioles → Alveoli

157. **Assertion (A):** In adult humans, venous blood has low pH as compared to arterial blood, except in pulmonary vein.
Reason (R): Venous blood has more CO_2 and H^+ concentration as compared to arterial blood, except in pulmonary vein.
In the light of above statements, select the **correct** option.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) Both (A) and (R) are false
(4) (A) is true but (R) is false

Space for Rough Work

158. Consider the following statements and select the **correct** option.

Statement (A): Breathing is defined as intake of O_2 rich air and removal of CO_2 rich air.

Statement (B): Cellular respiration is defined as oxidation of food to form CO_2, H_2O and energy.

Select the **correct** option.

(1) Both statements (A) and (B) are correct

(2) Only statement (A) is correct

(3) Only statement (B) is correct

(4) Both statements (A) and (B) are incorrect

159. Incomplete cartilaginous rings are absent in all of the following structures w.r.t. human respiratory system, **except**

(1) Initial bronchiole (2) Terminal bronchiole

(3) Alveoli (4) Alveolar ducts

160. Consider the given reaction.

$$H_2O + CO_2 \overset{'X'}{\rightleftharpoons} H_2CO_3 \overset{'X'}{\rightleftharpoons} HCO_3^- + H^+$$

Select the incorrect option w.r.t. enzyme 'X'.

(1) 'X' belongs to the class IV of enzymes

(2) 'X' facilitates the above reaction in forward direction in tissue

(3) 'X' facilitates the above reaction in backward direction in alveoli

(4) 'X' is abundantly present in RBCs as well as in WBCs

161. All of the following sets of respiratory volumes and capacities can be directly measured by a simple spirometer, **except**

(1) IRV, ERV (2) EC, TV, IRV

(3) VC, IC (4) FRC, TLC

162. Choose the **incorrect** statement w.r.t. humans.

(1) Normal expiration is a passive process.

(2) Inspiration is an active process.

(3) Thoracic chamber is laterally formed by diaphragm

(4) We cannot directly alter the pulmonary volume

163. Match column I and column II w.r.t. respiratory volumes in an adult human under normal conditions.

	Column I		**Column II**
a.	Respiratory minute volume	(i)	1000-1100 mL
b.	Expiratory reserve volume	(ii)	1100-1200 mL
c.	Residual volume	(iii)	6000-8000 mL
d.	Inspiratory reserve volume	(iv)	2500-3000 mL

Select the **correct** option.

(1) a(iv), b(i), c(ii), d(iii)

(2) a(iii), b(i), c(ii), d(iv)

(3) a(iii), b(i), c(iv), d(ii)

(4) a(i), b(ii), c(iii), d(iv)

164. **Assertion (A):** There is a negative pressure in the lungs with respect to atmospheric pressure during inspiration.

Reason (R): During inspiration, increase in the volume of thoracic chamber decreases the intra-pulmonary pressure to less than the atmospheric pressure.

Space for Rough Work

In the light of above statements, select the **correct** option w.r.t. humans.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) Both (A) and (R) are false

(4) (A) is true but (R) is false

165. Consider the following pulmonary volumes and capacities

a. VC = TLC – RV b. IC = TLC – FRC

c. FRC = ERV + RV d. TLC = RV + TV

Which of the above representations are correct?

(1) a, b and c only (2) b and c only

(3) b, c and d only (4) a and c only

166. Select the odd one w.r.t. nitrogenous excretory waste.

(1) Land snails

(2) Mammals

(3) Many terrestrial amphibians

(4) Marine fishes

167. During the process of haemodialysis in humans, the blood is usually drained from

(1) Any convenient blood vessel which carries deoxygenated blood from heart

(2) A convenient artery which carries oxygenated blood

(3) Pulmonary vein

(4) Pulmonary artery

168. Select the **correct** match.

(1) *Planaria* – Green glands

(2) *Pheretima* – Protonephridia

(3) *Periplaneta* – Malpighian tubules

(4) *Palaemon* – Nephridia

169. Which of the following is associated with the disorder caused due to hyposecretion of ADH?

(1) Hyperglycemia (2) Uremia

(3) Diuresis (4) Ketonuria

170. In a hypothetical situation, the loop of Henle are removed from all the juxtamedullary nephrons of human kidneys. This will lead to

(1) No urine formation

(2) Production of more diluted urine

(3) Production of more concentrated urine

(4) No change in osmolarity of urine

171. Which of the factors listed below are responsible for concentration of glomerular filtrate in juxtamedullary nephron?

a. The proximity between the Henle's loop and vasa recta

b. Flow of filtrate in opposite direction in limbs of loop of Henle

c. Long length of loop of Henle

d. Increasing osmolarity from inner medullary interstitium to renal cortex

Choose the **correct** option.

(1) a, b, c only

(2) b, c, d only

(3) c, d only

(4) a, c, d only

172. Conditional reabsorption of Na^+ and water takes place in

(1) Proximal convoluted tubule

(2) Distal convoluted tubule

(3) Descending limb of loop of Henle

(4) Ascending limb of loop of Henle

Space for Rough Work

173. The DCTs of many nephrons open into a straight tube called collecting duct, many of which converge and open into the _____ through medullary pyramids in the calyces.

Select the correct option to fill in the blank.

(1) Ureter

(2) Renal cortex

(3) Renal pelvis

(4) Renal column

174. Choose the **incorrect** match w.r.t. human excretory system.

(1)	Counter current mechanism	–	Henle's loop and vasa recta
(2)	Descending limb of loop of Henle	–	Completely impermeable to electrolytes
(3)	Distal convoluted tubule	–	Osmolarity of filtrate may be less than 300 $mOsmolL^{-1}$
(4)	Bowman's capsule	–	Contains epithelial cells called podocytes

175. Select the **correct** option to complete the analogy w.r.t. osmolarity of glomerular filtrate in different parts of nephrons in human.

Descending limb of loop of Henle: Hypertonic to blood plasma ::

Proximal convoluted tubule: _____ to blood plasma

(1) Isotonic

(2) Hypotonic

(3) Hypertonic

(4) Initially hypertonic then hypotonic

176. In humans, how many of the substances mentioned in the box below are present in glomerular blood but absent in fluid present in lumen of Bowman's capsule, under normal conditions?

Na^+, Glucose, Globulin, Fibrinogens, K^+, H^+

Choose the **correct** option.

(1) One

(2) Five

(3) Two

(4) Four

177. **Assertion (A):** Decrease in GFR can activate the JG cells to release renin.

Reason (R): Renin can inhibit the glomerular blood flow and thereby bring the GFR back to the normal by decreasing glomerular blood pressure.

In the light of above statements, select the **correct** option.

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(2) Both (A) and (R) are true and (R) is the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

178. In humans, the ascending limb of loop of Henle is _____ to water but allows transport of electrolytes _____.

Select the correct option to fill in the respective blanks.

(1) Impermeable, actively or passively

(2) Impermeable, only actively

(3) Permeable, only passively

(4) Permeable, only actively

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179. In counter current mechanism, NaCl is transported by the ascending limb of Henle's loop which is exchanged with the
(1) Descending limb of vasa recta
(2) Ascending limb of vasa recta
(3) Descending limb of Henle's loop
(4) Collecting duct

180. Which among the following is not a secretion of sweat glands?
(1) Urea
(2) Lactic acid
(3) NaCl
(4) Hydrocarbons

❑ ❑ ❑

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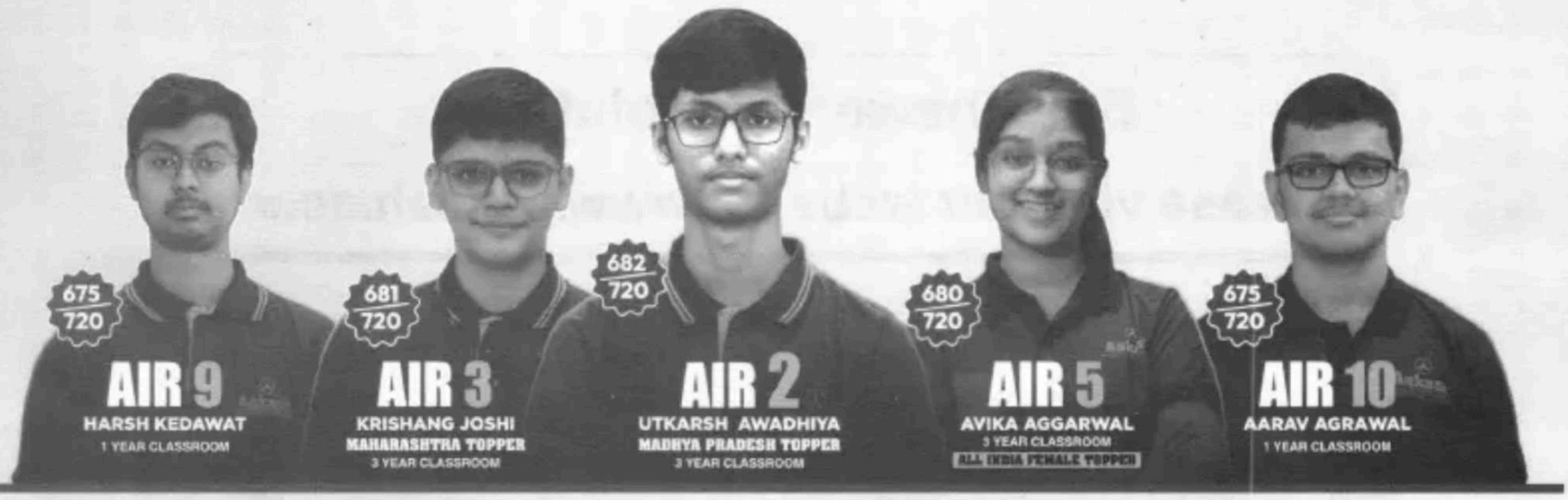

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ANSWERS

1. (2)	37. (2)	73. (4)	109. (3)	145. (1).
2. (4)	38. (4)	74. (1)	110. (3)	146. (2)
3. (3)	39. (4)	75. (3)	111. (4)	147. (3)
4. (1)	40. (2)	76. (4)	112. (3)	148. (3)
5. (4)	41. (1)	77. (1)	113. (4)	149. (2)
6. (2)	42. (4)	78. (4)	114. (3)	150. (1)
7. (3)	43. (4)	79. (2)	115. (2)	151. (1)
8. (2)	44. (4)	80. (4)	116. (2)	152. (1).
9. (3)	45. (1)	81. (1)	117. (2)	153. (2)
10. (4)	46. (4)	82. (2)	118. (3)	154. (3)
11. (1)	47. (1)	83. (2)	119. (4)	155. (4)
12. (2)	48. (4)	84 (1)	120. (3)	156. (4)
13. (2)	49. (2)	85 (4)	121. (3)	157. (1)
14. (2)	50. (1)	86. (2)	122. (1)	158. (1)
15. (1)	51. (2)	87. (3)	123. (2)	159. (1).
16. (1)	52. (2)	88. (3)	124. (1)	160. (1)
17. (1)	53. (3)	89. (2)	125. (2)	161. (3)
18. (3)	54. (1)	90. (1)	126. (4)	162. (2)
19. (2)	55. (2)	91. (2)	127. (4)	163. (1)
20. (4)	56. (2)	92. (3)	128. (3)	164. (1)
21. (3)	57. (3)	93. (3)	129. (3)	165. (1)
22. (1)	58. (1)	94. (3)	130. (2)	166. (4)
23. (2)	59. (4)	95. (4)	131. (2)	167. (4)
24. (4)	60. (3)	96. (2)	132. (2)	168. (4)
25. (3)	61. (3)	97. (1)	133. (2)	169. (1)
26. (3)	62. (4)	98. (3)	134. (3)	170. (1)
27. (1)	63. (2)	99. (2)	135. (3)	171. (4)
28. (3)	64. (4)	100. (1)	136. (4)	172. (2)
29. (4)	65. (1)	101. (1)	137. (1)	173. (1)
30. (3)	66. (1)	102. (1)	138. (1)	174. (1).
31. (2)	67. (2)	103. (4)	139. (3).	175. (4)
32. (3)	68. (3)	104. (2)	140. (3).	176. (2)
33. (3)	69. (2)	105. (3)	141. (1)	177. (4).
34. (4)	70. (2)	106. (3)	142. (2)	178. (3)
35. (1)	71. (4)	107. (1)	143. (3).	179. (1)
36. (3)	72. (4)	108. (3)	144. (2)	180. (3)

HINTS & SOLUTIONS

[PHYSICS]

1. Answer (2)

Hint: $\vec{A}\times\vec{B} = AB\sin\theta\,\hat{n}$

Sol.: $\vec{A}\times\vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & -5 \\ 3 & -2 & 5 \end{vmatrix}$

$= \hat{i}(15-10) - \hat{j}(10+15) + \hat{k}(-4-9)$

$= 5\hat{i} - 25\hat{j} - 13\hat{k}$

$|\vec{A}\times\vec{B}| = \sqrt{819}$ units

2. Answer (4)

Hint : a_t is along tangent and a_r is along radius.

Sol.: When the particle is moving in a vertical circle with the help of a string, the force towards the centre and the force along the tangent, vary from point to point and so both its radial and tangential acceleration also vary.

3. Answer (3)

Hint: $R^2 = P^2 + Q^2 + 2PQ\cos\theta$

Sol.: $P = (F_1 + F_2)$, $Q = (F_1 - F_2)$

$R = \sqrt{F_1^2 + F_2^2}$

$\cos\theta = \dfrac{R^2 - P^2 - Q^2}{2PQ}$

$= \dfrac{(F_1^2 + F_2^2) - (F_1 + F_2)^2 - (F_1 - F_2)^2}{2(F_1 + F_2)(F_1 - F_2)}$

$= -\left[\dfrac{(F_1^2 + F_2^2)}{2(F_1^2 - F_2^2)}\right] \Rightarrow \theta = \cos^{-1}\left[-\dfrac{(F_1^2 + F_2^2)}{2(F_1^2 - F_2^2)}\right]$

4. Answer (1)

Hint: $P = \vec{F}.\vec{v}$, $a = v\dfrac{dv}{dx}$

Sol.: $P = mav$ and $a = v\dfrac{dv}{dx}$

$v^2 dv = \dfrac{P}{m}dx$

$\int_u^v v^2\,dv = \dfrac{P}{m}\int_0^x dx$

$P = \dfrac{m(v^3 - u^3)}{3x} = \dfrac{2500(216-64)}{3\times 250}$

$= \dfrac{2500\times 152}{3\times 250} = 506.67\text{ W} \approx 507\text{ W}$

5. Answer (4)

Hint: $W_{cons} = -\Delta U$

Sol.: Potential energy is defined for conservative force field. Work done by system against conservative force results in increase in potential energy.

6. Answer (2)

Hint: $\alpha = \dfrac{\omega_f - \omega_i}{t}$ & $\theta = \omega_0 t + \dfrac{1}{2}\alpha t^2$

Sol.: $f_0 = 420$ rev/min

$\omega_0 = 2\pi \times 7 = 14\pi$ rad/s

At $t = 10$ s, $\omega = 0$

$\alpha = -\dfrac{14\pi}{10} = -\dfrac{7\pi}{5}$ rad/s^2

$\theta = \omega_0 t + \dfrac{1}{2}\alpha t^2$

$= 14\pi\times 10 - \dfrac{1}{2}\times\dfrac{7\pi}{5}\times 100$

$\theta = 70\pi$ rad

Number of revolutions $= \dfrac{\theta}{2\pi}$

$= \dfrac{70\pi}{2\pi} = 35$ revolutions

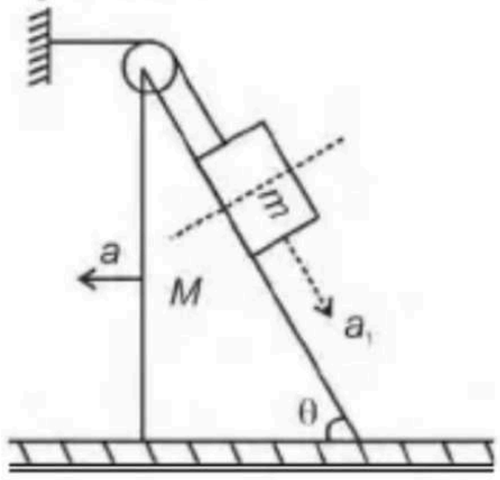
7. Answer (3)

Hint: Wedge constraint

(i) Along common normal, there is no relative motion.

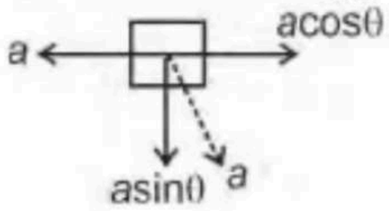
(ii) Perpendicular to common normal, motion is always relative.

Sol.:



$a_1 = a$ = acceleration of M w.r.t. ground = 2 m/s^2

acceleration of m w.r.t. $M = a_{m/M} = a = 2$ m/s^2



$$a_{m/g} = a_{m/M} + a_M$$

$$\left|\vec{a}_{m/g}\right| = \sqrt{(a - a\cos\theta)^2 + (a\sin\theta)^2}$$

$$= 2a\sin\frac{\theta}{2}$$

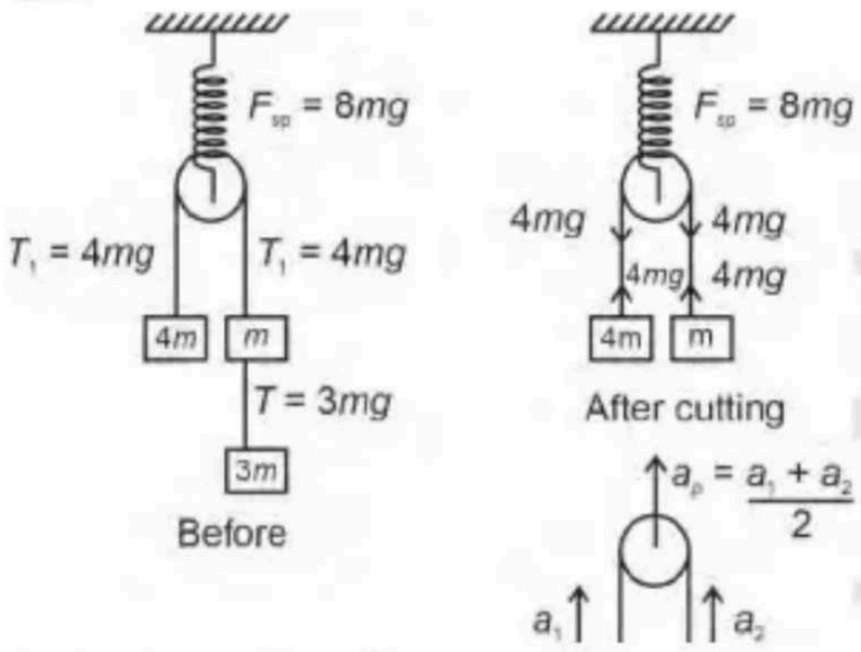
$$= 2 \times 2 \times \frac{1}{2} = 2\ \text{m/s}^2$$

8. Answer (2)

Hint: Initially system was at rest
Just after string is cut:

$$a_{pulley} = \frac{a_1 + a_2}{2}$$

Sol.:



Spring force, $F_{sp} = 8mg$

$a_1 = a_{4m} = 0$

$$a_2 = a_m = \frac{4mg - mg}{m} = 3g$$

$$a_p = \frac{0 + 3g}{2} = \frac{3g}{2}$$

9. Answer (3)

Hint : Use $m_1r_1 = m_2r_2$

Sol.: C.O.M. will be in IV quadrant

10. Answer (4)

Hint: Conservative force $F = -\frac{\partial U}{\partial r}$

Sol.: $F_x = -\frac{\partial U}{\partial x} = -3$

$$F_y = -\frac{\partial U}{\partial y} = -4$$

$$\vec{F} = F_x\hat{i} + F_y\hat{j}$$

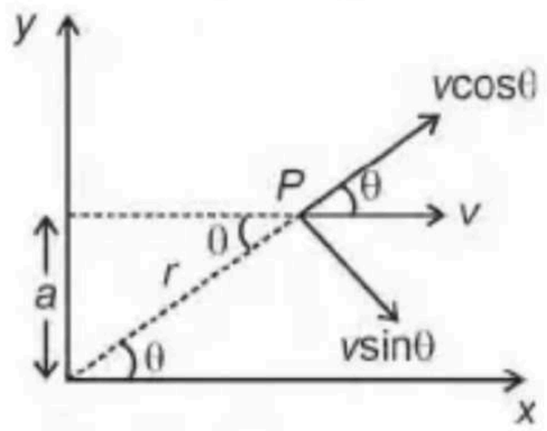
$$= -(3\hat{i} + 4\hat{j})\ \text{N}$$

$$|\vec{a}| = \frac{|\vec{F}|}{m} = \frac{\sqrt{9+16}}{5} = 1\ \text{m/s}^2$$

11. Answer (1)

Hint: Angular velocity $\omega = \frac{v_\perp}{r}$

Sol.: $\omega = \frac{v_\perp}{r} = \frac{v\sin\theta}{r}$



$$\frac{a}{r} = \sin\theta$$

$$r = \frac{a}{\sin\theta}$$

$$\omega = \frac{v\sin^2\theta}{a} = 2 \times \frac{1}{4 \times 2} = \frac{1}{4}\ \text{rad/s}$$

12. Answer (2)

Hint & Sol: Potential energy of a system can be positive, negative or zero.

- In circular motion, tension is perpendicular to velocity of block.

  Small work done, $dW = Tdx\cos90° = 0$

  $\Rightarrow$ Work done is zero

- Also, work done $= \int F.\,dx$

$\therefore$ Statement C is false

13. Answer (2)

Hint: In perfectly elastic head-on collision, kinetic energy transfer is maximum for $m_1 = m_2$

Sol: $v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)u_1,\ v_2 = \frac{2m_1}{(m_1 + m_2)}u_1$

Fraction of kinetic energy transferred

$$= \frac{(K_B)_{final}}{(K_A)_{initial}} = \frac{4m_1m_2}{(m_1 + m_2)^2}$$

This expression is maximum when

$m_1 = m_2$

14. Answer (2)

Hint: Power $P = \vec{F}.\vec{v}$

Sol.: $P = (2\hat{i} - 3\hat{j} + 4\hat{k}) \cdot (2\hat{i} + 3\hat{j} + 5\hat{k})$

$= 4 - 9 + 20 = 15\ \text{W}$

15. Answer (1)

Hint: Total acceleration is given by

$a_{total}^2 = a_t^2 + a_c^2$

Sol.: Since the tangential acceleration is constant:

$S = \frac{1}{4} \times 2\pi r = \frac{\pi r}{2}$

$S = ut + \frac{1}{2}at^2$

$\frac{\pi r}{2} = \frac{1}{2}at^2$

$t^2 = \frac{\pi r}{a}$

$v = a\sqrt{t^2} = a\sqrt{\frac{\pi r}{a}}$

$v = \sqrt{a\pi r}$

$a_c = \frac{v^2}{r} = \frac{a\pi r}{r} = a\pi$

$a_t = a$

$a_{total} = \sqrt{a_t^2 + a_c^2} = \sqrt{a^2 + a^2\pi}$

$= a\sqrt{1+\pi^2} = 2\sqrt{1+\pi^2}$

16. Answer (1)

Hint & Sol.: Under Newton's 3rd law, action and reaction are always equal and opposite. They always act on different bodies.

17. Answer (1)

Hint: $F \times t$ = impulse

& Impulse = change in momentum

Sol.:

(i) Impulse $= \frac{1}{2} \times 20 \times 20 = 200$ N-s

(ii) Impulse = $P_f - P_i \Rightarrow 200 = P_f - 10$

$P_f = 210$ kg-m/s

18. Answer (3)

Hint: Force $\vec{F} = \frac{d\vec{p}}{dt}$

Sol.: $F = \frac{2mu\cos\theta}{t}$

$= \frac{2 \times 500 \times 10^{-3} \times 40 \times 1}{0.4 \times 2} = 50$ N

19. Answer (2)

Hint: To loop-the-loop body should not lose contact with the track anywhere, so velocity $v \geq \sqrt{gr}$ at the topmost point of circular loop.

Sol.: By conservation of mechanical energy,

$\frac{1}{2}mv_{min}^2 = mg(h_{min} - 2r)$

$\frac{1}{2}m(gr) = mg(h_{min} - 2r)$

$\frac{1}{2}r = h_{min} - 2r$

$\therefore \quad h_{min} = \frac{r}{2} + 2r = \frac{5r}{2} = 2.5$ m

20. Answer (4)

Hint: For crossing the bridge in minimum time, it has to travel with maximum acceleration.

Sol.: a = maximum acceleration produced in car

$ma = \mu R$ (if acceleration is more than this, then tyres will slip)

$ma = \mu mg$

$a = \mu g = 10$ m/s^2

$u = 0$, $s = 500$ m, $a = 10$ m/s^2

$500 = \frac{1}{2} \times 10 \times t^2$

$t = 10$ s

21. Answer (3)

Hint: Work done (W) = $\Delta K \cdot E$

Sol.: $x^2 = 25t^3$

$x = 5t^{3/2}$

$v = \frac{dx}{dt} = 5 \times \frac{3}{2}t^{1/2}$

At $t = 0$, $v = 0$

$\Rightarrow$ Particle is at rest

$W = \frac{1}{2}mv_t^2 - \frac{1}{2}mv_{t=0}^2$

$W = \frac{1}{2}mv_t^2$

$= \frac{1}{2} \times 5 \times \left(\frac{15}{2}t^{1/2}\right)^2 = \frac{5 \times 225}{2 \times 4} \times t$

$\therefore \quad W \propto t$

22. Answer (1)

Hint: Work done $W = \vec{F} \cdot d\vec{s}$, $\hat{i} \cdot \hat{j} = 0$

Sol.: $W = \vec{F} \cdot \Delta\vec{x}$

$\Delta\vec{x} = (-3\hat{i})$ m

(a) $W = -15$ J

(b) W = zero

(c) $W = (-5\hat{i}) \cdot (-3\hat{j}) = 15$ J

(d) $W = (-8\hat{i} + 5\hat{j}) \cdot (-3\hat{i}) = 24$ J

23. Answer (2)

Hint: Use conservation of momentum and change in total kinetic energy to find work done by internal forces.

Sol.: $\vec{P}_i$ = zero

$\vec{P}_f = m_1\vec{u} + m_2\vec{v}$

$\vec{P}_i = \vec{P}_f$

$0 = m_1\vec{u} + m_2\vec{v}$

$\vec{v} = \dfrac{-m_1\vec{u}}{m_2}$

$K.E_i = 0$

$K.E_f = \dfrac{1}{2}m_1u^2 + \dfrac{1}{2}m_2v^2$

$K.E_f = \dfrac{1}{2}m_1u^2 + \dfrac{1}{2}m_2\left(\dfrac{m_1u}{m_2}\right)^2$

$= \dfrac{1}{2}m_1u^2 + \dfrac{1}{2}\dfrac{m_1^2u^2}{m_2}$

$K.E_f = \dfrac{1}{2}\left(\dfrac{m_1m_2 + m_1^2}{m_2}\right)u^2$

$\Delta K.E = \dfrac{1}{2}m_1u^2\left[1 + \dfrac{m_1}{m_2}\right]$

$= \dfrac{1}{2}\times 1\times 4\left[1 + \dfrac{1}{6}\right]$

$= 2\times\dfrac{7}{6} = \dfrac{7}{3}$ J

24. Answer (4)

Hint : Use momentum and energy conservation laws to verify post-collision velocities in each option.

Sol.: For elastic collision in option (3), $m_1 >> m_2$

for $u_1 \neq 0$, $u_2 = 0$

then, $v_1 = \left(\dfrac{m_1 - m_2}{m_1 + m_2}\right)u_1 \approx u_1$

$v_2 = \dfrac{2m_1u_1}{(m_1 + m_2)} \approx 2u_1$

For elastic collision in option (4), $m_1 = m_2$

for $u_1 \neq 0$, $u_2 = 0$,

then, $v_1 = 0$, $v_2 = u_1$

25. Answer (3)

Hint: Velocity of C.O.M. $\vec{v}_{cm} = \dfrac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1 + m_2}$

Sol.: $\vec{v}_{cm} = \dfrac{0.2\times 5\hat{i} + 0.5(10\hat{i} + 10\hat{j})}{0.2 + 0.5}$

$= \dfrac{\hat{i} + 5\hat{i} + 5\hat{j}}{0.7}$

$= \dfrac{6\hat{i} + 5\hat{j}}{0.7} = \dfrac{60\hat{i} + 50\hat{j}}{7}$ m/s

26. Answer (3)

Hint : Power due to a force in rotational motion is given by dot product of torque and angular velocity.

Sol.: $\vec{\tau} = \vec{r}\times\vec{F}$

$P = \vec{\tau}\cdot\vec{\omega}$

$P = (\vec{r}\times\vec{F})\cdot\vec{\omega}$

27. Answer (1)

Hint: Use conservation of angular momentum

Sol.: By conservation of angular momentum,

$I_1\omega_1 + I_2\omega_2 = (I_1 + I_2)\omega$

$\omega = \dfrac{2\times 4 + 4\times 2}{6} = \dfrac{16}{6} = \dfrac{8}{3}$ rad/s

Rotational kinetic energy = $\dfrac{1}{2}(I_1 + I_2)\omega^2$

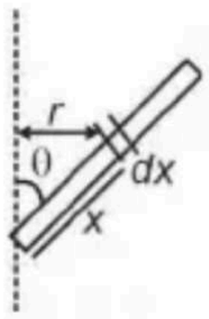
$= \dfrac{1}{2}\times 6\times\dfrac{64}{9} = \dfrac{64}{3}$ J

28. Answer (3)

Hint: Moment of inertia, $I = \int r^2\, dm$

Sol.: $r = x\sin\theta$

$I = \int r^2\, dm = \int_0^L x^2\sin^2\theta\left(\dfrac{M}{L}\right)dx$



$= \dfrac{ML^2\sin^2\theta}{3} = \dfrac{5\times 25\times 3}{4\times 3} = \dfrac{125}{4}$ kg m^2

29. Answer (4)

Hint: Torque $\vec{\tau} = \vec{r}\times\vec{F}$

Sol.: Torque is an axial vector and its direction is perpendicular to the plane containing $\vec{r}$ & $\vec{F}$.

$\vec{\omega}_{A/B} = \vec{\omega}_{B/A} = \vec{\omega}_{body}$

Angular velocity is property of entire rigid body.

$k_{annular\ disc} = \sqrt{\dfrac{1}{2}(R_1^2 + R_2^2)}$

30. Answer (3)

Hint: Angular impulse = change in angular momentum

Sol.: $\tau t = I\omega - 0$

$$\omega = \frac{\tau t}{I} = \frac{FRt}{I} = \frac{40\times1\times5}{4} = 50 \text{ rad/s}$$

31. Answer (2)

Hint: $\vec{F}_{net} = m\vec{a}$

$\vec{\tau} = \vec{r}\times\vec{F}$

Sol.: $F_{net} = -20+30-10+40-40 = \text{Zero}$

$\therefore$ No translatory motion.

Taking torque of these forces about centre of mass of the rod,

$|\vec{\tau}| = 30\times10+10\times10+40\times50-(20\times30)-(40\times30)$

$= 2400 - 1800 = 600$

$\therefore$ $\vec{\tau}_{net}$ is non-zero

32. Answer (3)

Hint: $I_{xx'} = I_0 + MR^2$ & $M = \lambda L$

Sol.: $I_{xx'} = \frac{1}{2}MR^2 + MR^2 = \frac{3}{2}MR^2$

$$= \frac{3}{2}(\lambda L)\left(\frac{L}{2\pi}\right)^2 = \frac{3\lambda L^3}{8\pi^2}$$

$$= \frac{3\times3\times10^{-3}\times8}{8\pi^2} = 9\times10^{-4} \text{ kg-m}^2$$

33. Answer (3)

Hint: Torque $\vec{\tau}_{net} = \frac{d\vec{L}}{dt}$

$$K_{rotational} = \frac{L^2}{2I}$$

Sol.: $\tau = 0$

$\frac{dL}{dt} = 0 \Rightarrow L = \text{constant}$

$L = I\omega$

If we decrease its radius, then I decreases

$\Rightarrow$ ω increases.

& $K_r = \frac{L^2}{2I}$

K_r increases as I decreases

34. Answer (4)

Hint & Sol.: Since the belt does not slide on the pulleys, magnitudes of velocity and acceleration of any point on the belt is same as that of any point on the periphery of either of the two pulleys.

$a_c = \alpha_A r_A = \alpha_B r_B$

$r_A = 0.25$ m, $r_B = 0.5$ m and $\alpha_A = 0.2$ rad/s^2

$a_c = 0.2\times0.25 = 0.05$ m/s^2

$$\alpha_B = \frac{a_c}{r_B} = \frac{0.05}{0.5} = 0.1 \text{ rad/s}^2$$

35. Answer (1)

Hint: Power output, $P = \tau\omega$

Sol.: $P = 200$ H.P

$= 200\times746 = 1.492\times10^5$ W

& $\omega = 6000$ rev/min $= 628$ rad/s

$$\tau = \frac{P}{\omega} = \frac{1.49\times10^5}{628} = 237.58 \text{ N-m}$$

36. Answer (3)

Hint: Coefficient of static friction, $\mu = \tan\theta$

Sol.: $\frac{1}{\sqrt{3}} = \tan\theta$

$$\theta = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$\theta = 30°$

37. Answer (2)

Hint: Retardation of block $a = \mu g$

Sol.: Use equation of motion,

$v^2 = u^2 + 2as$

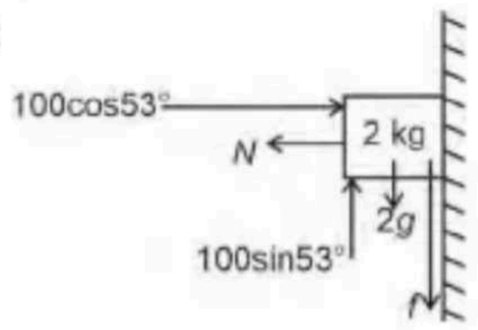
$0 = u^2 - 2\mu gs$

$$s = \frac{u^2}{2\mu g} = \frac{4\times4}{2\times3\times10} = \frac{16}{60} = \frac{4}{15}\text{m}$$

38. Answer (4)

Hint: Net force, $\vec{F}_{net} = m\vec{a}$

Sol.:



$$N = 100\cos53° = 100\times\frac{3}{5} = 60 \text{ N}$$

If there is no friction, then net vertical force on the block:

$R' = 100\sin53° - mg = 80 - 20 = 60$ N

$(f_s)_{max} = \mu_s N = 0.5\times60 = 30 \text{ N} < R'$

$\therefore$ Block moves and kinetic friction acts on it

$f_k = \mu_k N = 0.4\times60 = 24$ N

$$a = \frac{R' - f_k}{m} = \frac{60-24}{2} = \frac{36}{2} = 18 \text{ m/s}^2$$

39. Answer (4)

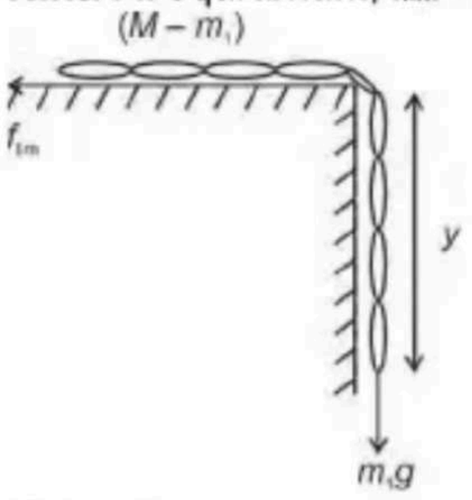
Hint: Force on block, $F = v\frac{dm}{dt}$

Sol.: $F = 5 \times 2 = 10$ N

$\Rightarrow \quad a = \frac{F}{m} = \frac{10}{3}$ m/s^2

40. Answer (2)

Hint: At equilibrium, $f_{lim} = m_1 g$



Sol.: $\mu N = m_1 g$

$\mu(M - m_1)g = m_1 g$

$y = \frac{\mu L}{(1+\mu)} = \frac{0.1 \times 4}{1.1} = 0.36$ m

41. Answer (1)

Hint: $\tan\theta = \frac{v^2}{rg}$

Sol.: As v increases, θ would also increase to maintain balance. If turn is sharp, then r is small and θ increases.

42. Answer (4)

Hint: Work done, $W = \int_{x_1}^{x_2} F \cdot dx$

Sol.: $W = \int_2^5 (6 + 2x)\, dx = (6x + x^2)\Big|_2^5$

$= [30 + 25] - [12 + 4] = 55 - 16 = 39$ J

43. Answer (4)

Hint: Work done by gravity, $W_g = -mg(h_2 - h_1)$

Work done on the ball by all the forces = ΔK.E.

Sol.: $W_g + W_{air} = \Delta$K.E.

$-mg(h_2 - h_1) + W_{air} = \frac{1}{2}m(v_2^2 - v_1^2)$

$-40 + W_{air} = -50$

$W_{air} = -10$ J

44. Answer (4)

Hint & Sol: A force is said to be conservative if work done by or against the force in moving a body from one position to another does not depend on path followed between these two positions.

Force of friction is a non-conservative force.

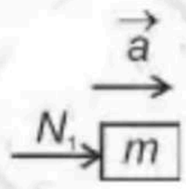
45. Answer (1)

Hint: Net force, $\vec{F}_{net} = M\vec{a}$

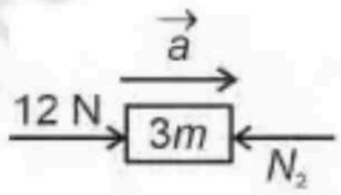
Sol.: Acceleration of the system of three masses = a

$= \frac{12}{6m} = \frac{2}{m}$ m/s^2

$N_1 = ma = 2$ N



$12 - N_2 = 3ma$



$N_2 = 6$ N

$\frac{N_1}{N_2} = \frac{1}{3}$

[CHEMISTRY]

46. Answer (4)

Hint: C_p and C_v are molar heat capacity at constant pressure and constant volume respectively. These are intensive properties.

Sol.: $C_p - C_v = R$; irrespective of the moles of an ideal gas.

47. Answer (1)

Hint: For irreversible isothermal process,

$W = -P_2(V_2 - V_1)$

Sol.: $W = -P_2(V_2 - V_1)$

$= -P_2\left(\frac{nRT}{P_2} - \frac{nRT}{P_1}\right)$

$= -nRTP_2\left(\frac{1}{P_2} - \frac{1}{P_1}\right)$

$= -2 \times 8.314 \times 298 \times 4\left(\frac{1}{4} - \frac{1}{1}\right) = 14.865$ kJ

48. Answer (4)

Hint: In expansion against vacuum,

$P_{ext} = 0$

Sol. $\therefore w = -P_{ext}(\Delta V) = 0$

49. Answer (2)

Hint: The functions which depend only on the path of the process are termed as path functions.

Sol.: Only heat and work are path functions.

50. Answer (1)

Hint: $\Delta G^\circ = -RT\ \ln K$

Sol.:

	$N_2O_4(g) \rightleftharpoons$	$2NO_2(g)$
t = 0	1	0
t = eq	(1 – 0.5)	1
	= 0.5	
x (Mole Fraction)	$\frac{1}{3}$	$\frac{2}{3}$
Partial pressure	1 atm	2 atm

$$K_p = \frac{(p_{NO_2})^2}{[p_{N_2O_4}]} = \frac{(2)^2}{1} = 4$$

$\Delta G^\circ = -RT\ \ln K$

$= -2.303\ RT\ \log 4$

$= -3.84$ kJ/mol

51. Answer (2)

Hint: ΔU does not discriminate between reversible and irreversible process, whereas ΔS_{total} does.

Sol.: Both for reversible and irreversible expansion of an ideal gas, under isothermal conditions,

$\Delta U = 0$

But $\Delta S_{total} = 0$, only for reversible process.

52. Answer (2)

Hint: $\Delta_r G^\ominus = \Delta_r H^\ominus - T\Delta_r S^\ominus$

$\Delta_r G^\ominus < 0$; reaction is spontaneous

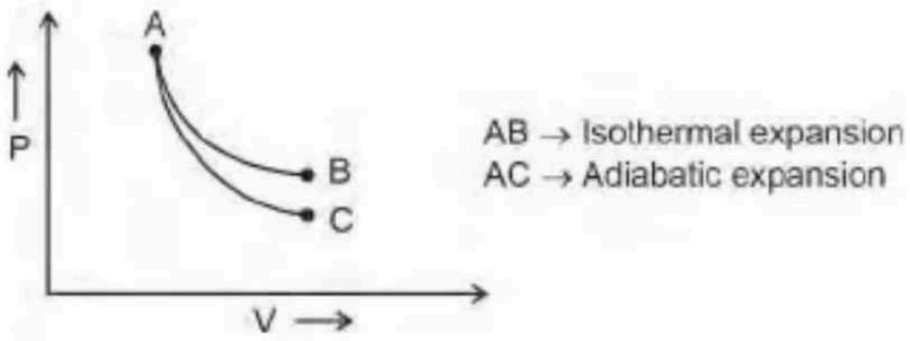
Sol.: $\Delta_r H^\ominus > 0$ and $\Delta_r S^\ominus < 0$

Hence $\Delta_r G^\ominus > 0$ at all temperature

53. Answer (3)

Hint: Cooling takes place upon adiabatic expansion of an ideal gas.

Sol.:



$\therefore\ T_A > T_C$

$T_A = T_B$

54. Answer (1)

Hint: Number of moles of gaseous product increases, entropy also increases.

Sol.: For $2NaHCO_3(s) \longrightarrow Na_2CO_3(s) + CO_2(g) + H_2O(g)$

$\Delta n_g = 2 - 0 = 2$

55. Answer (2)

Hint: Those properties which do not depend on the quantity or size of matter present are known as intensive properties

Sol.: Heat capacity is an extensive property while temperature is an intensive property.

Element	$\Delta_f H^\ominus$/(kJ mol^{-1})
S(s), rhombic	0
P(s), white	0

56. Answer (2)

Hint: For free expansion ($P_{ex} = 0$) w = 0

Sol.: For adiabatic process

q = 0.

$\Delta U = q + w$

$\Delta U = 0$

57. Answer (3)

Hint: $H_2(g) + Br_2(g) \longrightarrow 2\ HBr(g)$

$\Delta_r H = \Sigma B.E_{reactants} - \Sigma B.E_{products}$

Sol.: $-109 = B.E_{H_2} + B.E_{Br_2} - 2 \times B.E_{HBr}$

$-109 = x + 192 - 2(368)$

$-109 + 2(368) = x + 192$

$736 - 109 = x + 192$

$627 = x + 192$

$627 - 192 = x$

$435 = x$

58. Answer (1)

Hint: $\Delta S = nR\ln\frac{P_1}{P_2}$

Sol.: $\Delta S = 1 \times 8.314 \times 2.303\ \log 2$

$= 5.74\ J\ K^{-1}mol^{-1}$

59. Answer (4)

Hint: Due to better alignment of polymers, upon stretching of rubber entropy decreases.

Sol.: $C_6H_{12}O_6(s) + 6O_2(g) \rightarrow 6CO_2(g) + 6H_2O(l)$

$\Delta_c H^{\ominus} = 6 \times (-400) + 6 \times (-300) - (-1300)$

$= -2900\ kJ\ mol^{-1}$

Combustion of 180 gm of glucose, heat released is $= \frac{2900}{18} = 16.11\ kJ\ g^{-1}$

60. Answer (3)

Hint: $\gamma = \frac{C_p}{C_v}$

Sol.: For diatomic gas,

$C_p = \frac{7}{2}R$

$C_v = \frac{5}{2}R$

$\gamma = \frac{C_p}{C_v} = 1.40$

61. Answer (3)

Hint & Sol.:

$\left(\frac{1}{2}A \rightarrow B\right) \times 2$; $\Delta H = 150 \times 2$ ...(i)

$3B \rightarrow 2C + D$; $\Delta H = 125$ ...(ii)

Reverse of $E + A \longrightarrow 2D$ and multiplying by 2 we get

$4D \longrightarrow 2E + 2A$; $\Delta H = (-350) \times 2$... (iii)

add eq. (i), (ii) and (iii) we get

$3D + B \longrightarrow 2C + 2E + A$

$\therefore \Delta H = 300 + 125 - 350 \times 2$

$= 425 - 700 = -275\ kJ\ mol^{-1}$

62. Answer (4)

Hint & Sol.:

$2A(g) \rightleftharpoons B(g) + 2C(g)$ ...(i) ; K_1

$\frac{3}{2}A(g) \rightleftharpoons \frac{3}{4}B(g) + \frac{3}{2}C(g)$ ...(ii) ; K_2

Equation (ii) is $\frac{3}{4}$ times of equation (i)

$\therefore K_2 = (K_1)^{\frac{3}{4}}$

63. Answer (2)

Hint: $K_p = K_c(RT)^{\Delta n_g}$

Sol.: For $K_p = K_c$

$\because \Delta n_g = 0$

In $2C(s) + O_2(g) \rightleftharpoons 2CO(g)$

$\Delta n_g \neq 0$

$\therefore K_p \neq K_c$

64. Answer (4)

Hint: $A(s) \longrightarrow \underset{P_1}{B(g)} + \underset{P_1 + P_2}{C(g)}$

$D(s) \longrightarrow \underset{P_1 + P_2}{C(g)} + \underset{P_2}{E(g)}$

Sol.: $x = (P_1)(P_1 + P_2)$

$y = (P_2)(P_1 + P_2)$

$P_T = P_C + P_B + P_E$

$= P_1 + P_2 + P_1 + P_2 = 2(P_1 + P_2)$

$x + y = P_1^2 + P_1P_2 + P_2^2 + P_1P_2$

$x + y = (P_1 + P_2)^2$

$\therefore \boxed{2\sqrt{x+y} = P_T}$

65. Answer (1)

Hint: $2A \rightleftharpoons B + C$

$Q_c = \frac{[B][C]}{[A]^2} = \frac{(2)(2)}{\left(\frac{1}{2}\right)^2} = \frac{4}{\frac{1}{4}} = 4 \times 4 = 16$

Sol.: $\log k_c = \frac{-\Delta G^{\circ}}{2.303 \times RT} = \frac{-2500}{2.303 \times 8.314 \times 300}$

$= -\frac{2500}{5744} = -0.436$

$K_c = 0.36$

Since $Q_c > K_c$; Backward direction

66. Answer (1)

Hint & Sol.:

Volumes increases, number of moles decreases then

$\therefore$ Reaction shifts to a direction where number of gaseous moles increase

67. Answer (2)

Hint: Change in concentration of solid compounds do not affect the equilibrium

Sol.: No change in equilibrium is observed upon addition of an inert gas at constant volume because there is no change the partial pressure or the molar concentration of the substance involved in the reaction.

68. Answer (3)

Hint & Sol.: Volume gets halved and pressure gets doubled.

So to keep K_p constant, α decreases.

69. Answer (2)

Hint & Sol.: $H_2C_2O_4$ reacts with Fe^{3+} ions to form a stable complex $[Fe(C_2O_4)_3]^{3-}$ and $HgCl_2$ reacts with SCN^- ions to form $[Hg(SCN)_4]^{2-}$ and shifts reactions in backward direction.

70. Answer (2)

Hint: According to Bronsted-Lowry theory, acid is a substance that is capable of donating a hydrogen ion H^+ only.

Sol.: Conjugate base of $HN_3 = N_3^-$

71. Answer (4)

Hint: Bronsted bases are capable of accepting of proton

Sol.: Bronsted base are $C_2O_4^{2-}$ and PO_4^{3-}

72. Answer (4)

Hint: $\Delta G = \Delta H - T\Delta S$

Sol.: $\Delta H = \Delta U + \Delta n_g RT$

$$\Delta H = 1.1 + \frac{2\times 8.314\times 300}{1000}$$

$= 1.1 + 4.98 = 6.08$ kJ

$$\Delta G = 6.08 - \frac{300\times 11}{1000}$$

$= 6.08 - 3.3 = 2.78$ kJ

73. Answer (4)

Hint: Solution of weak acid with its salt of strong base is an acidic buffer

Sol.:

$$CH_3COOH + NaOH \rightleftharpoons CH_3COONa + H_2O$$

t_{eq} $1 - 0.5$ 0.5

$$pH = pK_a + \log\frac{[Salt]}{[Acid]}$$

$pH = pK_a$

74. Answer (1)

Hint:

	HCl	+ CH_3COONa	$\rightleftharpoons$ CH_3COOH	+ NaCl
t = 0	5 mmol	10 mmol	–	–
t_{eq}	–	5 mmol	5 mmol	5 mmol

$$pH = pK_a + \log\frac{[Salt]}{[Acid]}$$

pH = 4.76, acidic buffer

Sol.:

	NH_4OH	+ HCl	$\rightleftharpoons$ NH_4Cl	+ H_2O
t = 0	20 mmol	10 mmol	–	–
t_{eq}	10	–	10	10

$$pH = pK_b + \log\frac{[Salt]}{[Base]}$$

= 4.75, basic buffer

75. Answer (3)

Hint & Sol.: $K_a = \frac{[H^+][A^-]}{[HA]}$

$$10^{-5} = \frac{[H^+][3HA]}{[HA]};\ [H^+] = 3.3\times 10^{-6}$$

$pH = -\log(3.3\times 10^{-6}) = 5.48$

76. Answer (4)

Hint: Salt of strong base and weak acid is basic in nature.

Sol.: Calcium cyanide is a salt of strong base and weak acid.

77. Answer (1)

Hint : $C_m = \frac{q}{n\Delta T}$

Sol: $q = \frac{24\times 54}{27}\times 20 = 960$ J

78. Answer (4)

Hint & Sol: pH for the salt of weak acid and weak base :

$$pH = 7 + \frac{pK_a}{2} - \frac{pK_b}{2}$$

$$= 7 + \frac{4.77}{2} - \frac{3.27}{2} = 7.75$$

79. Answer (2)

Hint: $C_5H_5N + H_2O \rightleftharpoons C_5H_5\overset{+}{N}H + OH^-$

Sol.: $\alpha = \sqrt{\frac{K_b}{c}}$

$$= \sqrt{\frac{1.7}{2}\times 10^{-8}}$$

$\alpha = 0.92\times 10^{-4}$

$\%\alpha = 0.0092\%$

80. Answer (4)

Hint & Sol.: For a polyprotic acid, it is more difficult to remove a positively charged proton from a negative ion.

81. Answer (1)

Hint: $pOH = pK_b + \log \frac{[NH_4^+]}{[NH_3]}$

Sol.: pH + pOH = 14

pOH = 14 – 8.26 = 5.74

$5.74 = 4.74 + \log \frac{[NH_4^+]}{0.2}$

$[NH_4^+] = 2\ M$

Moles of NH_4Cl = 2 moles

Amount = 2 × 53.5 g mol^{-1} = 107 g

82. Answer (2)

Hint: Higher the solubility, lower will be the precipitation

Sol.: K_{sp} [AgI] = $[Ag^+]$ $[I^-]$

$[I^-] = 8.5 \times 10^{-16}$ M

K_{sp} $[Hg_2I_2] = [[Hg_2^{2+}][I^-]^2$

$[I^-] = 5 \times 10^{-13}$ M

$[I^-]$ for AgI is smaller than for Hg_2I_2

∴ At 5×10^{-13} M almost complete precipitation of AgI takes place

83. Answer (2)

Hint : An acid is a species which accepts electron pair.

Sol.: BF_3 is a Lewis acid but CH_4 is not a Lewis acid

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) + heat;$

The equilibrium shift in backward direction by increasing temperature.

84 Answer (1)

Hint & Sol.: NaCl won't undergo hydrolysis as both ions comes from strong acid and strong base. Hence, the solution will be neutral

85 Answer (4)

Hint & Sol.: Cations which react with water to give weak base undergo cationic hydrolysis

86. Answer (2)

Hint:

$2CH_3COOH + Ca(OH)_2 \rightleftharpoons (CH_3COO)_2Ca + 2H_2O$

$[CH_3COO^-] = 2 \times 0.05 = 0.1$

Sol.:

$pH = 7 + \frac{pK_a}{2} + \frac{\log C}{2}$

$= 7 + \frac{4.74}{2} + \frac{1}{2}\log(10^{-1})$

pH = 7 + 2.37 – 0.5 = 8.87

87. Answer (3)

Hint

Sol.: $K_h = \frac{K_w}{K_b}$

$10^{-14} = 5.5 \times 10^{-10} \times K_b$

$K_b = \frac{10^{-14}}{5.5 \times 10^{-10}}$

$= 0.18 \times 10^{-4} = 1.8 \times 10^{-5}$

88. Answer (3)

Hint: For HCN, $h = \sqrt{\frac{K_h}{C}}$

Sol.: $pH = 7 + \frac{1}{2}[pK_a + \log C]$

89. Answer (2)

Hint: $Ba(OH)_2 \rightleftharpoons Ba^{2+} + 2OH^-$

pH = 11

pOH = 3

$[OH^-] = 10^{-3}$

Sol.:

$Ba(OH)_2 \rightleftharpoons Ba^{2+} + 2OH^-$

$\frac{10^{-3}}{2}$ 10^{-3}

$K_{sp} = 0.5 \times 10^{-3} \times 10^{-6}$

$= 0.5 \times 10^{-9}$

$= 5 \times 10^{-10}$

90. Answer (1)

Hint & Sol.: $Ag_2CrO_4 \rightleftharpoons 2Ag^+ + CrO_4^{2-}$

Solubility higher, lower will be the precipitation.

Order of solubility is

AgI < AgBr < AgCl < Ag_2CrO_4

[BOTANY]

91. Answer (2)

Hint: Basidiomycetes, produce sexual spores exogenously

Sol.: *Agaricus* belongs to basidiomycetes.

92. Answer (3)

Hint: Two kingdom classification was given by C. Linnaeus.

Sol.: Five kingdom classification was given by R.H Whittaker.

93. Answer (3)

Hint: Methanogens are responsible for production of biogas.

Sol.: Archaebacteria differ from other bacteria in having a different cell wall structure. Methanogens are capable of converting CO_2, methanol and formic acid into methane.

94. Answer (3)

Hint: Mycoplasma is the smallest living cell.

Sol.: Mycoplasma do not have cell wall. Cell membranes are not made of peptidoglycan.

95. Answer (4)

Hint: Deuteromycetes are also known as imperfect fungi.

Sol.: *Agaricus* (Mushroom) belongs to basidiomycetes.

96. Answer (2)

Hint: Viruses — Mosaic formation in plants

Sol.: Bacteria — Citrus canker

Viroids — Potato spindle tuber disease

Prions — Mad cow disease in cattle

97. Answer (1)

Hint: Viruses are non-cellular organisms.

Sol.: Viruses did not find a place in classification since they are not considered truly 'living'.

98. Answer (3)

Hint: Cyanobacteria have chlorophyll a similar to higher plants.

Sol.: Euglenoids have pigments identical to higher plants. Except chlorophyll a, other pigments of cyanobacteria differ from higher plants.

99. Answer (2)

Hint: The given diagram is of *Paramecium*.

Sol.: *Paramecium* possess cilia instead of pseudopodia. Dinoflagellates possess cell wall having cellulose plates on the outer surface.

100. Answer (1)

Hint: Asexual spores are generally not found in the members of basidiomycetes.

Sol.: Asexual spores found in *Penicillium* is conidia. Zygospore is a sexual spore and it is formed by fusion of gametes.

101. Answer (1)

Hint: Yeast belongs to Ascomycetes.

Sol.: Yeast unicellular member of ascomycetes and do not form ascocarp. Dinoflagellates are mostly marine.

102. Answer (1)

Hint: In racemose type of inflorescence, the main axis continues to grow.

Sol.: Belladonna shows cymose inflorescence rest all of the examples showing racemose inflorescence.

103. Answer (4)

Hint: Members of Solanaceae have actinomorphic flowers.

Sol.: Ashwagandha being Solanaceae member has actinomorphic flowers.

104. Answer (2)

Hint: Root hairs are present in the zone of maturation

Sol.: Some cells from the region of maturation form very fine and delicate thread like structures called root hair.

105. Answer (3)

Hint: Sunflower and China rose show alternate phyllotaxy.

Sol.: Opposite phyllotaxy is seen in *Calotropis*.

106. Answer (3)

Hint: This type of aestivation is found in *Cassia*.

Sol.: In imbricate aestivation, margin of petals or sepals overlap each other but not in a particular direction.

107. Answer (1)

Hint:

- Monoadelphous stamens — China rose

Sol.:

- Diadelphous stamens – Pea
- Polyadelphous stamens – Citrus
- Polyandrous stamens – Mustard

108. Answer (3)

Hint: *Aspergillus* being a sac fungus has septate hyphae.

Sol.: *Mucor* – produce sexual spores by meiosis.

Colletotrichum – Lacks sexual reproduction and fruiting bodies, it asexually reproduce by conidia.

Puffballs – Lacks asexual reproduction

109. Answer (3)

Hint: Fumigatory plant, tobacco is included in Solanaceae family.

Sol.: Fruit type is capsule or berry. They have five sepals that are persistent.

110. Answer (3)

Hint: Phyllotaxy is the pattern of leaves on stem

Sol.: The arrangement of veins and veinlets in the lamina or leaf blade is termed as venation.

111. Answer (4)

Hint: Root provides anchorage to the plant parts.

Sol.: Stem bears and support leaves, flowers and fruits.

112. Answer (3)

Hint: Mustard is a member of Brassicaceae while *Solanum* is a member of Solanaceae family.

Sol.:

Features	**Mustard**	***Solanum***
Phyllotaxy	Alternate	Alternate
Floral symmetry	Actinomorphic	Actinomorphic
Type of flower based on position of gynoecium	Hypogynous	Hypogynous
Type of placentation	Parietal	Axile

In mustard stamens have different length of filaments.

113. Answer (4)

Hint: In basal placentation, the placenta develops at the base of the ovary

Sol.: Marigold have basal placentation.

114. Answer (3)

Hint: In mango and coconut, the fruit is known as a drupe

Sol.: Mango and coconut develop from monocarpellary superior ovaries.

115. Answer (2)

Hint: In epipetalous condition stamens are attached to petals.

Sol.: $\overset{\frown}{C\ A}$ represents epipetalous stamens.

116. Answer (2)

Hint: Dinoflagellates are mostly marine and photosynthetic

Sol.: The cell wall of dinoflagellates has stiff cellulose plates on the outer surface.

117. Answer (2)

Hint: Kingdom Plantae show autotrophic mode of nutrition.

Sol.: Multicellular organism with loose tissue organisation having heterotrophic mode of nutrition is fungi.

118. Answer (3)

Hint: *Agaricus* (mushroom) is an edible fungi

Sol.: *Ustilago* and *Puccinia* are smut and rust fungi respectively.

119. Answer (4)

Hint: Sexual reproduction occurs by ascospores in ascomycetes class of fungi.

Sol.: Asexual reproduction in fungi occurs by asexual spores called conidia or sporangiospores.

120. Answer (3)

Hint: Viruses are inert outside their specific host cell.

Sol.: W.M. Stanley showed that viruses could be crystallised and crystals consist largely of protein. Viruses are obligate parasites.

121. Answer (3)

Hint: Epiphyllous condition of stamens is found in members of Liliaceae.

Sol.: When stamens are attached to the petals, they are called epipetalous as in brinjal or epiphyllous when attached to the perianth as in flower of lily.

122. Answer (1)

Hint: Pea is a leguminous plant.

Sol.: Leaf base becomes swollen in Pea

Tricarpellary and syncarpous gynoecium – *(Asparagus)*

Presence of persistent sepal – Ashwagandha

Parietal placentation – *Argemone*

123. Answer (2)

Hint: The seed coat has two layers, the outer testa and inner tegmen

Sol.: A large shield shaped cotyledon found in maize seeds is known as scutellum. Above the hilum is a small pore called the micropyle.

124. Answer (1)

Hint: In non-endospermic seed, endosperm is not present in the mature seed

Sol.: Pea, Bean and Gram are examples of non-endospermic seeds.

125. Answer (2)

Hint: Marginal placentation is seen in pea

Sol.: Axile placentation is seen in China rose, tomato and lemon. *Primrose* has free central placentation.

126. Answer (4)

Hint: Spine is a modification of leaf.

Sol.: Spines in cactus are modified leaves.

127. Answer (4)

Hint: Mustard belongs to Brassicaceae family while China rose belongs to malvaceae family

Sol.:

	Feature	Mustard	China Rose
a.	Placentation	Parietal	Axile
b.	Floral symmetry	Actinomorphic	Actinomorphic
c.	Adhesion of stamen	Free stamens	Epipetalous
d.	Aestivation of corolla	Valvate	Twisted
e.	Union of carpels	Syncarpous	Syncarpous
f.	Type of fruit	Siliqua	Capsule
g.	Number of carpels	Two	Five

128. Answer (3)

Hint: Heterotrophic bacteria are helpful in making curd from milk and fixing nitrogen in legume

Sol.: Some of cyanobacteria can fix atmospheric nitrogen in specialised cells called heterocysts. They lack flagella.

129. Answer (3)

Hint: Six Kingdom classification was given by Carl Woese

Sol.: In six Kingdom classification, kingdom fungi and animalia have organisms with heterotrophic mode of nutrition only.

130. Answer (2)

Hint: T.O. Diener discovered a new infectious agent that was smaller than viruses known as viroid.

Sol.: Pseudopodia are locomotory and feeding structure in amoeboid protozoans.

131. Answer (2)

Hint: Hypogynous flowers have superior ovary.

Sol.: Plum, peach and rose – Half inferior ovary Guava, cucumber, ray florets of sunflower – Inferior ovary. Rest all have superior ovary.

132. Answer (2)

Hint: Bicarpellary and bilocular ovary is seen in the members of solanaceae family

Sol.: Epipetalous stamens and axile placentation are seen in tomato.

133. Answer (2)

Hint: Testa is the outer protective covering of seed

Sol.: Coleoptile is a sheath enclosing plumule in monocot seed. The outer covering of endosperm separates the embryo by a proteinaceous layer called aleurone layer.

134. Answer (3)

Hint: *Entamoeba* is an amoeboid protozoan.

Sol.: Amoeboid protozoans have pseudopodia for movement and capturing the prey. Marine forms of amoeboid protozoans have silica shells on their surface.

135. Answer (3)

Hint: A bud is present in the axil of petiole in both simple and compound leaf

Sol.: In pinnately compound leaves, a number of leaflets are present on rachis.

[ZOOLOGY]

136. Answer (4)

Hint: Secretion of sebaceous gland

Sol.: Sweat produced by the sweat glands is a watery fluid containing NaCl, small amounts of urea, lactic acid, *etc*. Though the primary function of sweat is to facilitate a cooling effect on body surface, it also helps to remove some of the wastes.

Sebaceous glands eliminate certain substances like sterols, hydrocarbons and waxes through sebum.

137. Answer (1)

Hint: Close proximity between Henle's loop and vasa recta

Sol.: NaCl transported by the ascending limb of Henle's loop is exchanged with the descending limb of vasa recta. NaCl is returned to the interstitium by the ascending portion of vasa recta. The proximity between Henle's loop and vasa recta as well as counter current in them help in maintaining an increasing osmolarity towards inner medullary interstitium *i.e.*, from 300 mOsmol L^{-1} to 1200 mOsmol L^{-1}. This gradient is mainly caused by NaCl and urea.

138. Answer (1)

Hint: The ascending limb is impermeable to water

Sol.: The ascending limb of loop of Henle is impermeable to water but allows transport of electrolytes actively or passively. Therefore, as the concentrated filtrate pass upward, it gets diluted due to the passage of electrolytes to the medullary fluid.

139. Answer (3)

Hint: Renin stimulates glomerular blood flow

Sol.: A fall in GFR can activate JG cells to release renin which converts angiotensinogen in blood to angiotensin I and further to angiotensin II. Angiotensin II, being a powerful vasoconstrictor, increases the glomerular blood pressure and thereby bring the GFR back to normal.

140. Answer (3)

Hint: Glomerular filtrate is proteinless plasma

Sol.: Blood is filtered so finely through filtration slits that almost all the constituents of the plasma except the proteins pass onto the lumen of the Bowman's capsule. Therefore, it is considered as a process of ultra filtration.
Globulins and fibrinogens are proteins.

141. Answer (1)
Hint: 70-80 per cent of electrolytes and water are reabsorbed by PCT.
Sol.: Glomerular filtrate in PCT is isotonic to blood plasma. Nearly all of the essential nutrients and 70-80 per cent of electrolytes and water are reabsorbed by this segment. Fluid present in descending limb of loop of Henle is hypertonic to the blood plasma.

142. Answer (2)
Hint: Concentrating segment.
Sol.: The ascending limb of loop of Henle is impermeable to water but allows transport of electrolytes actively or passively. Descending limb of loop of Henle is almost impermeable to electrolytes. Counter current mechanism operates in Henle's loop and vasa recta. DCT has hypotonic glomerular filtrate with respect to blood plasma. The epithelial cells of Bowman's capsule called podocytes are arranged in an intricate manner.

143. Answer (3)
Hint: Broad funnel shaped space
Sol.: The DCTs of many nephrons join to form a common collecting duct many of which converge and ultimately open into the renal pelvis through the medullary pyramids in the calyces. Inside the kidney, there are two zones, an outer cortex and an inner medulla. In humans, the excretory system consists of one pair of kidneys, a pair of ureters, a urinary bladder and a urethra. Renal columns are invagination of cortex into medulla.

144. Answer (2)
Hint: Part of nephron present in cortex of kidney
Sol.: Conditional reabsorption of Na^+ and water takes place in DCT. DCT is also capable of reabsorption of HCO_3^- and selective secretion of hydrogen and potassium ions and NH_3 to maintain the pH and sodium-potassium balance in blood.

145. Answer (1)
Hint: Osmolarity increases from renal cortex to renal medulla.
Sol.: The flow of filtrate in the two limbs of Henle's loop and vasa recta is in opposite direction and thus forms a counter current. The flow of blood through the two limbs of vasa recta is also in a counter current pattern. The proximity between the Henle's loop and vasa recta, as well as the counter current in them help in maintaining an increasing osmolarity towards the inner medullary interstitium.

146. Answer (2)
Hint: Counter current mechanism becomes impaired.
Sol.: Loop of Henle, along with vasa recta of juxtamedullary nephrons participates in counter current mechanism that leads to creation of concentration gradient in the medullary interstitium. So, more dilute urine will be produced upon removal of loop of Henle as less water is reabsorbed

147. Answer (3)
Hint: Exclude conditions associated with diabetes mellitus.
Sol.: Presence of glucose (glycosuria) and ketone bodies (ketonuria) in urine are indicative of diabetes mellitus. Malfunctioning of kidneys can lead to accumulation of urea in blood, a condition called uremia. An impairment affecting synthesis or release of ADH results in a diminished ability of the kidney to conserve water leading to water loss and dehydration (diuresis).

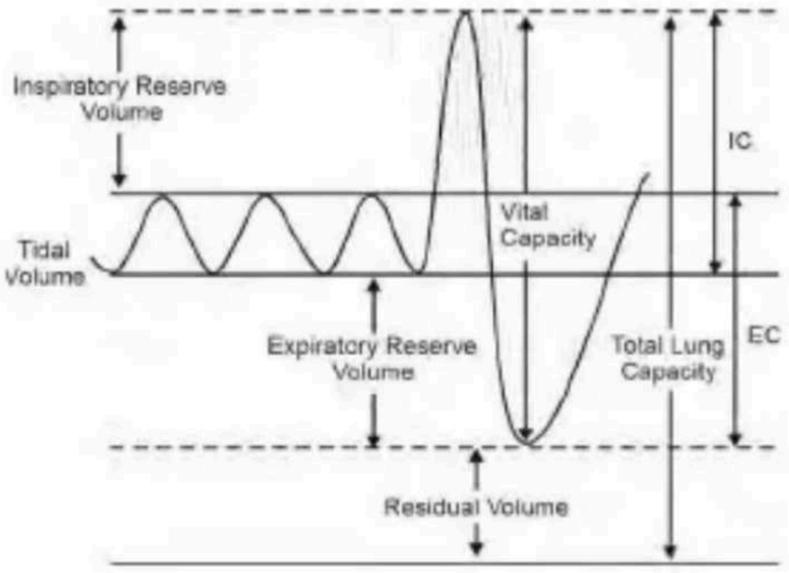
148. Answer (3)
Hint: Insects have Malpighian tubules.
Sol.: • Protonephridia or flame cells are excretory structures in platyhelminths. (Flatworms *e.g. Planaria*)
- Nephridia are tubular excretory structures of earthworms (*Pheretima*) and other annelids.
- Malpighian tubules are the excretory structures of most of the insects including cockroaches (*Periplaneta*).
- Antennal glands or green glands perform the excretory function in crustaceans like prawns (*Palaemon*).

149. Answer (2)
Hint: Blood vessel which carries blood away from heart.
Sol.: During the process of haemodialysis, the blood drained from a convenient artery is pumped into a dialysing unit called artificial kidney. Blood drained from convenient artery is pumped into a dialysing unit after adding an anticoagulant like heparin.

150. Answer (1)
Hint: Least toxic form of nitrogenous waste
Sol.: Land snails-Uricotelic.
Mammals, many terrestrial amphibians and marine fishes mainly excrete urea, hence are ureotelic animals.

151. Answer (1)
Hint: VC = ERV+TV+IRV

Sol.:EC = TV+ ERV
EC = FRC – RV+TV
TLC = RV+ERV+TV+IRV
VC = ERV+TV+IRV
IC = IRV+TV



152. Answer (1)

Hint: Intra-pulmonary pressure is less than atmospheric pressure.

Sol.: Inspiration can occur if the pressure within the lungs *i.e.* intra-pulmonary pressure is less than the atmospheric pressure *i.e,* there is a negative pressure in the lungs with respect to atmospheric pressure.

153. Answer (2)

Hint: Respiratory minute volume = Breathing rate × tidal volume.

Sol.: A healthy human breathes 12-16 times/min in resting conditions. Tidal volume is approximately 500 mL.

So, Respiratory minute volume (RMV) =

TV × Respiratory rate

= 500 × (12-16)

= 6000-8000 mL per minute

Inspiratory reserve volume = 2500-3000 mL

Expiratory reserve volume = 1000-1100 mL

Residual volume = 1100-1200 mL

154. Answer (3)

Hint: Dome shaped structure

Sol.: Normal expiration is a passive process as relaxation of muscle occurs. The thoracic chamber is formed dorsally by vertebral column, ventrally by sternum, laterally by ribs and on the lower side by dome-shaped diaphragm.

155. Answer (4)

Hint: Spirometer do not measure residual volume

Sol.: The volume of air involved in breathing movements can be estimated by using a spirometer which helps in clinical assessment of pulmonary function. FRC, TLC and RV cannot be measured with the help of a simple spirometer.

156. Answer (4)

Hint: Carbonic anhydrase

Sol.: Enzyme 'X' is carbonic anhydrase which belongs to class IV category of enzymes. Carbonic anhydrase accelerates the reaction rate by about 10 million times. Carbonic anhydrase facilitates reaction in forward direction in tissue due to high pCO_2, low pO_2 and more H^+. Reaction moves in backward direction at alveoli due to high pO_2, low pCO_2 and less H^+. Carbonic anhydrase is not present in WBCs.

157. Answer (1)

Hint: Incomplete cartilaginous rings are absent in exchange part.

Sol.: The trachea, primary, secondary and tertiary bronchi and initial bronchioles are supported by incomplete cartilaginous rings. These are absent in terminal bronchioles, alveoli and their ducts (alveolar ducts).

158. Answer (1)

Hint: Inspiration and expiration is included in breathing

Sol.: Breathing involves two stages: inspiration of O_2 rich atmospheric air and expiration of CO_2 rich alveolar air.

Cellular respiration is defined as oxidation of food to form CO_2, H_2O and energy.

159. Answer (1)

Hint: Venous blood is the deoxygenated blood.

Sol.: Venous blood has more CO_2 and more H^+ which leads to low pH as compared to arterial blood. Pulmonary vein carries oxygenated blood.

160. Answer (1)

Hint: Nasal chamber opens into pharynx

Sol.: The nasal chamber opens into the pharynx, a portion of which is the common passage for food and air. The pharynx opens through the larynx region into the trachea. Trachea divides at the level of 5th thoracic vertebra into a right and left primary bronchi. Each bronchi undergoes repeated division to form the secondary and tertiary bronchi and initial bronchioles ending up in very thin terminal bronchioles. Each terminal bronchiole gives rise to a number of very thin irregular-walled and vascularised bag-like structures called alveoli.

161. Answer (3)

Hint: Medulla oblongata regulates respiration.

Sol.: Aortic arch and carotid artery are peripheral chemoreceptors for regulation of breathing. They can recognise change in CO_2 and H^+ concentration and send necessary signals to the rhythm centre (Medulla oblongata).

162. Answer (2)

Hint: Observed during forceful expiration.

Sol.: Inspiration is initiated by the contraction of diaphragm which increases the volume of thoracic chamber in the antero-posterior axis. The contraction of external inter-costal muscles causes an increase in the volume of the thoracic chamber. We have the ability to increase the strength of inspiration and expiration with the help of additional muscles in abdomen. Contraction of internal inter-costal muscles occur during forceful expiration.

163. Answer (1)

Hint: Collects deoxygenated blood from upper parts of the body

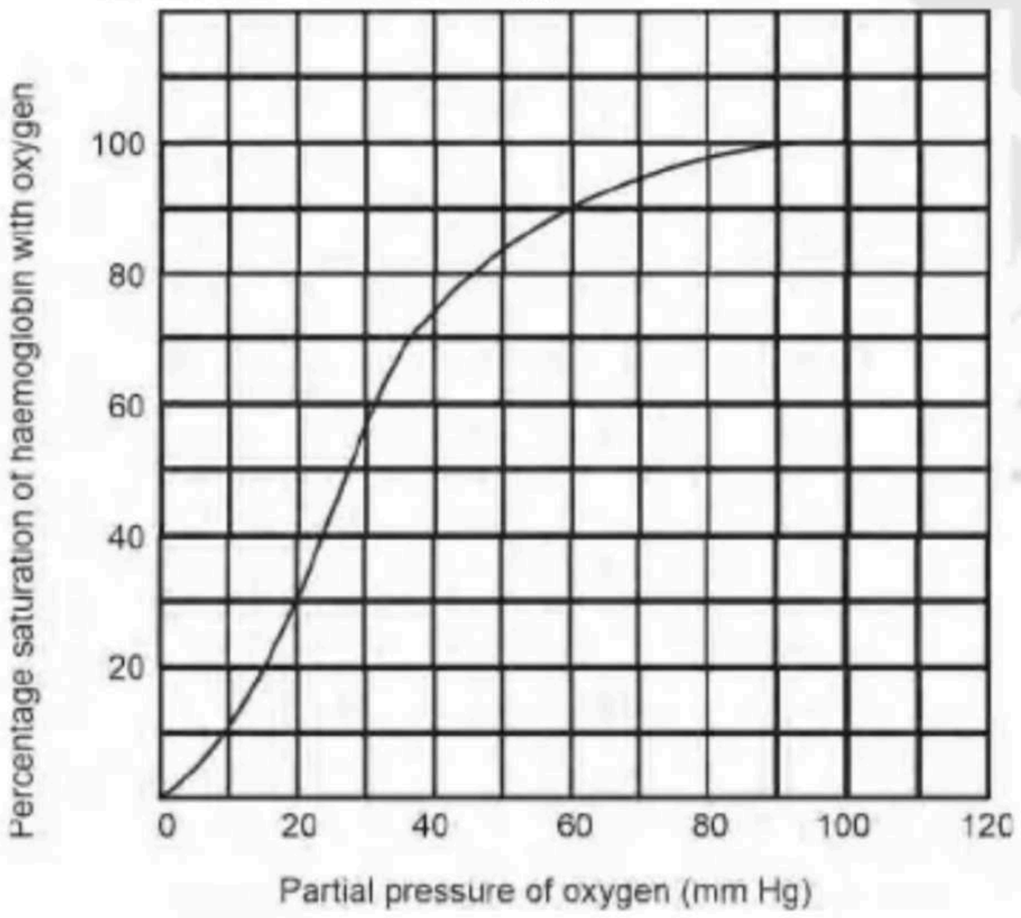
Sol.: Superior vena cava and inferior vena cava carry deoxygenated blood.

Respiratory Gas	Atmospheric Air	Alveoli	Blood (Deoxy-genated)	Blood (Oxygenated)	Tissues
O_2	159	104	40	95	40
CO_2	0.3	40	45	40	45

164. Answer (1)

Hint: Numerical value of urea excreted out per day.

Sol.: The partial pressure of oxygen is approximately 25-30 mm Hg when haemoglobin is 50% saturated with oxygen.



165. Answer (1)

Hint: Higher pCO_2 dissociates oxyhaemoglobin

Sol.: A sigmoid curve is obtained when percentage saturation of haemoglobin with O_2 is plotted against the pO_2. This curve is called the oxygen dissociation curve. At the tissue site where low pO_2, high pCO_2, high H^+ concentration and higher temperature exist, the conditions are favourable for dissociation of oxygen from the oxyhaemoglobin.

166. Answer (4)

Hint: It is more than 10

Sol.: A healthy human breathes 12-16 times per minute.

- A healthy individual has 12-16 gms of haemoglobin in every 100mL of blood.
- Neutrophils are the most abundant cell (60-65 per cent) of the total WBCs.
- Monocytes are 6-8 per cent of the total WBCs.
- 90-92 per cent of plasma is water and proteins contribute 6-8 per cent of it.

167. Answer (4)

Hint: CO_2 is more soluble than O_2

Sol.: Exchange of O_2 and CO_2 between blood and tissues is mainly based on pressure/concentration gradient.

Solubility of gases as well as the thickness of membranes involved in diffusion are some of the important factors that affect rate of diffusion of gases.

168. Answer (4)

Hint: Leucocytes have short life span

Sol.: In healthy human, heart rate is 72 times per 60 sec. If we reverse the ratio, we will get duration of cardiac cycle. Eosinophils (2-3 per cent) resist infections and they have bilobed nucleus. Leucocytes are also known as white blood cells (WBCs).

Leucocytes are generally short lived.

169. Answer (1)

Hint: Cardiac output = heart rate × stroke volume

Sol.: Stroke volume = end diastolic volume – end systolic volume

= 150 mL – 50 mL

Stroke volume = 100 mL

Cardiac output = heart rate × stroke volume

$$\text{Heart rate} = \frac{\text{Cardiac output}}{\text{Stroke volume}} = \frac{7200\ \text{mL}}{100\ \text{mL}}$$

Heart rate = 72 beats per minute

170. Answer (1)

Hint: Pulmonary vein has oxygenated blood

Sol.: Tunica media is thicker in arteries as compared to veins, so their lumen is narrower. In arteries, blood flows always away from heart while in vein it flows towards the heart.

171. Answer (4)

Hint: Secreted from mast cells.

Sol.: Certain factors released by the tissue, at the site of injury can initiate coagulation. Ca^{+2}, fibrinogen and prothrombin are involved in the cascade that leads to blood clotting. Serotonin is a vasoconstrictor.

172. Answer (2)

Hint: Type of connective tissue

Sol.: The atrium and the ventricle of the same side are separated by a thick fibrous tissue called the atrio-ventricular septum. A thin, muscular wall called the inter-atrial septum separates the right and left atria.

173. Answer (1)

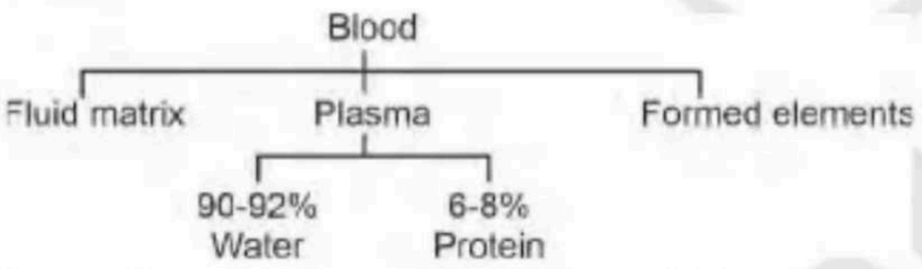
Hint: Universal recipient

Sol.: Persons with AB^{+ve} blood group do not have anti-A and anti-B antibodies in their plasma. They have A, B and Rh antigens on the RBC surface. Rh antigen will be absent on the RBCs of a AB^{-ve} individual.

174. Answer (1)

Hint: Fights pathogen

Sol.: Plasma without clotting factors is called serum.



Thus, formed elements do not constitute plasma.

175. Answer (4)

Hint: Smooth muscle fibres taper at the both ends

Sol.: Human heart has four chambers, two relatively small upper chambers called atria and two larger lower chambers called ventricles. A specialised cardiac musculature called the nodal tissue is also distributed in the heart.

Heart has cardiac muscle fibres.

Organ that stores urine is urinary bladder and it has smooth muscle fibres.

Cardiac muscle fibres	**Smooth muscle fibres**
• Branched	• Unbranched
• Cylindrical	• Fusiform
• Striated	• Non-striated

176. Answer (2)

Hint: First heart sound is produced at the beginning of ventricular systole.

Sol.: Lub is first heart sound which is associated with closure of atrio-ventricular valves. Dub is second heart sound which is associated with closure of semilunar valves. Increased blood flow into ventricles by about 30 per cent occurs during the atrial systole.

177. Answer (4)

Hint: Left ventricle pumps blood into aorta

Sol.: Blood pressure is high in arteries as compared to vein. A special coronary system of blood vessels is present in our body exclusively for the circulation of blood to and from the cardiac musculature.

178. Answer (3)

Hint: Angina is acute chest pain.

Sol.:

- Coronary artery disease, often referred to as atherosclerosis, affect the vessels that supply blood to the heart muscle.
- Angina is also called angina pectoris. A symptom of acute chest pain appears when not enough oxygen is reaching the heart muscle.
- Heart failure is sometimes called congestive heart failure because congestion of lungs is one of the main symptoms of this disease.
- In cardiac arrest, heart stops beating.

179. Answer (1)

Hint: Cardiac output = heart rate × stroke volume

Sol.: Athletes due to continuous training, have greater stroke volume than that of ordinary individuals. Human can alter the stroke volume and cardiac output. Athletes have higher stroke volume to maintain normal cardiac output. If repeated checks of blood pressure of an individual is 140/90 mm Hg or higher, it shows hypertension. Atherosclerosis affects the vessels that supply blood to the heart.

180. Answer (3)

Hint: Each artery and vein consists of three layers

Sol.: Basically, each artery and vein consists of three layers. Inner lining, the tunica intima, a middle layer tunica media and external layer tunica externa. The tunica media is comparatively thin in the veins. Pulmonary veins carries oxygenated blood while pulmonary artery carries deoxygenated blood. Partial pressure of O_2 in the pulmonary vein is 95 mm Hg.

❑ ❑ ❑

All India Aakash Test Series for NEET - 2026

TEST - 5 (Code-B)

Click here for Code-A Sol.

Test Date : 28/12/2025

ANSWERS

1.	(1)	37.	(3)	73.	(2)	109.	(2)	145.	(4)
2.	(4)	38.	(2)	74.	(2)	110.	(2)	146.	(1)
3.	(4)	39.	(3)	75.	(1)	111.	(2)	147.	(1)
4.	(4)	40.	(2)	76.	(4)	112.	(3)	148.	(4)
5.	(1)	41.	(4)	77.	(2)	113.	(4)	149.	(4)
6.	(2)	42.	(1)	78.	(4)	114.	(3)	150.	(4)
7.	(4)	43.	(3)	79.	(1)	115.	(4)	151.	(1)
8.	(4)	44.	(4)	80.	(4)	116.	(3)	152.	(1)
9.	(2)	45.	(2)	81.	(3)	117.	(3)	153.	(1)
10.	(3)	46.	(4)	82.	(1)	118.	(3)	154.	(2)
11.	(1)	47.	(1)	83.	(4)	119.	(1)	155.	(3)
12.	(4)	48.	(4)	84.	(4)	120.	(3)	156.	(1)
13.	(3)	49.	(2)	85.	(4)	121.	(3)	157.	(1)
14.	(3)	50.	(1)	86.	(2)	122.	(2)	158.	(1)
15.	(2)	51.	(2)	87.	(2)	123.	(4)	159.	(1)
16.	(3)	52.	(2)	88.	(3)	124.	(1)	160.	(4)
17.	(4)	53.	(3)	89.	(2)	125.	(1)	161.	(4)
18.	(3)	54.	(1)	90.	(1)	126.	(1)	162.	(3)
19.	(1)	55.	(2)	91.	(3)	127.	(2)	163.	(2)
20.	(3)	56.	(2)	92.	(3)	128.	(3)	164.	(1)
21.	(3)	57.	(3)	93.	(2)	129.	(1)	165.	(1)
22.	(4)	58.	(1)	94.	(2)	130.	(2)	166.	(1)
23.	(2)	59.	(4)	95.	(2)	131.	(4)	167.	(2)
24.	(1)	60.	(3)	96.	(2)	132.	(3)	168.	(3)
25.	(3)	61.	(3)	97.	(3)	133.	(3)	169.	(3)
26.	(4)	62.	(4)	98.	(3)	134.	(3)	170.	(2)
27.	(2)	63.	(2)	99.	(4)	135.	(2)	171.	(1)
28.	(3)	64.	(4)	100.	(4)	136.	(3)	172.	(2)
29.	(1)	65.	(1)	101.	(2)	137.	(1)	173.	(3)
30.	(1)	66.	(1)	102.	(1)	138.	(3)	174.	(2)
31.	(1)	67.	(2)	103.	(2)	139.	(4)	175.	(1)
32.	(2)	68.	(3)	104.	(1)	140.	(2)	176.	(3)
33.	(2)	69.	(3)	105.	(3)	141.	(4)	177.	(3)
34.	(2)	70.	(2)	106.	(3)	142.	(1)	178.	(1)
35.	(1)	71.	(4)	107.	(4)	143.	(1)	179.	(1)
36.	(4)	72.	(1)	108.	(3)	144.	(2)	180.	(4)

HINTS & SOLUTIONS

[PHYSICS]

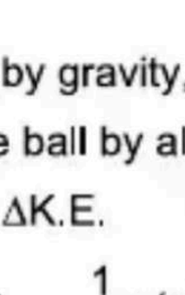
1. Answer (1)

Hint: Net force, $\vec{F}_{net} = M\bar{a}$

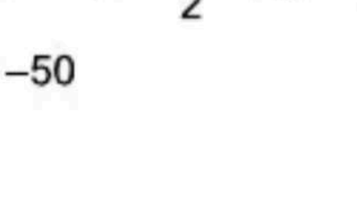
Sol.: Acceleration of the system of three masses = a

$= \frac{12}{6m} = \frac{2}{m}$ m/s^2

$N_1 = ma = 2$ N



$12 - N_2 = 3ma$



$N_2 = 6$ N

$\frac{N_1}{N_2} = \frac{1}{3}$

2. Answer (4)

Hint & Sol: A force is said to be conservative if work done by or against the force in moving a body from one position to another does not depend on path followed between these two positions.

Force of friction is a non-conservative force.

3. Answer (4)

Hint: Work done by gravity, $W_g = -mg(h_2 - h_1)$

Work done on the ball by all the forces = ΔK.E.

Sol.: $W_g + W_{air} = \Delta$K.E.

$-mg(h_2 - h_1) + W_{air} = \frac{1}{2}m(v_2^2 - v_1^2)$

$-40 + W_{air} = -50$

$W_{air} = -10$ J

4. Answer (4)

Hint: Work done, $W = \int_{x_1}^{x_2} F \cdot dx$

Sol.: $W = \int_2^5 (6 + 2x)\, dx = (6x + x^2)\Big|_2^5$

$= [30 + 25] - [12 + 4] = 55 - 16 = 39$ J

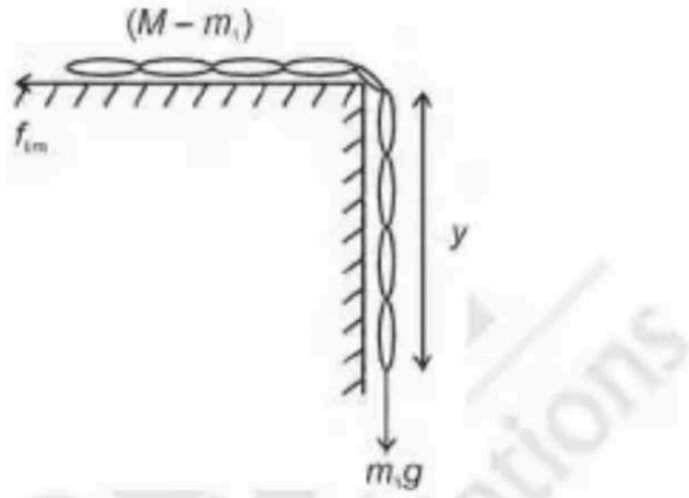
5. Answer (1)

Hint: $\tan\theta = \frac{v^2}{rg}$

Sol.: As v increases, θ would also increase to maintain balance. If turn is sharp, then r is small and θ increases

6. Answer (2)

Hint: At equilibrium, $f_{lim} = m_1 g$



Sol.: $\mu N = m_1 g$

$\mu(M - m_1)g = m_1 g$

$y = \frac{\mu L}{(1+\mu)} = \frac{0.1 \times 4}{1.1} = 0.36$ m

7. Answer (4)

Hint: Force on block, $F = v\frac{dm}{dt}$

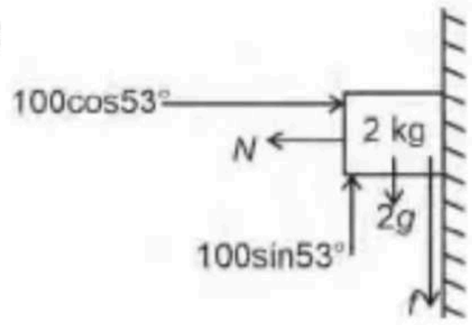
Sol.: $F = 5 \times 2 = 10$ N

$\Rightarrow \quad a = \frac{F}{m} = \frac{10}{3}$ m/s^2

8. Answer (4)

Hint: Net force, $\vec{F}_{net} = m\vec{a}$

Sol.:



$N = 100\cos 53° = 100 \times \frac{3}{5} = 60$ N

If there is no friction, then net vertical force on the block:

$R' = 100 \sin 53° - mg = 80 - 20 = 60$ N

$(f_s)_{max} = \mu_s N = 0.5 \times 60 = 30$ N $< R'$

$\therefore$ Block moves and kinetic friction acts on it

$f_k = \mu_k N = 0.4 \times 60 = 24$ N

$$a = \frac{R' - f_k}{m} = \frac{60 - 24}{2} = \frac{36}{2} = 18 \text{ m/s}^2$$

9. Answer (2)

Hint: Retardation of block $a = \mu g$

Sol.: Use equation of motion,

$v^2 = u^2 + 2as$

$0 = u^2 - 2\mu gs$

$$s = \frac{u^2}{2\mu g} = \frac{4 \times 4}{2 \times 3 \times 10} = \frac{16}{60} = \frac{4}{15} \text{m}$$

10. Answer (3)

Hint: Coefficient of static friction,

$\mu = \tan\theta$

Sol.: $\frac{1}{\sqrt{3}} = \tan\theta$

$$\theta = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$\theta = 30°$

11. Answer (1)

Hint: Power output, $P = \tau\omega$

Sol.: $P = 200$ H.P

$= 200 \times 746 = 1.492 \times 10^5$ W

& $\omega = 6000$ rev/min $= 628$ rad/s

$$\tau = \frac{P}{\omega} = \frac{1.49 \times 10^5}{628} = 237.58 \text{ N-m}$$

12. Answer (4)

Hint & Sol.: Since the belt does not slide on the pulleys, magnitudes of velocity and acceleration of any point on the belt is same as that of any point on the periphery of either of the two pulleys

$a_c = \alpha_A r_A = \alpha_B r_B$

$r_A = 0.25$ m, $r_B = 0.5$ m and $\alpha_A = 0.2$ rad/s^2

$a_c = 0.2 \times 0.25 = 0.05$ m/s^2

$$\alpha_B = \frac{a_c}{r_B} = \frac{0.05}{0.5} = 0.1 \text{ rad/s}^2$$

13. Answer (3)

Hint: Torque $\vec{\tau}_{net} = \frac{d\vec{L}}{dt}$

$K_{rotational} = \frac{L^2}{2I}$

Sol.: $\tau = 0$

$\frac{dL}{dt} = 0 \Rightarrow L =$ constant

$L = I\omega$

If we decrease its radius, then I decreases

$\Rightarrow$ ω increases.

& $K_r = \frac{L^2}{2I}$

K_r increases as I decreases

14. Answer (3)

Hint: $I_{xx'} = I_0 + MR^2$ & $M = \lambda L$

Sol.: $I_{xx'} = \frac{1}{2}MR^2 + MR^2 = \frac{3}{2}MR^2$

$$= \frac{3}{2}(\lambda L)\left(\frac{L}{2\pi}\right)^2 = \frac{3\lambda L^3}{8\pi^2}$$

$$= \frac{3 \times 3 \times 10^{-3} \times 8}{8\pi^2} = 9 \times 10^{-4} \text{ kg-m}^2$$

15. Answer (2)

Hint: $\vec{F}_{net} = m\vec{a}$

$\vec{\tau} = \vec{r} \times \vec{F}$

Sol.: $F_{net} = -20 + 30 - 10 + 40 - 40 =$ Zero

$\therefore$ No translatory motion.

Taking torque of these forces about centre of mass of the rod,

$|\vec{\tau}| = 30 \times 10 + 10 \times 10 + 40 \times 50 - (20 \times 30) - (40 \times 30)$

$= 2400 - 1800 = 600$

$\therefore$ $\vec{\tau}_{net}$ is non-zero

16. Answer (3)

Hint: Angular impulse = change in angular momentum

Sol.: $\tau t = I\omega - 0$

$$\omega = \frac{\tau t}{I} = \frac{FRt}{I} = \frac{40 \times 1 \times 5}{4} = 50 \text{ rad/s}$$

17. Answer (4)

Hint: Torque $\vec{\tau} = \vec{r} \times \vec{F}$

Sol.: Torque is an axial vector and its direction is perpendicular to the plane containing $\vec{r}$ & $\vec{F}$.

$\vec{\omega}_{A/B} = \vec{\omega}_{B/A} = \vec{\omega}_{body}$

Angular velocity is property of entire rigid body.

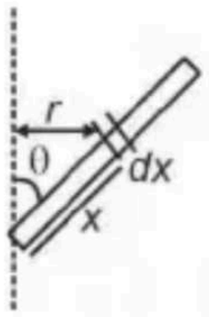
$$k_{\text{annular disc}} = \sqrt{\frac{1}{2}(R_1^2 + R_2^2)}$$

18. Answer (3)

Hint: Moment of inertia, $I = \int r^2\, dm$

Sol.: $r = x \sin\theta$

$$I = \int r^2\, dm = \int_0^L x^2 \sin^2\theta \left(\frac{M}{L}\right) dx$$



$$= \frac{ML^2 \sin^2\theta}{3} = \frac{5 \times 25 \times 3}{4 \times 3} = \frac{125}{4}\ \text{kg m}^2$$

19. Answer (1)

Hint: Use conservation of angular momentum

Sol.: By conservation of angular momentum

$I_1\omega_1 + I_2\omega_2 = (I_1 + I_2)\omega$

$$\omega = \frac{2 \times 4 + 4 \times 2}{6} = \frac{16}{6} = \frac{8}{3}\ \text{rad/s}$$

Rotational kinetic energy = $\frac{1}{2}(I_1 + I_2)\omega^2$

$$= \frac{1}{2} \times 6 \times \frac{64}{9} = \frac{64}{3}\ \text{J}$$

20. Answer (3)

Hint : Power due to a force in rotational motion is given by dot product of torque and angular velocity.

Sol.: $\vec{\tau} = \vec{r} \times \vec{F}$

$P = \vec{\tau} \cdot \vec{\omega}$

$P = (\vec{r} \times \vec{F}) \cdot \vec{\omega}$

21. Answer (3)

Hint: Velocity of C.O.M. $\vec{v}_{cm} = \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1 + m_2}$

Sol.: $\vec{v}_{cm} = \frac{0.2 \times 5\hat{i} + 0.5(10\hat{i} + 10\hat{j})}{0.2 + 0.5}$

$$= \frac{\hat{i} + 5\hat{i} + 5\hat{j}}{0.7}$$

$$= \frac{6\hat{i} + 5\hat{j}}{0.7} = \frac{60\hat{i} + 50\hat{j}}{7}\ \text{m/s}$$

22. Answer (4)

Hint : Use momentum and energy conservation laws to verify post-collision velocities in each option.

Sol.: For elastic collision in option (3), $m_1 >> m_2$

for $u_1 \neq 0$, $u_2 = 0$

then, $v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) u_1 \approx u_1$

$$v_2 = \frac{2m_1u_1}{(m_1 + m_2)} \approx 2u_1$$

For elastic collision in option (4), $m_1 = m_2$

for $u_1 \neq 0$, $u_2 = 0$,

then, $v_1 = 0$, $v_2 = u_1$

23. Answer (2)

Hint: Use conservation of momentum and change in total kinetic energy to find work done by internal forces.

Sol.: $\vec{P}_i$ = zero

$\vec{P}_f = m_1\vec{u} + m_2\vec{v}$

$\vec{P}_i = \vec{P}_f$

$0 = m_1\vec{u} + m_2\vec{v}$

$\vec{v} = \frac{-m_1\vec{u}}{m_2}$

$K.E_i = 0$

$K.E_f = \frac{1}{2}m_1u^2 + \frac{1}{2}m_2v^2$

$K.E_f = \frac{1}{2}m_1u^2 + \frac{1}{2}m_2\left(\frac{m_1u}{m_2}\right)^2$

$= \frac{1}{2}m_1u^2 + \frac{1}{2}\frac{m_1^2u^2}{m_2}$

$$K.E_f = \frac{1}{2}\left(\frac{m_1m_2 + m_1^2}{m_2}\right)u^2$$

$$\Delta K.E = \frac{1}{2}m_1u^2\left[1+\frac{m_1}{m_2}\right]$$

$$= \frac{1}{2}\times 1\times 4\left[1+\frac{1}{6}\right]$$

$$= 2\times\frac{7}{6} = \frac{7}{3}\text{ J}$$

24. Answer (1)

Hint: Work done $W = \vec{F}\cdot d\vec{s}$, $\hat{i}\cdot\hat{j} = 0$

Sol.: $W = \vec{F}\cdot\Delta\vec{x}$

$\Delta\vec{x} = (-3\hat{i})$ m

(a) $W = -15$ J

(b) $W =$ zero

(c) $W = (-5\hat{i})\cdot(-3\hat{j}) = 15$ J

(d) $W = (-8\hat{i}+5\hat{j})\cdot(-3\hat{i}) = 24$ J

25. Answer (3)

Hint: Work done $(W) = \Delta K\cdot E$

Sol.: $x^2 = 25t^3$

$x = 5t^{3/2}$

$$v = \frac{dx}{dt} = 5\times\frac{3}{2}t^{1/2}$$

At $t = 0$, $v = 0$

$\Rightarrow$ Particle is at rest

$$W = \frac{1}{2}mv_t^2 - \frac{1}{2}mv_{t=0}^2$$

$$W = \frac{1}{2}mv_t^2$$

$$= \frac{1}{2}\times 5\times\left(\frac{15}{2}t^{1/2}\right)^2 = \frac{5\times 225}{2\times 4}\times t$$

$\therefore\ W\propto t$

26. Answer (4)

Hint: For crossing the bridge in minimum time, it has to travel with maximum acceleration.

Sol.: a = maximum acceleration produced in car

$ma = \mu R$ (if acceleration is more than this, then tyres will slip)

$ma = \mu mg$

$a = \mu g = 10$ m/s^2

$u = 0$, $s = 500$ m, $a = 10$ m/s^2

$$500 = \frac{1}{2}\times 10\times t^2$$

$t = 10$ s

27. Answer (2)

Hint: To loop-the-loop body should not lose contact with the track anywhere, so velocity $v \geq \sqrt{gr}$ at the topmost point of circular loop.

Sol.: By conservation of mechanical energy,

$$\frac{1}{2}mv_{min}^2 = mg(h_{min} - 2r)$$

$$\frac{1}{2}m(gr) = mg(h_{min} - 2r)$$

$$\frac{1}{2}r = h_{min} - 2r$$

$$\therefore\ h_{min} = \frac{r}{2} + 2r = \frac{5r}{2} = 2.5\text{ m}$$

28. Answer (3)

Hint: Force $\vec{F} = \frac{d\vec{p}}{dt}$

Sol.: $F = \frac{2mu\cos\theta}{t}$

$$= \frac{2\times 500\times 10^{-3}\times 40\times 1}{0.4\times 2} = 50\text{ N}$$

29. Answer (1)

Hint: $F\times t$ = impulse

& Impulse = change in momentum

Sol.:

(i) Impulse $= \frac{1}{2}\times 20\times 20 = 200$ N-s

(ii) Impulse $= P_f - P_i \Rightarrow 200 = P_f - 10$

$P_f = 210$ kg-m/s

30. Answer (1)

Hint & Sol.: Under Newton's 3rd law, action and reaction are always equal and opposite. They always act on different bodies.

31. Answer (1)

Hint: Total acceleration is given by

$a_{total}^2 = a_t^2 + a_c^2$

Sol.: Since the tangential acceleration is constant:

$S = \frac{1}{4} \times 2\pi r = \frac{\pi r}{2}$

$S = ut + \frac{1}{2}at^2$

$\frac{\pi r}{2} = \frac{1}{2}at^2$

$t^2 = \frac{\pi r}{a}$

$v = a\sqrt{t^2} = a\sqrt{\frac{\pi r}{a}}$

$v = \sqrt{a\pi r}$

$a_c = \frac{v^2}{r} = \frac{a\pi r}{r} = a\pi$

$a_t = a$

$a_{total} = \sqrt{a_t^2 + a_c^2} = \sqrt{a^2 + a^2\pi}$

$= a\sqrt{1+\pi^2} = 2\sqrt{1+\pi^2}$

32. Answer (2)

Hint: Power $P = \vec{F}.\vec{v}$

Sol.: $P = (2\hat{i} - 3\hat{j} + 4\hat{k}) \cdot (2\hat{i} + 3\hat{j} + 5\hat{k})$

$= 4 - 9 + 20 = 15$ W

33. Answer (2)

Hint: In perfectly elastic head-on collision, kinetic energy transfer is maximum for $m_1 = m_2$

Sol: $v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)u_1$, $v_2 = \frac{2m_1}{(m_1 + m_2)}u_1$

Fraction of kinetic energy transferred

$= \frac{(K_B)_{final}}{(K_A)_{initial}} = \frac{4m_1 m_2}{(m_1 + m_2)^2}$

This expression is maximum when

$m_1 = m_2$

34. Answer (2)

Hint & Sol: Potential energy of a system can be positive, negative or zero.

- In circular motion, tension is perpendicular to velocity of block.

Small work done, $dW = Tdx\cos 90° = 0$

$\Rightarrow$ Work done is zero

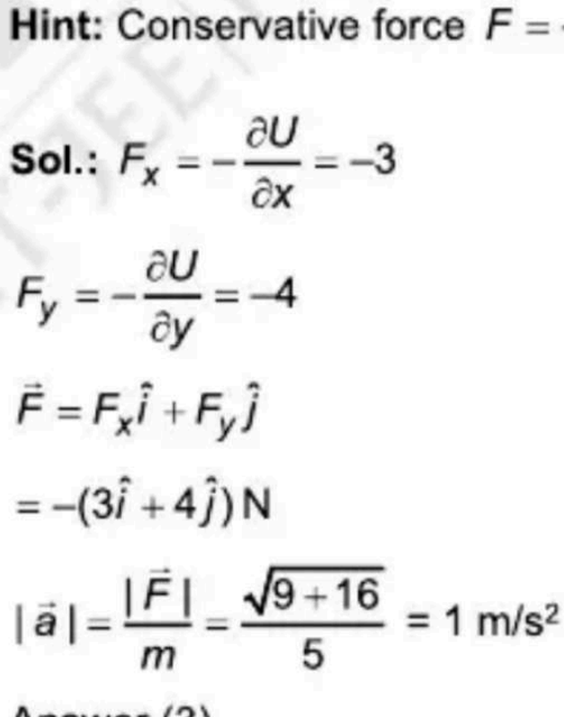
- Also, work done $= \int F.dx$

$\therefore$ Statement C is false

35. Answer (1)

Hint: Angular velocity $\omega = \frac{v_\perp}{r}$

Sol.: $\omega = \frac{v_\perp}{r} = \frac{v\sin\theta}{r}$



$\frac{a}{r} = \sin\theta$

$r = \frac{a}{\sin\theta}$

$\omega = \frac{v\sin^2\theta}{a} = 2 \times \frac{1}{4 \times 2} = \frac{1}{4}$ rad/s

36. Answer (4)

Hint: Conservative force $F = -\frac{\partial U}{\partial r}$

Sol.: $F_x = -\frac{\partial U}{\partial x} = -3$

$F_y = -\frac{\partial U}{\partial y} = -4$

$\vec{F} = F_x\hat{i} + F_y\hat{j}$

$= -(3\hat{i} + 4\hat{j})$ N

$|\vec{a}| = \frac{|\vec{F}|}{m} = \frac{\sqrt{9+16}}{5} = 1$ m/s^2

37. Answer (3)

Hint : Use $m_1r_1 = m_2r_2$

Sol.: C.O.M. will be in IV quadrant

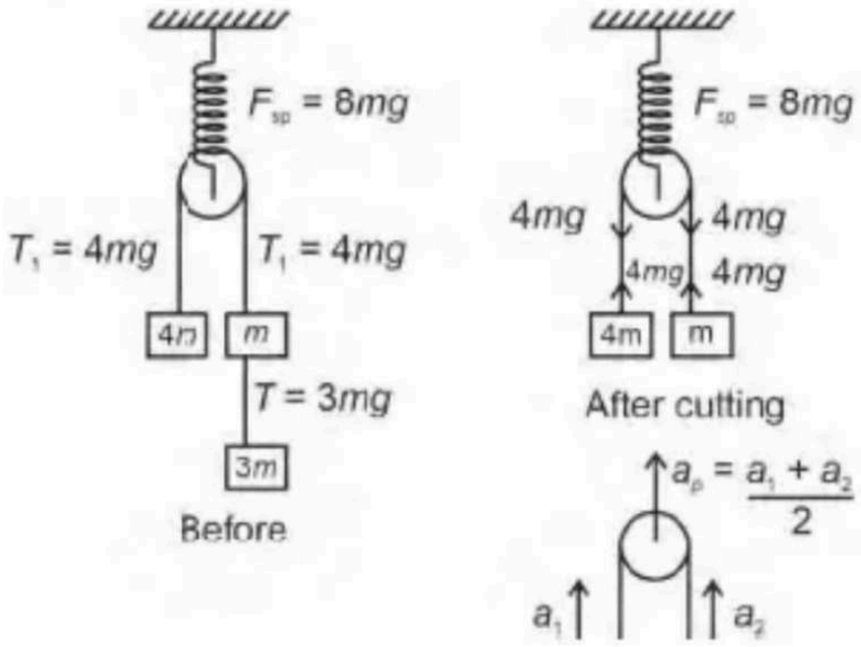
38. Answer (2)

Hint: Initially system was at rest

Just after string is cut:

$a_{pulley} = \frac{a_1 + a_2}{2}$

Sol.:



Spring force, $F_{sp} = 8mg$

$a_1 = a_{4m} = 0$

$$a_2 = a_m = \frac{4mg - mg}{m} = 3g$$

$$a_p = \frac{0 + 3g}{2} = \frac{3g}{2}$$

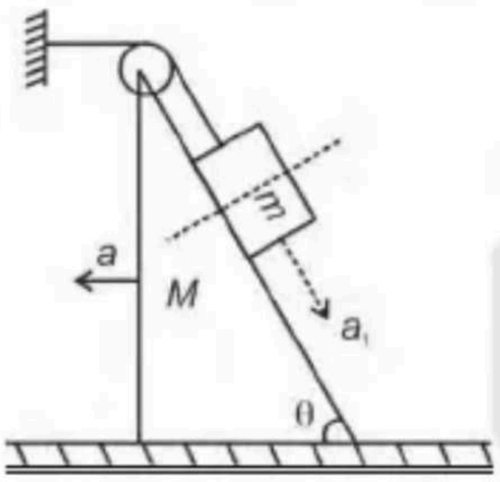
39. Answer (3)

Hint: Wedge constraint

(i) Along common normal, there is no relative motion.

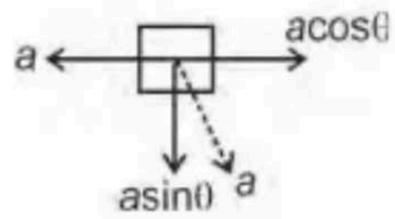
(ii) Perpendicular to common normal, motion is always relative.

Sol.:



$a_1 = a$ = acceleration of M w.r.t. ground = 2 m/s²

acceleration of m w.r.t. $M = a_{m/M} = a = 2$ m/s²



$$\vec{a}_{m/g} = \vec{a}_{m/M} + \vec{a}_M$$

$$\left|\vec{a}_{m/g}\right| = \sqrt{(a - a\cos\theta)^2 + (a\sin\theta)^2}$$

$$= 2a\sin\frac{\theta}{2}$$

$$= 2 \times 2 \times \frac{1}{2} = 2\ \text{m/s}^2$$

40. Answer (2)

Hint: $\alpha = \frac{\omega_f - \omega_i}{t}$ & $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$

Sol.: $f_0 = 420$ rev/min

$\omega_0 = 2\pi \times 7 = 14\pi$ rad/s

At $t = 10$ s, $\omega = 0$

$$\alpha = -\frac{14\pi}{10} = -\frac{7\pi}{5}\ \text{rad/s}^2$$

$$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$$

$$= 14\pi \times 10 - \frac{1}{2} \times \frac{7\pi}{5} \times 100$$

$\theta = 70\pi$ rad

Number of revolutions = $\frac{\theta}{2\pi}$

$$= \frac{70\pi}{2\pi} = 35 \text{ revolutions}$$

41. Answer (4)

Hint: $W_{cons} = -\Delta U$

Sol.: Potential energy is defined for conservative force field. Work done by system against conservative force results in increase in potential energy.

42. Answer (1)

Hint: $P = \vec{F}.\vec{v}, a = v\frac{dv}{dx}$

Sol.: $P = mav$ and $a = v\frac{dv}{dx}$

$$v^2 dv = \frac{P}{m}dx$$

$$\int_u^v v^2\, dv = \frac{P}{m}\int_0^x dx$$

$$P = \frac{m(v^3 - u^3)}{3x} = \frac{2500(216 - 64)}{3 \times 250}$$

$$= \frac{2500 \times 152}{3 \times 250} = 506.67\ \text{W} \approx 507\ \text{W}$$

43. Answer (3)

Hint: $R^2 = P^2 + Q^2 + 2PQ\cos\theta$

Sol.: $P = (F_1 + F_2)$, $Q = (F_1 - F_2)$

$$R = \sqrt{F_1^2 + F_2^2}$$

$$\cos\theta = \frac{R^2 - P^2 - Q^2}{2PQ}$$

$$= \frac{(F_1^2 + F_2^2) - (F_1 + F_2)^2 - (F_1 - F_2)^2}{2(F_1 + F_2)(F_1 - F_2)}$$

$$= -\left[\frac{(F_1^2 + F_2^2)}{2(F_1^2 - F_2^2)}\right] \Rightarrow \theta = \cos^{-1}\left[-\frac{(F_1^2 + F_2^2)}{2(F_1^2 - F_2^2)}\right]$$

44. Answer (4)

Hint : a_t is along tangent and a_r is along radius.

Sol.: When the particle is moving in a vertical circle with the help of a string, the force towards the centre and the force along the tangent, vary from point to point and so both its radial and tangential acceleration also vary.

45. Answer (2)

Hint: $\vec{A} \times \vec{B} = AB\sin\theta\,\hat{n}$

Sol.: $\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & -5 \\ 3 & -2 & 5 \end{vmatrix}$

$= \hat{i}(15 - 10) - \hat{j}(10 + 15) + \hat{k}(-4 - 9)$

$= 5\hat{i} - 25\hat{j} - 13\hat{k}$

$|\vec{A} \times \vec{B}| = \sqrt{819}$ units

[CHEMISTRY]

46. Answer (4)

Hint: C_p and C_v are molar heat capacity at constant pressure and constant volume respectively. These are intensive properties.

Sol.: $C_p - C_v = R$; irrespective of the moles of an ideal gas.

47. Answer (1)

Hint: For irreversible isothermal process,

$W = -P_2(V_2 - V_1)$

Sol.: $W = -P_2(V_2 - V_1)$

$$= -P_2\left(\frac{nRT}{P_2} - \frac{nRT}{P_1}\right)$$

$$= -nRTP_2\left(\frac{1}{P_2} - \frac{1}{P_1}\right)$$

$$= -2 \times 8.314 \times 298 \times 4\left(\frac{1}{4} - \frac{1}{1}\right) = 14.865 \text{ kJ}$$

48. Answer (4)

Hint: In expansion against vacuum,

$P_{ext} = 0$

Sol. $\therefore w = -P_{ext}(\Delta V) = 0$

49. Answer (2)

Hint: The functions which depend only on the path of the process are termed as path functions.

Sol.: Only heat and work are path functions.

50. Answer (1)

Hint: $\Delta G^\circ = -RT\ln K$

Sol.:

	$N_2O_4(g) \rightleftharpoons$	$2NO_2(g)$
t = 0	1	0
t = eq	(1 – 0.5) = 0.5	1
x (Mole Fraction)	$\frac{1}{3}$	$\frac{2}{3}$
Partial pressure	1 atm	2 atm

$$K_p = \frac{(p_{NO_2})^2}{[p_{N_2O_4}]} = \frac{(2)^2}{1} = 4$$

$\Delta G^\circ = -RT\ln K$

$= -2.303\,RT\log 4$

$= -3.84$ kJ/mol

51. Answer (2)

Hint: ΔU does not discriminate between reversible and irreversible process, whereas ΔS_{total} does.

Sol.: Both for reversible and irreversible expansion of an ideal gas, under isothermal conditions,

$\Delta U = 0$

But $\Delta S_{total} = 0$, only for reversible process.

52. Answer (2)

Hint: $\Delta_r G^\ominus = \Delta_r H^\ominus - T\Delta_r S^\ominus$

$\Delta_r G^\ominus < 0$; reaction is spontaneous

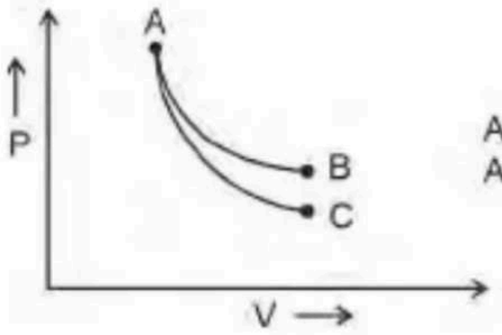
Sol.: $\Delta_r H^\ominus > 0$ and $\Delta_r S^\ominus < 0$

Hence $\Delta_r G^\ominus > 0$ at all temperature

53. Answer (3)

Hint: Cooling takes place upon adiabatic expansion of an ideal gas.

Sol.:



AB → Isothermal expansion
AC → Adiabatic expansion

$\therefore$ $T_A > T_C$

$T_A = T_B$

54. Answer (1)

Hint: Number of moles of gaseous product increases, entropy also increases.

Sol.: For $2NaHCO_3(s) \longrightarrow Na_2CO_3(s) + CO_2(g) + H_2O(g)$

$\Delta n_g = 2 - 0 = 2$

55. Answer (2)

Hint: Those properties which do not depend on the quantity or size of matter present are known as intensive properties

Sol.: Heat capacity is an extensive property while temperature is an intensive property.

Element	$\Delta_f H^\ominus$/(kJ mol^{-1})
S(s), rhombic	0
P(s), White	0

56. Answer (2)

Hint: For free expansion ($P_{ex} = 0$) w = 0

Sol.: For adiabatic process

q = 0.

$\Delta U = q + w$

$\Delta U = 0$

57. Answer (3)

Hint: $H_2(g) + Br_2(g) \longrightarrow 2HBr(g)$

$\Delta_r H = \Sigma B.E_{reactants} - \Sigma B.E_{products}$

Sol.: $-109 = B.E_{H_2} + B.E_{Br_2} - 2 \times B.E_{HBr}$

$-109 = x + 192 - 2(368)$

$-109 + 2(368) = x + 192$

$736 - 109 = x + 192$

$627 = x + 192$

$627 - 192 = x$

$435 = x$

58. Answer (1)

Hint: $\Delta S = nR\ln\frac{P_1}{P_2}$

Sol.: $\Delta S = 1 \times 8.314 \times 2.303 \log 2$

$= 5.74 \text{ J K}^{-1}\text{mol}^{-1}$

59. Answer (4)

Hint: Due to better alignment of polymers, upon stretching of rubber entropy decreases.

Sol.: $C_6H_{12}O_6(s) + 6O_2(g) \rightarrow 6CO_2(g) + 6H_2O(l)$

$\Delta_c H^\ominus = 6 \times (-400) + 6 \times (-300) - (-1300)$

$= -2900 \text{ kJ mol}^{-1}$

Combustion of 180 gm of glucose, heat released is $= \frac{2900}{18} = 16.11 \text{ kJ g}^{-1}$

60. Answer (3)

Hint: $\gamma = \frac{C_p}{C_v}$

Sol.: For diatomic gas,

$C_p = \frac{7}{2}R$

$C_v = \frac{5}{2}R$

$\gamma = \frac{C_p}{C_v} = 1.40$

61. Answer (3)

Hint & Sol.:

$\left(\frac{1}{2}A \rightarrow B\right) \times 2$; $\Delta H = 150 \times 2$ ...(i)

$3B \rightarrow 2C + D$; $\Delta H = 125$ ...(ii)

Reverse of $E + A \longrightarrow 2D$ and multiplying by 2 we get

$4D \longrightarrow 2E + 2A$; $\Delta H = (-350) \times 2$... (iii)

add eq. (i), (ii) and (iii) we get

$3D + B \longrightarrow 2C + 2E + A$

$\therefore \Delta H = 300 + 125 - 350 \times 2$

$= 425 - 700 = -275$ kJ mol^{-1}

62. Answer (4)

Hint & Sol.:

$2A(g) \rightleftharpoons B(g) + 2C(g)$ ...(i) ; K_1

$\frac{3}{2}A(g) \rightleftharpoons \frac{3}{4}B(g) + \frac{3}{2}C(g)$ ...(ii) ; K_2

Equation (ii) is $\frac{3}{4}$ times of equation (i)

$\therefore K_2 = (K_1)^{\frac{3}{4}}$

63. Answer (2)

Hint: $K_p = K_c(RT)^{\Delta n_g}$

Sol.: For $K_p = K_c$

$\because \Delta n_g = 0$

In $2C(s) + O_2(g) \rightleftharpoons 2CO(g)$

$\Delta n_g \neq 0$

$\therefore K_p \neq K_c$

64. Answer (4)

Hint: $A(s) \longrightarrow \underset{P_1}{B(g)} + \underset{P_1+P_2}{C(g)}$

$D(s) \longrightarrow \underset{P_1+P_2}{C(g)} + \underset{P_2}{E(g)}$

Sol.: $x = (P_1)(P_1 + P_2)$

$y = (P_2)(P_1 + P_2)$

$P_T = P_C + P_B + P_E$

$= P_1 + P_2 + P_1 + P_2 = 2(P_1 + P_2)$

$x + y = P_1^2 + P_1P_2 + P_2^2 + P_1P_2$

$x + y = (P_1 + P_2)^2$

$\therefore \boxed{2\sqrt{x+y} = P_T}$

65. Answer (1)

Hint: $2A \rightleftharpoons B + C$

$Q_c = \frac{[B][C]}{[A]^2} = \frac{(2)(2)}{\left(\frac{1}{2}\right)^2} = \frac{4}{\frac{1}{4}} = 4 \times 4 = 16$

Sol.: $\log k_c = \frac{-\Delta G^\circ}{2.303 \times RT} = \frac{-2500}{2.303 \times 8.314 \times 300}$

$= -\frac{2500}{5744} = -0.436$

$K_c = 0.36$

Since $Q_c > K_c$; Backward direction

66. Answer (1)

Hint & Sol.: $Ag_2CrO_4 \rightleftharpoons 2Ag^+ + CrO_4^{2-}$

Solubility higher, lower will be the precipitation.

Order of solubility is

$AgI < AgBr < AgCl < Ag_2CrO_4$

67. Answer (2)

Hint: $Ba(OH)_2 \rightleftharpoons Ba^{2+} + 2OH^-$

pH = 11

pOH = 3

$[OH^-] = 10^{-3}$

Sol.:

$Ba(OH)_2 \rightleftharpoons Ba^{2+} + 2OH^-$

$\frac{10^{-3}}{2}$ 10^{-3}

$K_{sp} = 0.5 \times 10^{-3} \times 10^{-6}$

$= 0.5 \times 10^{-9}$

$= 5 \times 10^{-10}$

68. Answer (3)

Hint: For HCN, $h = \sqrt{\frac{K_h}{C}}$

Sol.: $pH = 7 + \frac{1}{2}[pK_a + \log C]$

69. Answer (3)

Hint

Sol.: $K_h = \frac{K_w}{K_b}$

$10^{-14} = 5.5 \times 10^{-10} \times K_b$

$$K_b = \frac{10^{-14}}{5.5\times10^{-10}}$$

$= 0.18 \times 10^{-4} = 1.8 \times 10^{-5}$

70. Answer (2)

Hint:

$2CH_3COOH + Ca(OH)_2 \rightleftharpoons (CH_3COO)_2Ca + 2H_2O$

$[CH_3COO^-] = 2 \times 0.05 = 0.1$

Sol.:

$$pH = 7 + \frac{pK_a}{2} + \frac{\log C}{2}$$

$$= 7 + \frac{4.74}{2} + \frac{1}{2}\log(10^{-1})$$

pH = 7 + 2.37 – 0.5 = 8.87

71 Answer (4)

Hint & Sol.: Cations which react with water to give weak base undergo cationic hydrolysis

72 Answer (1)

Hint & Sol.: NaCl won't undergo hydrolysis as both ions comes from strong acid and strong base. Hence, the solution will be neutral

73. Answer (2)

Hint : An acid is a species which accepts electron pair.

Sol.: BF_3 is a Lewis acid but CH_4 is not a Lewis acid

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) + \text{heat};$

The equilibrium shift in backward direction by increasing temperature.

74. Answer (2)

Hint: Higher the solubility, lower will be the precipitation

Sol.: K_{sp} [AgI] = $[Ag^+]$ $[I^-]$

$[I^-] = 8.5 \times 10^{-16}$ M

K_{sp} $[Hg_2I_2] = [[Hg_2^{2+}][I^-]^2$

$[I^-] = 5 \times 10^{-13}$ M

$[I^-]$ for AgI is smaller than for Hg_2I_2

∴ At 5×10^{-13} M almost complete precipitation of AgI takes place

75. Answer (1)

Hint: $pOH = pK_b + \log \frac{[NH_4^+]}{[NH_3]}$

Sol.: pH + pOH = 14

pOH = 14 – 8.26 = 5.74

$$5.74 = 4.74 + \log \frac{[NH_4^+]}{0.2}$$

$[NH_4^+] = 2$ M

Moles of NH_4Cl = 2 moles

Amount = 2×53.5 g mol^{-1} = 107 g

76. Answer (4)

Hint & Sol.: For a polyprotic acid, it is more difficult to remove a positively charged proton from a negative ion.

77. Answer (2)

Hint: $C_5H_5N + H_2O \rightleftharpoons C_5H_5\overset{+}{N}H + OH^-$

Sol.: $\alpha = \sqrt{\frac{K_b}{c}}$

$$= \sqrt{\frac{1.7}{2}\times10^{-8}}$$

$\alpha = 0.92 \times 10^{-4}$

$\%\alpha = 0.0092\%$

78. Answer (4)

Hint & Sol: pH for the salt of weak acid and weak base :

$$pH = 7 + \frac{pK_a}{2} - \frac{pK_b}{2}$$

$$= 7 + \frac{4.77}{2} - \frac{3.27}{2} = 7.75$$

79. Answer (1)

Hint : $C_m = \frac{q}{n\Delta T}$

Sol: $q = \frac{24\times54}{27}\times20 = 960$ J

80. Answer (4)

Hint: Salt of strong base and weak acid is basic in nature.

Sol.: Calcium cyanide is a salt of strong base and weak acid.

81. Answer (3)

Hint & Sol.: $K_a = \frac{[H^+][A^-]}{[HA]}$

$10^{-5} = \frac{[H^+][3HA]}{[HA]}$; $[H^+] = 3.3 \times 10^{-6}$

$pH = -\log(3.3 \times 10^{-6}) = 5.48$

82. Answer (1)

Hint:

	HCl +	CH_3COONa $\rightleftharpoons$	CH_3COOH +	$NaCl$
t = 0	5 mmol	10 mmol	–	–
t_{eq}	–	5 mmol	5 mmol	5 mmol

$pH = pK_a + \log \frac{[Salt]}{[Acid]}$

pH = 4.76, acidic buffer

Sol.:

	NH_4OH +	HCl $\rightleftharpoons$	NH_4Cl +	H_2O
t = 0	20 mmol	10 mmol	–	–
t_{eq}	10	–	10	10

$pH = pK_b + \log \frac{[Salt]}{[Base]}$

= 4.75, basic buffer

83. Answer (4)

Hint: Solution of weak acid with its salt of strong base is an acidic buffer

Sol.:

$CH_3COOH + NaOH \rightleftharpoons CH_3COONa + H_2O$

t_{eq} $1 - 0.5$ 0.5

$pH = pK_a + \log \frac{[Salt]}{[Acid]}$

$pH = pK_a$

84. Answer (4)

Hint: $\Delta G = \Delta H - T\Delta S$

Sol.: $\Delta H = \Delta U + \Delta n_g RT$

$\Delta H = 1.1 + \frac{2 \times 8.314 \times 300}{1000}$

= 1.1 + 4.98 = 6.08 kJ

$\Delta G = 6.08 - \frac{300 \times 11}{1000}$

= 6.08 – 3.3 = 2.78 kJ

85. Answer (4)

Hint: Bronsted bases are capable of accepting of proton

Sol.: Bronsted base are $C_2O_4^{2-}$ and PO_4^{3-}

86. Answer (2)

Hint: According to Bronsted-Lowry theory, acid is a substance that is capable of donating a hydrogen ion H^+ only.

Sol.: Conjugate base of $HN_3 = N_3^-$

87. Answer (2)

Hint & Sol.: $H_2C_2O_4$ reacts with Fe^{3+} ions to form a stable complex $[Fe(C_2O_4)_3]^{3-}$ and $HgCl_2$ reacts with SCN^- ions to form $[Hg(SCN)_4]^{2-}$ and shifts reactions in backward direction.

88. Answer (3)

Hint & Sol.: Volume gets halved and pressure gets doubled.

So to keep K_p constant, α decreases.

89. Answer (2)

Hint: Change in concentration of solid compounds do not affect the equilibrium

Sol.: No change in equilibrium is observed upon addition of an inert gas at constant volume because there is no change the partial pressure or the molar concentration of the substance involved in the reaction.

90. Answer (1)

Hint & Sol.:

Volumes increases, number of moles decreases then

∴ Reaction shifts to a direction where number of gaseous moles increase

[BOTANY]

91. Answer (3)

Hint: A bud is present in the axil of petiole in both simple and compound leaf

Sol.: In pinnately compound leaves, a number of leaflets are present on rachis.

92. Answer (3)

Hint: *Entamoeba* is an amoeboid protozoan.

Sol.: Amoeboid protozoans have pseudopodia for movement and capturing the prey. Marine forms of amoeboid protozoans have silica shells on their surface.

93. Answer (2)

Hint: Testa is the outer protective covering of seed

Sol.: Coleoptile is a sheath enclosing plumule in monocot seed. The outer covering of endosperm separates the embryo by a proteinaceous layer called aleurone layer.

94. Answer (2)

Hint: Bicarpellary and bilocular ovary is seen in the members of solanaceae family

Sol.: Epipetalous stamens and axile placentation are seen in tomato.

95. Answer (2)

Hint: Hypogynous flowers have superior ovary.

Sol.: Plum, peach and rose – Half inferior ovary

Guava, cucumber, ray florets of sunflower – Inferior ovary. Rest all have superior ovary.

96. Answer (2)

Hint: T.O. Diener discovered a new infectious agent that was smaller than viruses known as viroid.

Sol.: Pseudopodia are locomotory and feeding structure in amoeboid protozoans.

97. Answer (3)

Hint: Six Kingdom classification was given by Carl Woese

Sol.: In six Kingdom classification, kingdom fungi and animalia have organisms with heterotrophic mode of nutrition only.

98. Answer (3)

Hint: Heterotrophic bacteria are helpful in making curd from milk and fixing nitrogen in legume

Sol.: Some of cyanobacteria can fix atmospheric nitrogen in specialised cells called heterocysts. They lack flagella.

99. Answer (4)

Hint: Mustard belongs to Brassicaceae family while China rose belongs to malvaceae family

Sol.:

	Feature	Mustard	China Rose
a.	Placentation	Parietal	Axile
b.	Floral symmetry	Actinomorphic	Actinomorphic
c.	Adhesion of stamen	Free stamens	Epipetalous
d.	Aestivation of corolla	Valvate	Twisted
e.	Union of carpels	Syncarpous	Syncarpous
f.	Type of fruit	Siliqua	Capsule
g.	Number of carpels	Two	Five

100. Answer (4)

Hint: Spine is a modification of leaf.

Sol.: Spines in cactus are modified leaves.

101. Answer (2)

Hint: Marginal placentation is seen in pea

Sol.: Axile placentation is seen in China rose, tomato and lemon. *Primrose* has free central placentation.

102. Answer (1)

Hint: In non-endospermic seed, endosperm is not present in the mature seed

Sol.: Pea, Bean and Gram are examples of non-endospermic seeds.

103. Answer (2)

Hint: The seed coat has two layers, the outer testa and inner tegmen

Sol.: A large shield shaped cotyledon found in maize seeds is known as scutellum. Above the hilum is a small pore called the micropyle.

104. Answer (1)

Hint: Pea is a leguminous plant.

Sol.: Leaf base becomes swollen in Pea

Tricarpellary and syncarpous gynoecium – *(Asparagus)*

Presence of persistent sepal – Ashwagandha

Parietal placentation – *Argemone*

105. Answer (3)

Hint: Epiphyllous condition of stamens is found in members of Liliaceae.

Sol.: When stamens are attached to the petals,

they are called epipetalous as in brinjal or epiphyllous when attached to the perianth as in flower of lily.

106. Answer (3)

Hint: Viruses are inert outside their specific host cell.

Sol.: W.M. Stanley showed that viruses could be crystallised and crystals consist largely of protein. Viruses are obligate parasites.

107. Answer (4)

Hint: Sexual reproduction occurs by ascospores in ascomycetes class of fungi.

Sol.: Asexual reproduction in fungi occurs by asexual spores called conidia or sporangiospores.

108. Answer (3)

Hint: *Agaricus* (mushroom) is an edible fungi

Sol.: *Ustilago* and *Puccinia* are smut and rust fungi respectively.

109. Answer (2)

Hint: Kingdom Plantae show autotrophic mode of nutrition.

Sol.: Multicellular organism with loose tissue organisation having heterotrophic mode of nutrition is fungi.

110. Answer (2)

Hint: Dinoflagellates are mostly marine and photosynthetic

Sol.: The cell wall of dinoflagellates has stiff cellulose plates on the outer surface.

111. Answer (2)

Hint: In epipetalous condition stamens are attached to petals.

Sol.: C $\overset{\frown}{A}$ represents epipetalous stamens.

112. Answer (3)

Hint: In mango and coconut, the fruit is known as a drupe

Sol.: Mango and coconut develop from monocarpellary superior ovaries.

113. Answer (4)

Hint: In basal placentation, the placenta develops at the base of the ovary

Sol.: Marigold have basal placentation.

114. Answer (3)

Hint: Mustard is a member of Brassicaceae while *Solanum* is a member of Solanaceae family.

Sol.:

Features	Mustard	*Solanum*
Phyllotaxy	Alternate	Alternate
Floral symmetry	Actinomorphic	Actinomorphic
Type of flower based on position of gynoecium	Hypogynous	Hypogynous
Type of placentation	Parietal	Axile

In mustard stamens have different length of filaments.

115. Answer (4)

Hint: Root provides anchorage to the plant parts.

Sol.: Stem bears and support leaves, flowers and fruits.

116. Answer (3)

Hint: Phyllotaxy is the pattern of leaves on stem

Sol.: The arrangement of veins and veinlets in the lamina or leaf blade is termed as venation.

117. Answer (3)

Hint: Fumigatory plant, tobacco is included in Solanaceae family.

Sol.: Fruit type is capsule or berry. They have five sepals that are persistent.

118. Answer (3)

Hint: *Aspergillus* being a sac fungus has septate hyphae.

Sol.: *Mucor* – produce sexual spores by meiosis.

Colletotrichum – Lacks sexual reproduction and fruiting bodies, it asexually reproduce by conidia.

Puffballs – Lacks asexual reproduction

119. Answer (1)

Hint:

- Monoadelphous stamens — China rose

Sol.:

- Diadelphous stamens – Pea
- Polyadelphous stamens – Citrus
- Polyandrous stamens – Mustard

120. Answer (3)

Hint: This type of aestivation is found in *Cassia*.

Sol.: In imbricate aestivation, margin of petals or sepals overlap each other but not in a particular direction.

121. Answer (3)

Hint: Sunflower and China rose show alternate phyllotaxy.

Sol.: Opposite phyllotaxy is seen in *Calotropis*.

122. Answer (2)

Hint: Root hairs are present in the zone of maturation

Sol.: Some cells from the region of maturation form very fine and delicate thread like structures called root hair.

123. Answer (4)

Hint: Members of Solanaceae have actinomorphic flowers.

Sol.: Ashwagandha being Solanaceae member has actinomorphic flowers.

124. Answer (1)

Hint: In racemose type of inflorescence, the main axis continues to grow.

Sol.: Belladonna shows cymose inflorescence rest all of the examples showing racemose inflorescence.

125. Answer (1)

Hint: Yeast belongs to Ascomycetes.

Sol.: Yeast unicellular member of ascomycetes and do not form ascocarp. Dinoflagellates are mostly marine.

126. Answer (1)

Hint: Asexual spores are generally not found in the members of basidiomycetes.

Sol.: Asexual spores found in *Penicillium* is conidia. Zygospore is a sexual spore and it is formed by fusion of gametes.

127. Answer (2)

Hint: The above given diagram is of *Paramecium*.

Sol.: *Paramecium* possess cilia instead of pseudopodia. Dinoflagellates possess cell wall having cellulose plates on the outer surface.

128. Answer (3)

Hint: Cyanobacteria have chlorophyll a similar to higher plants.

Sol.: Euglenoids have pigments identical to higher plants. Except chlorophyll a, other pigments of cyanobacteria differ from higher plants.

129. Answer (1)

Hint: Viruses are non-cellular organisms.

Sol.: Viruses did not find a place in classification since they are not considered truly 'living'.

130. Answer (2)

Hint: Viruses — Mosaic formation in plants

Sol.: Bacteria — Citrus canker

Viroids — Potato spindle tuber disease

Prions — Mad cow disease in cattle

131. Answer (4)

Hint: Deuteromycetes are also known as imperfect fungi.

Sol.: *Agaricus* (Mushroom) belongs to basidiomycetes.

132. Answer (3)

Hint: Mycoplasma is the smallest living cell.

Sol.: Mycoplasma do not have cell wall. Cell membranes are not made of peptidoglycan.

133. Answer (3)

Hint: Methanogens are responsible for production of biogas.

Sol.: Archaebacteria differ from other bacteria in having a different cell wall structure. Methanogens are capable of converting CO_2, methanol and formic acid into methane.

134. Answer (3)

Hint: Two kingdom classification was given by C. Linnaeus.

Sol.: Five kingdom classification was given by R.H Whittaker.

135. Answer (2)

Hint: Basidiomycetes, produce sexual spores exogenously

Sol.: *Agaricus* belongs to basidiomycetes.

[ZOOLOGY]

136. Answer (3)

Hint: Each artery and vein consists of three layers

Sol.: Basically, each artery and vein consists of three layers. Inner lining, the tunica intima, a middle layer tunica media and external layer tunica externa. The tunica media is comparatively thin in the veins. Pulmonary veins carries oxygenated blood while pulmonary artery carries deoxygenated blood. Partial pressure of O_2 in the pulmonary vein is 95 mm Hg.

137. Answer (1)

Hint: Cardiac output = heart rate × stroke volume

Sol.: Athletes due to continuous training, have greater stroke volume than that of ordinary individuals. Human can alter the stroke volume and cardiac output. Athletes have higher stroke volume to maintain normal cardiac output. If repeated checks of blood pressure of an individual is 140/90 mm Hg or higher, it shows hypertension. Atherosclerosis affects the vessels that supply blood to the heart.

138. Answer (3)

Hint: Angina is acute chest pain.

Sol.:

- Coronary artery disease, often referred to as atherosclerosis, affect the vessels that supply blood to the heart muscle.
- Angina is also called angina pectoris. A symptom of acute chest pain appears when not enough oxygen is reaching the heart muscle.
- Heart failure is sometimes called congestive heart failure because congestion of lungs is one of the main symptoms of this disease.
- In cardiac arrest, heart stops beating.

139. Answer (4)

Hint: Left ventricle pumps blood into aorta

Sol.: Blood pressure is high in arteries as compared to vein. A special coronary system of blood vessels is present in our body exclusively for the circulation of blood to and from the cardiac musculature.

140. Answer (2)

Hint: First heart sound is produced at the beginning of ventricular systole.

Sol.: Lub is first heart sound which is associated with closure of atrio-ventricular valves. Dub is second heart sound which is associated with closure of semilunar valves. Increased blood flow into ventricles by about 30 per cent occurs during the atrial systole.

141. Answer (4)

Hint: Smooth muscle fibres taper at the both ends

Sol.: Human heart has four chambers, two relatively small upper chambers called atria and two larger lower chambers called ventricles. A specialised cardiac musculature called the nodal tissue is also distributed in the heart.

Heart has cardiac muscle fibres.

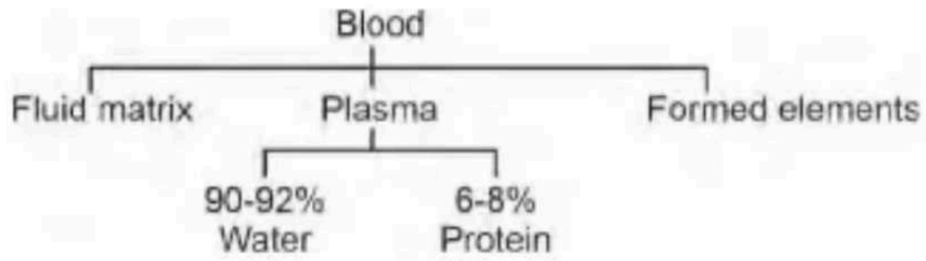
Organ that stores urine is urinary bladder and it has smooth muscle fibres.

Cardiac muscle fibres	**Smooth muscle fibres**
• Branched	• Unbranched
• Cylindrical	• Fusiform
• Striated	• Non-striated

142. Answer (1)

Hint: Fights pathogen

Sol.: Plasma without clotting factors is called serum.



Thus, formed elements do not constitute plasma.

143. Answer (1)

Hint: Universal recipient

Sol.: Persons with AB^{+ve} blood group do not have anti-A and anti-B antibodies in their plasma. They have A, B and Rh antigens on the RBC surface. Rh antigen will be absent on the RBCs of a AB^{-ve} individual.

144. Answer (2)

Hint: Type of connective tissue

Sol.: The atrium and the ventricle of the same side are separated by a thick fibrous tissue called the atrio-ventricular septum. A thin, muscular wall called the inter-atrial septum separates the right and left atria.

145. Answer (4)

Hint: Secreted from mast cells.

Sol.: Certain factors released by the tissue, at the site of injury can initiate coagulation. Ca^{+2}, fibrinogen and prothrombin are involved in the cascade that leads to blood clotting. Serotonin is a vasoconstrictor.

146. Answer (1)

Hint: Pulmonary vein has oxygenated blood

Sol.: Tunica media is thicker in arteries as compared to veins, so their lumen is narrower. In arteries, blood flows always away from heart while in vein it flows towards the heart.

147. Answer (1)

Hint: Cardiac output = heart rate X stroke volume

Sol.: Stroke volume = end diastolic volume – end systolic volume

= 150 mL – 50 mL

Stroke volume = 100 mL

Cardiac output = heart rate × stroke volume

$$\text{Heart rate} = \frac{\text{Cardiac output}}{\text{Stroke volume}} = \frac{7200\ \text{mL}}{100\ \text{mL}}$$

Heart rate = 72 beats per minute

148. Answer (4)

Hint: Leucocytes have short life span

Sol.: In healthy human, heart rate is 72 times per 60 sec. If we reverse the ratio, we will get duration of cardiac cycle. Eosinophils (2-3 per cent) resist infections and they have bilobed nucleus. Leucocytes are also known as white blood cells (WBCs).

Leucocytes are generally short lived.

149. Answer (4)

Hint: CO_2 is more soluble than O_2

Sol.: Exchange of O_2 and CO_2 between blood and tissues is mainly based on pressure/concentration gradient.

Solubility of gases as well as the thickness of membranes involved in diffusion are some of the important factors that affect rate of diffusion of gases.

150. Answer (4)

Hint: It is more than 10

Sol.: A healthy human breathes 12-16 times per minute.

- A healthy individual has 12-16 gms of haemoglobin in every 100mL of blood.
- Neutrophils are the most abundant cell (60-65 per cent) of the total WBCs.
- Monocytes are 6-8 per cent of the total WBCs.
- 90-92 per cent of plasma is water and proteins contribute 6-8 per cent of it.

151. Answer (1)

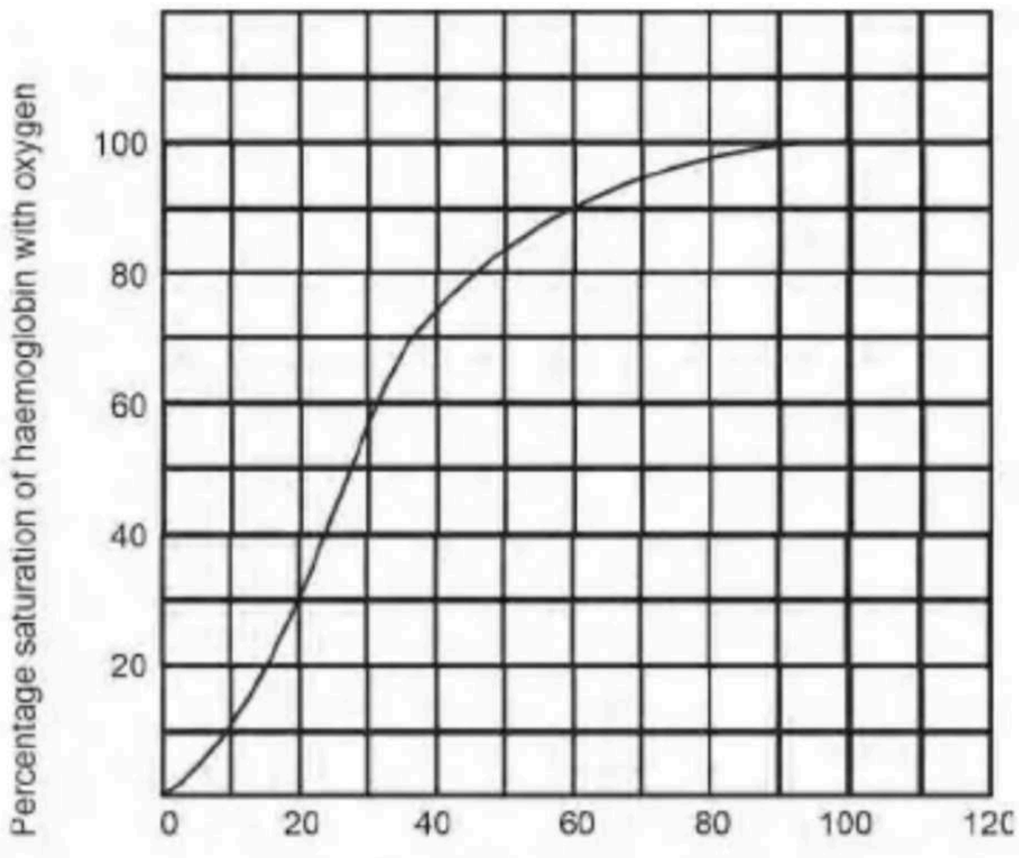
Hint: Higher pCO_2 dissociates oxyhaemoglobin

Sol.: A sigmoid curve is obtained when percentage saturation of haemoglobin with O_2 is plotted against the pO_2. This curve is called the oxygen dissociation curve. At the tissue site where low pO_2, high pCO_2, high H^+ concentration and higher temperature exist, the conditions are favourable for dissociation of oxygen from the oxyhaemoglobin.

152. Answer (1)

Hint: Numerical value of urea excreted out per day.

Sol.: The partial pressure of oxygen is approximately 25-30 mm Hg when haemoglobin is 50% saturated with oxygen.



153. Answer (1)

Hint: Collects deoxygenated blood from upper parts of the body

Sol.: Superior vena cava and inferior vena cava carry deoxygenated blood.

Respiratory Gas	Atmospheric Air	Alveoli	Blood (Deoxy-genated)	Blood (Oxygenated)	Tissues
O_2	159	104	40	95	40
CO_2	0.3	40	45	40	45

154. Answer (2)

Hint: Observed during forceful expiration.

Sol.: Inspiration is initiated by the contraction of diaphragm which increases the volume of thoracic chamber in the antero-posterior axis. The contraction of external inter-costal muscles causes an increase in the volume of the thoracic chamber. We have the ability to increase the strength of inspiration and expiration with the help of additional muscles in abdomen. Contraction of internal inter-costal muscles occur during forceful expiration.

155. Answer (3)

Hint: Medulla oblongata regulates respiration.

Sol.: Aortic arch and carotid artery are peripheral chemoreceptors for regulation of breathing. They can recognise change in CO_2 and H^+ concentration and send necessary signals to the rhythm centre (Medulla oblongata).

156. Answer (1)

Hint: Nasal chamber opens into pharynx

Sol.: The nasal chamber opens into the pharynx, a portion of which is the common passage for food and air. The pharynx opens through the larynx region into the trachea. Trachea divides at the level of 5th thoracic vertebra into a right and left primary bronchi. Each bronchi undergoes repeated division to form the secondary and tertiary bronchi and initial bronchioles ending up in very thin terminal bronchioles. Each terminal bronchiole gives rise to a number of very thin irregular-walled and vascularised bag-like structures called alveoli.

157. Answer (1)

Hint: Venous blood is the deoxygenated blood.

Sol.: Venous blood has more CO_2 and more H^+ which leads to low pH as compared to arterial blood. Pulmonary vein carries oxygenated blood.

158. Answer (1)

Hint: Inspiration and expiration is included in breathing

Sol.: Breathing involves two stages: inspiration of O_2 rich atmospheric air and expiration of CO_2 rich alveolar air.

Cellular respiration is defined as oxidation of food to form CO_2, H_2O and energy.

159. Answer (1)

Hint: Incomplete cartilaginous rings are absent in exchange part.

Sol.: The trachea, primary, secondary and tertiary bronchi and initial bronchioles are supported by incomplete cartilaginous rings. These are absent in terminal bronchioles, alveoli and their ducts (alveolar ducts).

160. Answer (4)

Hint: Carbonic anhydrase

Sol.: Enzyme 'X' is carbonic anhydrase which belongs to class IV category of enzymes. Carbonic anhydrase accelerates the reaction rate by about 10 million times. Carbonic anhydrase facilitates reaction in forward direction in tissue due to high pCO_2, low pO_2 and more H^+. Reaction moves in backward direction at alveoli due to high pO_2, low pCO_2 and less H^+. Carbonic anhydrase is not present in WBCs.

161. Answer (4)

Hint: Spirometer do not measure residual volume

Sol.: The volume of air involved in breathing movements can be estimated by using a spirometer which helps in clinical assessment of pulmonary function. FRC, TLC and RV cannot be measured with the help of a simple spirometer.

162. Answer (3)

Hint: Dome shaped structure

Sol.: Normal expiration is a passive process as relaxation of muscle occurs. The thoracic chamber is formed dorsally by vertebral column, ventrally by sternum, laterally by ribs and on the lower side by dome-shaped diaphragm.

163. Answer (2)

Hint: Respiratory minute volume = Breathing rate × tidal volume.

Sol.: A healthy human breathes 12-16 times/min in resting conditions. Tidal volume is approximately 500 mL.

So, Respiratory minute volume (RMV) =

TV × Respiratory rate

= 500 × (12-16)

= 6000-8000 mL per minute

Inspiratory reserve volume = 2500-3000 mL

Expiratory reserve volume = 1000-1100 mL

Residual volume = 1100-1200 mL

164. Answer (1)

Hint: Intra-pulmonary pressure is less than atmospheric pressure.

Sol.: Inspiration can occur if the pressure within the lungs *i.e.* intra-pulmonary pressure is less than the atmospheric pressure *i.e,* there is a negative pressure in the lungs with respect to atmospheric pressure.

165. Answer (1)

Hint: VC = ERV+TV+IRV

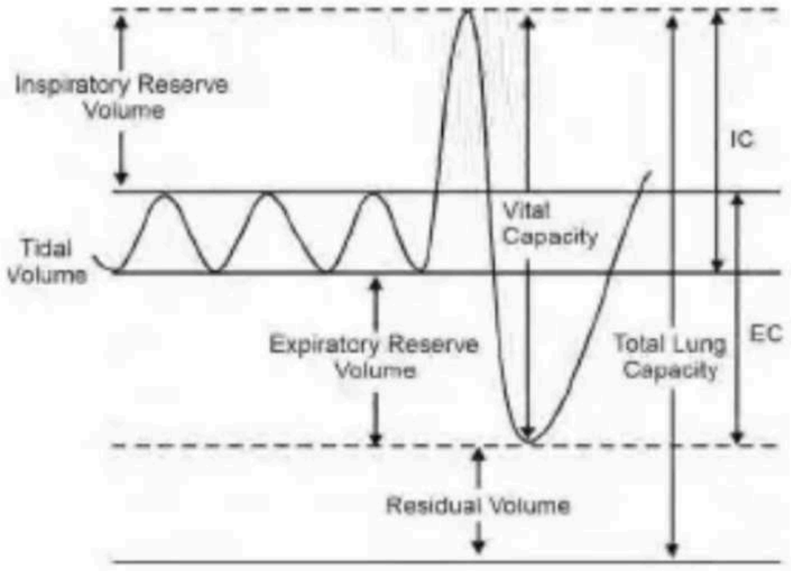
Sol.:EC = TV+ ERV

EC = FRC-RV+TV

TLC = RV+ERV+TV+IRV

VC = ERV+TV+IRV

IC = IRV+TV



166. Answer (1)

Hint: Least toxic form of nitrogenous waste

Sol.: Land snails-Uricotelic.

Mammals, many terrestrial amphibians and marine fishes mainly excrete urea, hence are ureotelic animals.

167. Answer (2)

Hint: Blood vessel which carries blood away from heart.

Sol.: During the process of haemodialysis, the blood drained from a convenient artery is pumped into a dialysing unit called artificial kidney. Blood drained from convenient artery is pumped into a dialysing unit after adding an anticoagulant like heparin.

168. Answer (3)

Hint: Insects have Malpighian tubules.

Sol.: • Protonephridia or flame cells are excretory structures in platyhelminths. (Flatworms *e.g. Planaria*)

- Nephridia are tubular excretory structures of earthworms (*Pheretima*) and other annelids.
- Malpighian tubules are the excretory structures of most of the insects including cockroaches (*Periplaneta*).
- Antennal glands or green glands perform the excretory function in crustaceans like prawns (*Palaemon*).

169. Answer (3)

Hint: Exclude conditions associated with diabetes mellitus.

Sol.: Presence of glucose (glycosuria) and ketone bodies (ketonuria) in urine are indicative of diabetes mellitus. Malfunctioning of kidneys can lead to accumulation of urea in blood, a condition called uremia. An impairment affecting synthesis or release of ADH results in a diminished ability of the kidney to conserve water leading to water loss and dehydration (diuresis).

170. Answer (2)

Hint: Counter current mechanism becomes impaired.

Sol.: Loop of Henle, along with vasa recta of juxtamedullary nephrons participates in counter current mechanism that leads to creation of concentration gradient in the medullary interstitium. So, more dilute urine will be produced upon removal of loop of Henle as less water is reabsorbed

171. Answer (1)

Hint: Osmolarity increases from renal cortex to renal medulla.

Sol.: The flow of filtrate in the two limbs of Henle's loop and vasa recta is in opposite direction and thus forms a counter current. The flow of blood through the two limbs of vasa recta is also in a counter current pattern. The proximity between the Henle's loop and vasa recta, as well as the counter current in them help in maintaining an increasing osmolarity towards the inner medullary interstitium.

172. Answer (2)

Hint: Part of nephron present in cortex of kidney

Sol.: Conditional reabsorption of Na^+ and water takes place in DCT. DCT is also capable of reabsorption of HCO_3^- and selective secretion of hydrogen and potassium ions and NH_3 to maintain the pH and sodium-potassium balance in blood.

173. Answer (3)

Hint: Broad funnel shaped space

Sol.: The DCTs of many nephrons join to form a common collecting duct many of which converge and ultimately open into the renal pelvis through the medullary pyramids in the calyces. Inside the kidney, there are two zones, an outer cortex and an inner medulla. In humans, the excretory system consists of one pair of kidneys, a pair of ureters, a urinary bladder and a urethra. Renal columns are invagination of cortex into medulla.

174. Answer (2)

Hint: Concentrating segment.

Sol.: The ascending limb of loop of Henle is impermeable to water but allows transport of electrolytes actively or passively. Descending limb of loop of Henle is almost impermeable to electrolytes. Counter current mechanism operates in Henle's loop and vasa recta. DCT has hypotonic glomerular filtrate with respect to blood plasma. The epithelial cells of Bowman's capsule called podocytes are arranged in an intricate manner.

175. Answer (1)

Hint: 70-80 per cent of electrolytes and water are reabsorbed by PCT.

Sol.: Glomerular filtrate in PCT is isotonic to blood plasma. Nearly all of the essential nutrients and 70-80 per cent of electrolytes and water are reabsorbed by this segment. Fluid present in descending limb of loop of Henle is hypertonic to the blood plasma.

176. Answer (3)

Hint: Glomerular filtrate is proteinless plasma

Sol.: Blood is filtered so finely through filtration slits that almost all the constituents of the plasma except the proteins pass onto the lumen of the Bowman's capsule. Therefore, it is considered as a process of ultra filtration.

Globulins and fibrinogens are proteins.

177. Answer (3)

Hint: Renin stimulates glomerular blood flow

Sol.: A fall in GFR can activate JG cells to release renin which converts angiotensinogen in blood to angiotensin I and further to angiotensin II. Angiotensin II, being a powerful vasoconstrictor, increases the glomerular blood pressure and thereby bring the GFR back to normal.

178. Answer (1)

Hint: The ascending limb is impermeable to water

Sol.: The ascending limb of loop of Henle is impermeable to water but allows transport of electrolytes actively or passively. Therefore, as the concentrated filtrate pass upward, it gets diluted due to the passage of electrolytes to the medullary fluid.

179. Answer (1)

Hint: Close proximity between Henle's loop and vasa recta

Sol.: NaCl transported by the ascending limb of Henle's loop is exchanged with the descending limb of vasa recta. NaCl is returned to the interstitium by the ascending portion of vasa recta. The proximity between Henle's loop and vasa recta as well as counter current in them help in maintaining an increasing osmolarity towards inner medullary interstitium *i.e.*, from 300 mOsmol L^{-1} to 1200 mOsmol L^{-1}. This gradient is mainly caused by NaCl and urea.

180. Answer (4)

Hint: Secretion of sebaceous gland

Sol.: Sweat produced by the sweat glands is a watery fluid containing NaCl, small amounts of urea, lactic acid, *etc*. Though the primary function of sweat is to facilitate a cooling effect on body surface, it also helps to remove some of the wastes.

Sebaceous glands eliminate certain substances like sterols, hydrocarbons and waxes through sebum.

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